HOOVER INSTITUTION | Stanford University

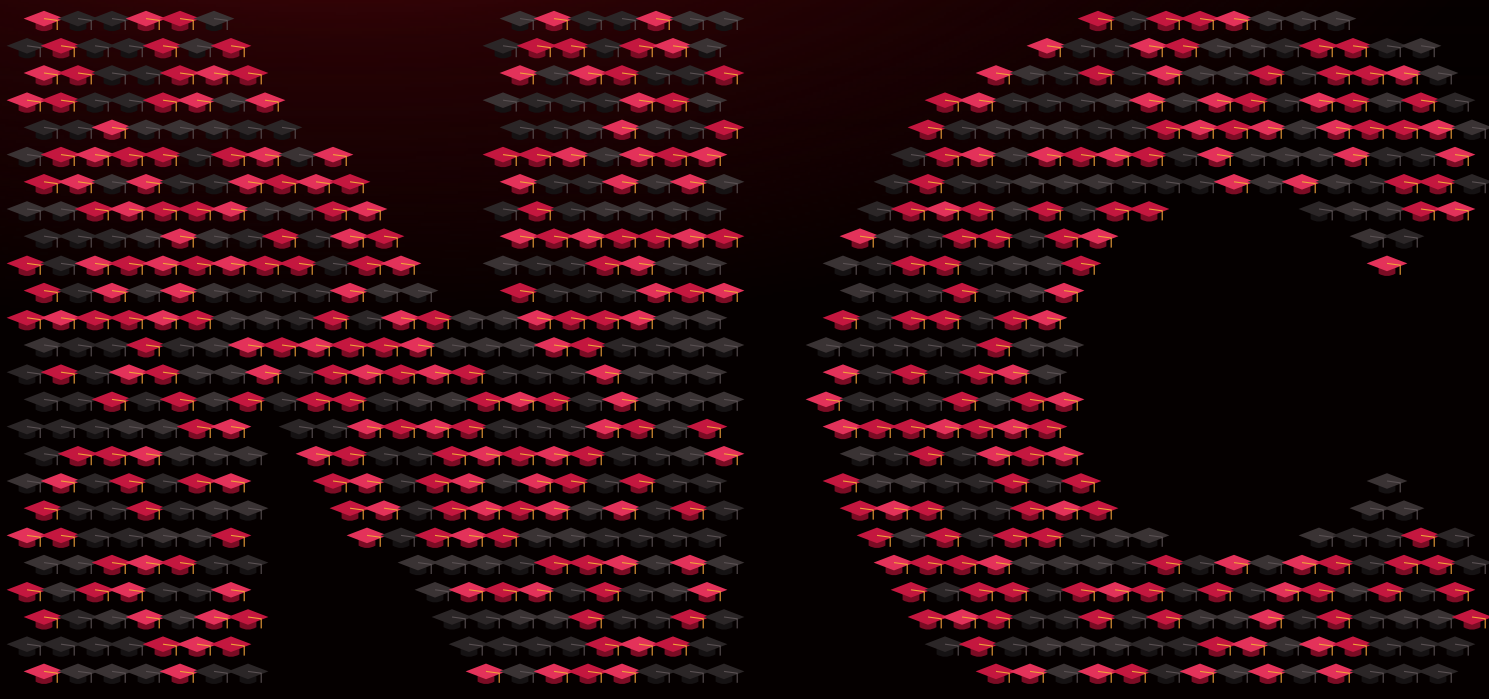
HAI Stanford University Human-Centered Artificial Intelligence

NOVEMBER 2026

The 2026 North Carolina Talent Profile

Emerson Johnston
Amy Zegart

The bright caps, **49 in every 100**, mark the international share of North Carolina's STEM master's degrees.



This page intentionally left blank.

HOOVER INSTITUTION | Stanford University

HAI Stanford University Human-Centered Artificial Intelligence

The 2026 North Carolina Talent Profile

By Emerson Johnston and Amy Zegart

NOVEMBER 2026

A publication jointly produced by the Hoover Institution and the Institute for Human-Centered Artificial Intelligence at Stanford University.

Table of Contents

OVERVIEW

At a Glance 6

SECTIONS

Section 1 International Dependence 8
Section 2 STEM Degrees 20
Section 3 Where Students Come From 26
Section 4 K-12 Preparation 34
Section 5 Graduate Research 36
Section 6 Do Graduates Stay? 38
Section 7 Research Funding 40
Section 8 H-1B Jobs and Green Cards 42
Section 9 STEM Workforce 46
Section 10 Patents 48
Section 11 Competitors 50
Section 12 Fields of Study 52
Section 13 Students and Money 62
Section 14 Student Outcomes 70
Section 15 Faculty 74
Section 16 Institutions 78

BACK MATTER

About these Data 82
By the Numbers 84
Sources 94

Authors

Emerson Victoria Johnston is a research assistant with the Hoover Institution's Technology Policy Accelerator. A Knight-Hennessy Scholar, she holds a master's in international policy from Stanford University, and is on leave from its master's program in the history of science, technology, environment, and medicine.

Dr. Amy Zegart is the Morris Arnold and Nona Jean Cox Senior Fellow at the Hoover Institution, director of Hoover's Technology Policy Accelerator, and a senior fellow and associate director at Stanford University's Institute for Human-Centered Artificial Intelligence.

Data come from the Integrated Postsecondary Education Data System (IPEDS), the Department of Homeland Security's SEVIS by the Numbers, the National Assessment of Educational Progress, NSF's Survey of Graduate Students and Postdoctorates and Survey of Earned Doctorates, the Department of Labor's Office of Foreign Labor Certification, the USCIS H-1B Employer Data Hub, the Census Bureau's American Community Survey and USPTO patent data via PatentsView, processed in Python; every figure and table is generated from those sources, and each one names its source.

About This Profile

This profile is a companion to *The 2026 US Talent Report: Benchmarking America's STEM Pipeline Amid the Great Talent Competition*. It applies the report's measures to North Carolina: the students the state develops at home, those it draws from abroad, and how both compare with the nation and with the states closest to it in size.

Degrees, enrollment, prices, outcomes and faculty come from IPEDS, the federal census of U.S. colleges and universities, harmonized across 1980–2025. Countries of citizenship come from SEVIS by the Numbers, the Department of Homeland Security's monthly count of active F-1 and M-1 students. Each figure names its source. Peer states are the four closest in total fall enrollment.

How to Cite this Profile

Emerson V. Johnston and Amy Zegart, “The 2026 North Carolina Talent Profile,” Technology Policy Accelerator, Hoover Institution, and the Institute for Human-Centered Artificial Intelligence, Stanford University, Stanford, CA, November 2026.

Public Data and Tools

The data behind every chart come from federal sources: the Integrated Postsecondary Education Data System (IPEDS), published by the National Center for Education Statistics, as the backbone, together with NAEP, NSF's surveys of graduate students, doctorates and research funding, USAspending.gov, DHS and ICE SEVIS data, the Department of Labor, USCIS, the Census Bureau's American Community Survey and Post-Secondary Employment Outcomes, and USPTO PatentsView. Each figure names its source, and the Sources page lists all of them. An interactive version of this profile accompanies the report.

To share any charts and figures, proper attribution is required. This profile is licensed under Attribution-NoDerivatives 4.0 International (CC BY-ND 4.0). Please cite as: The 2026 North Carolina Talent Profile.

AT A GLANCE

What North Carolina's numbers say

1. **International students earn 49% of North Carolina's STEM master's degrees,** 3.1 points below the U.S. (52%) in 2025, ranking 20th of 51 on dependence. The share is 39% for STEM research doctorates (U.S. 43%) and 4.3% for STEM bachelor's (U.S. 6.6%).
2. **North Carolina awarded 2.42K STEM master's degrees to international students in 2025,** 2.1% of the U.S. total and 17th of 51 among states. Since 2015 the count changed +94%, against 2.2× nationally.
3. **The heaviest dependence is in engineering technology,** where international students earned 66% of master's degrees (533 degrees; U.S. 54%). 2 STEM fields with at least 50 master's graduates are majority-international.
4. **Among states of similar size,** North Carolina ranks 3rd of 5 on the international share of STEM master's: North Carolina 49%, Virginia 39%, Georgia 44%, Ohio 58%, Pennsylvania 54%.
5. **79% of North Carolina's first-time students in fall 2024 were state residents** and 1.7% came from abroad. More students arrived from other states (16K) than residents left (12.8K), a net gain of 3.2K; the top destination for those leaving was Virginia.

49%

of the STEM master's degrees North Carolina awards go to international students (U.S. 52%).

SCORECARD

Nine numbers against the nation

582K #10
Students enrolled, 2024
U.S. **19.7M** · +3.4% since 2015

2.42K #17
International STEM master's, 2025
U.S. **117K** · +94% since 2015

49% #20
International share of STEM master's, 2025
U.S. **52%** · +11.4 pp since 2015

14% #24
International share of grad students, 2024
U.S. **17%** · +1.8 pp since 2015

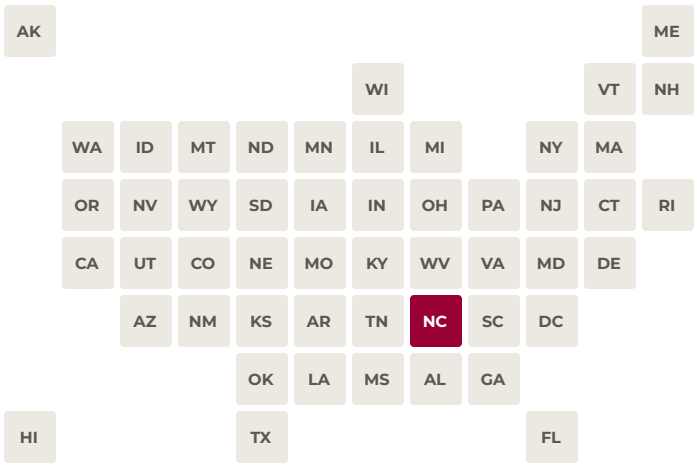
26% #13
STEM share of bachelor's, 2025
U.S. **24%** · +6.4 pp since 2015

66% #22
Six-year graduation rate, 2024
U.S. **65%** · +5.0 pp since 2015

$15.6K #9
Appropriations per student, 2024
U.S. **$11.4K** · +12% since 2015

79% #11
First-time students from in state, 2024
U.S. **–** · −3.0 pp since 2016

$95.1K #28
Average faculty salary, 2024
U.S. **$106K** · −6.3% since 2016



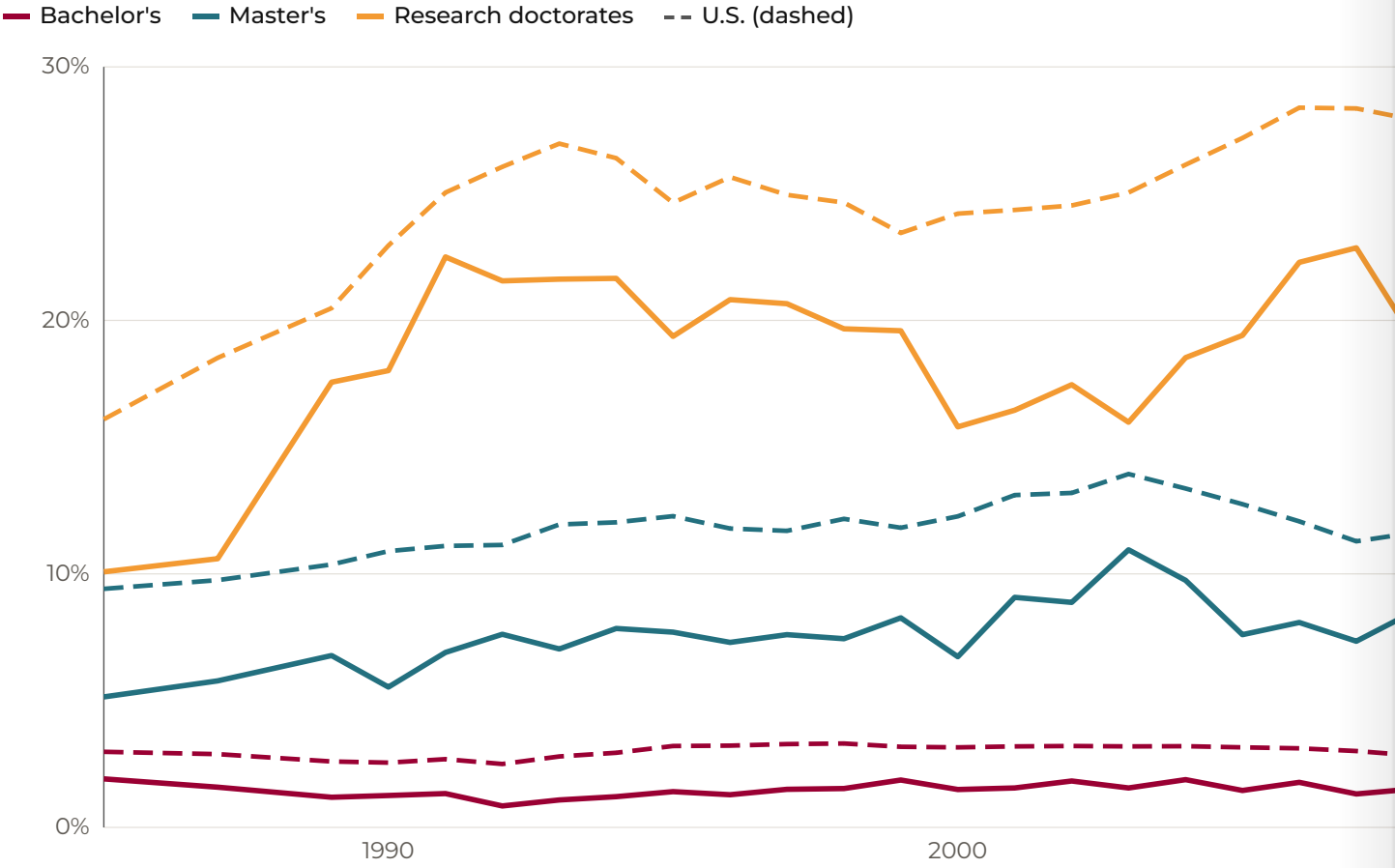
Badges give North Carolina's rank among the 50 states and DC (1 = highest). Changes are since 2015.

SECTION 1

International Dependence

North Carolina ranks 20th of 51 on the international share of STEM master's, at 49% in 2025 against 52% for the U.S.

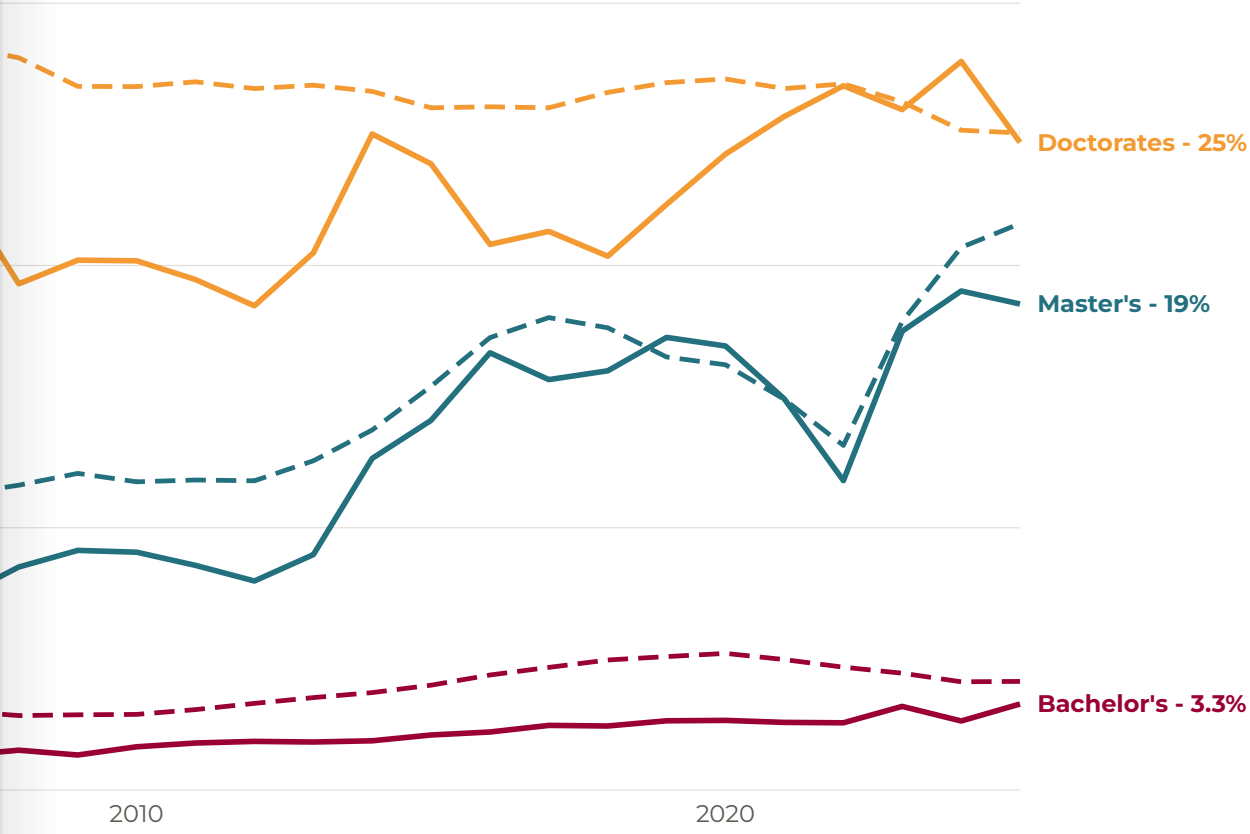
Figure 1 What Share of North Carolina's Degrees Go to International Students?



International share of degrees awarded, all fields; state solid, U.S. dashed. Data from NCES, IPEDS Completions, 1985–2025

International students make up 14% of North Carolina's graduate enrollment and 2.1% of its undergraduates. Dependence climbs with each degree: 3.3% of bachelor's, 19% of master's and 25% of research doctorates go to international students.

In STEM the shares are higher still: 4.3% of STEM bachelor's, 49% of STEM master's and 39% of STEM doctorates. The state awards 2.1% of all U.S. international STEM master's, 17th of 51.



5.

14%

Int. share of grad students, 2024

U.S. **17%** · **24th** of 51 · +1.8 pp since 2015

19%

Int. share of all master's, 2025

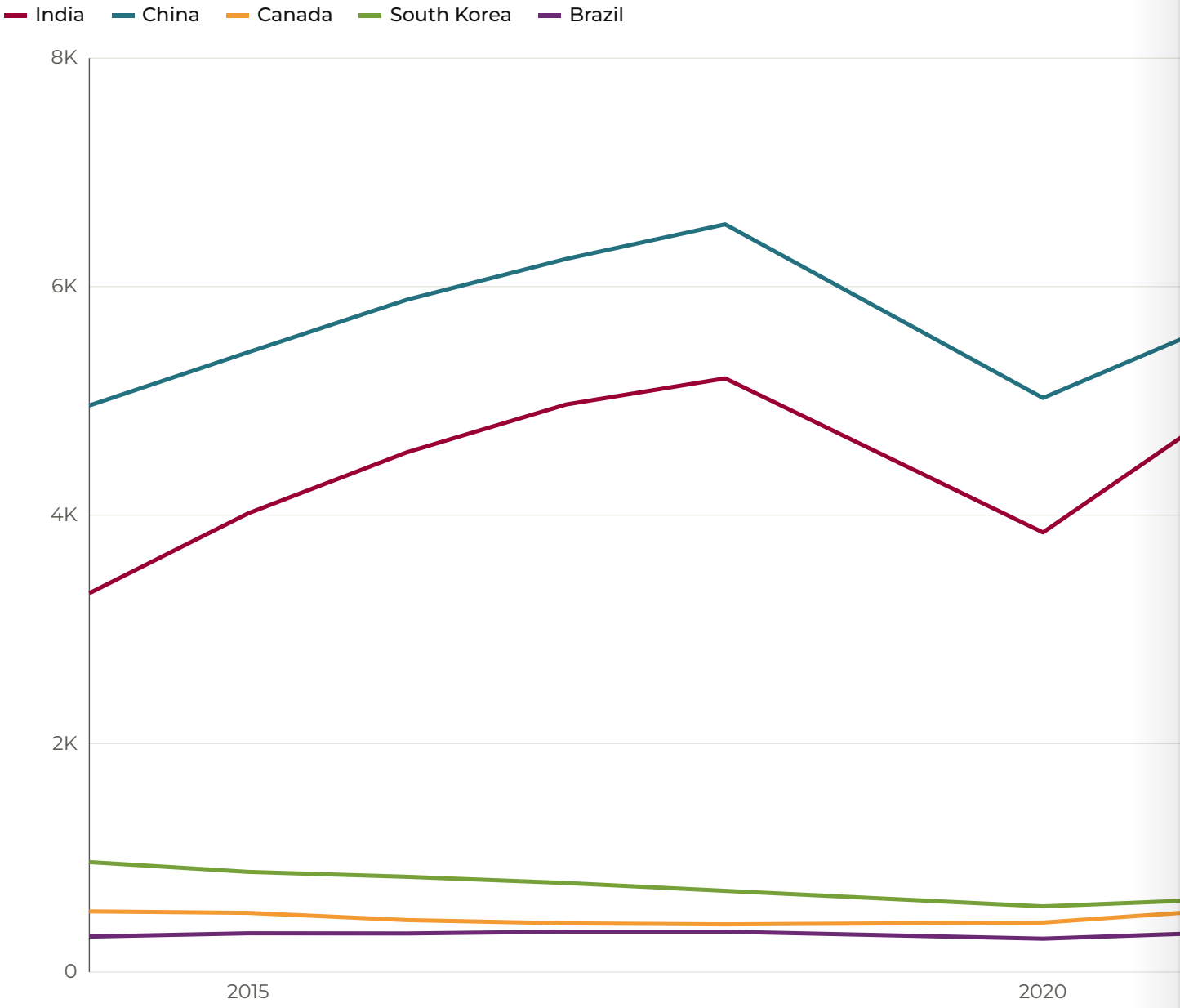
U.S. **22%** · **21st** of 51 · +4.4 pp since 2015

49%

Int. share of STEM master's, 2025

U.S. **52%** · **20th** of 51 · +11.4 pp since 2015

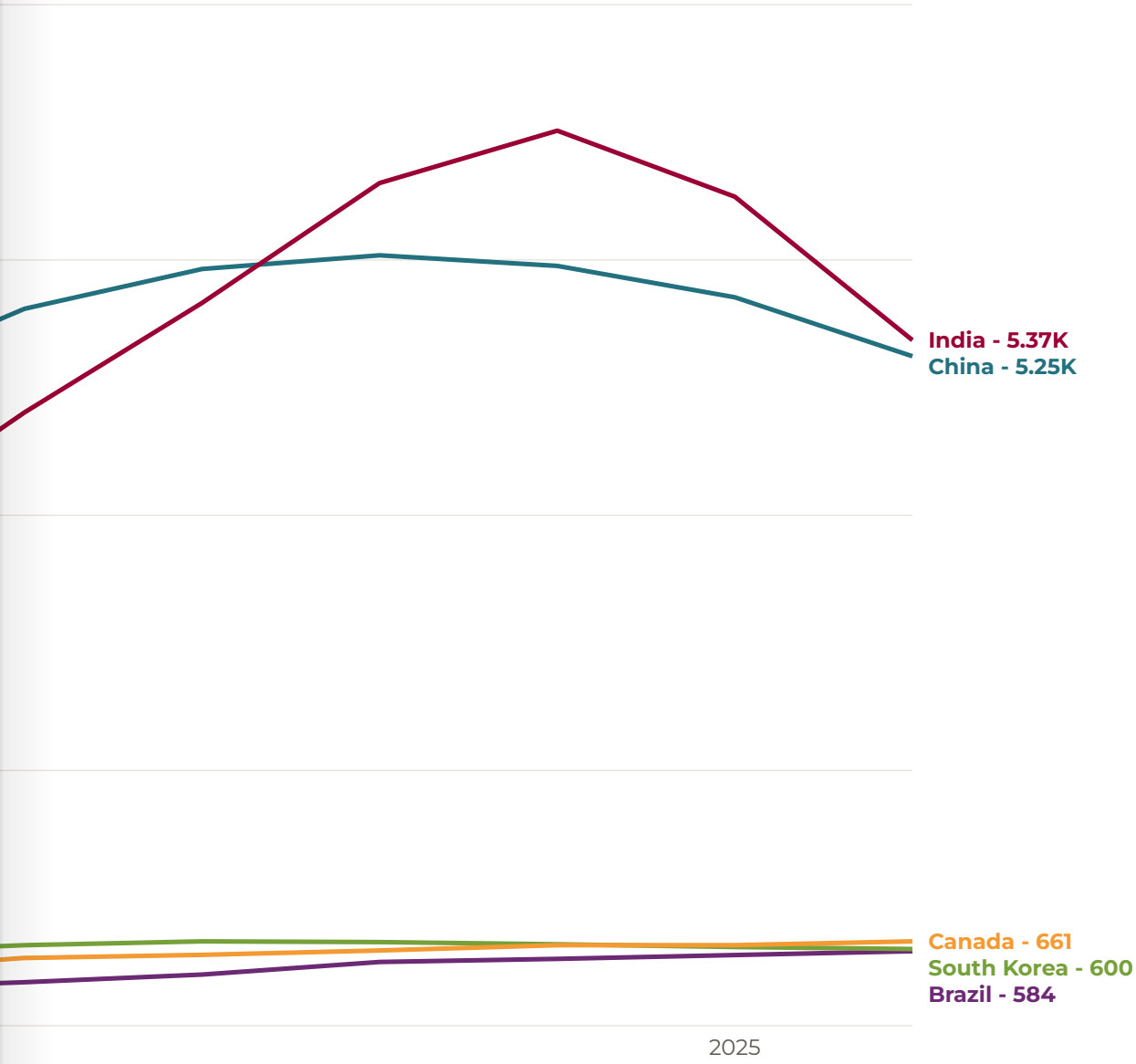
Figure 2 Where Do North Carolina's International Students Come From?



Active F-1 and M-1 students by country of citizenship, the five largest in October 2026; one fall snapshot per year (the mor

In October 2026, North Carolina had 23.2K active international students in SEVIS, 16th of 51. India accounted for 23% of them and China for 23%, against 27% and 18% nationally.

Students came from 183 countries, but the two largest supplied 46%. The state's total is up 22% since 2015, when it stood at 19K.



nth closest to November), October 2014–October 2026. All education levels. Data from DHS, SEVIS by the Numbers.

23.2K

Active international students, October 2026

16th of 51 · U.S. **1.19M**

23%

From India

U.S. **27%**

76%

In STEM fields

U.S. **63%**

SECTION 1 · INTERNATIONAL DEPENDENCE

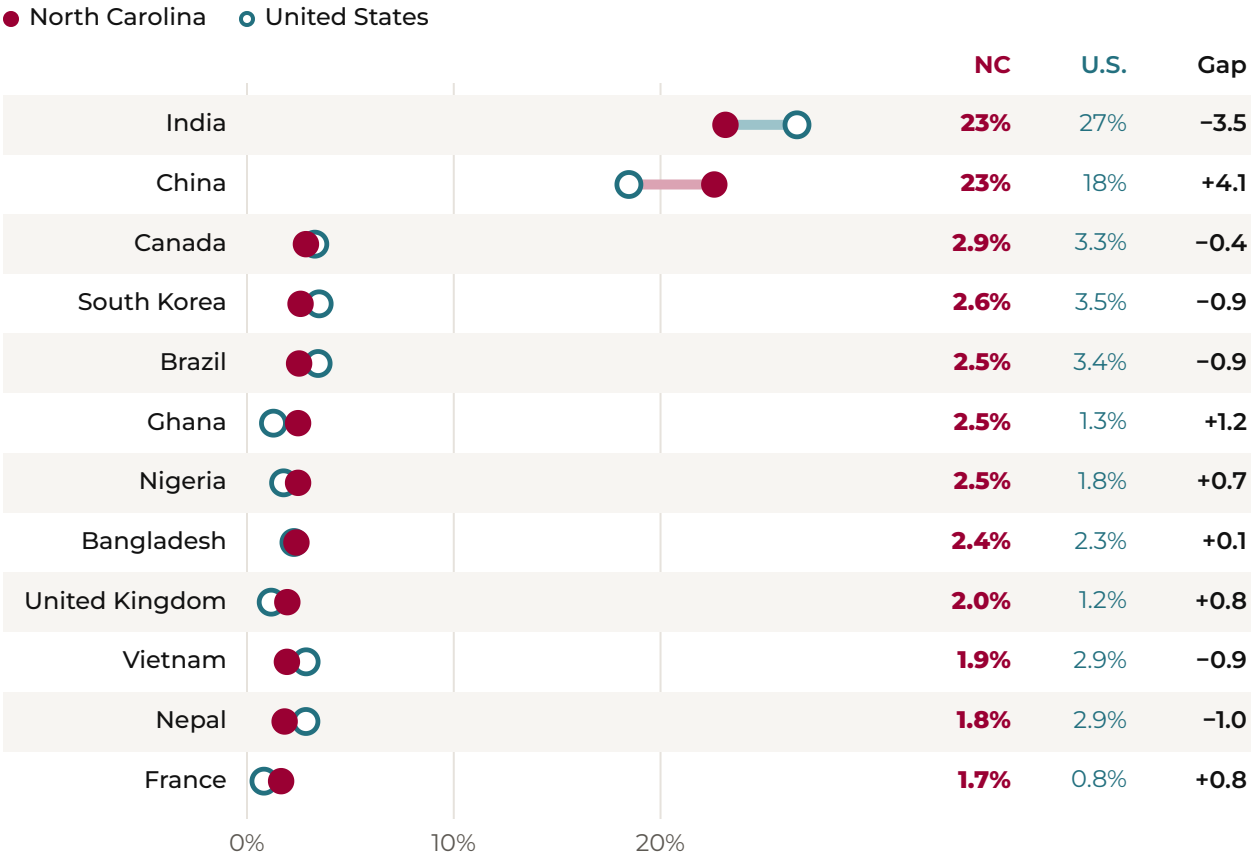
How Does North Carolina's Mix of Countries Compare With the Nation's?

In October 2026, North Carolina had **23.2K** active F-1 and M-1 students, 16th among the 50 states and DC and **1.9%** of the U.S. total. They came from 183 countries.

India supplied **23%** of them, against 27% nationally, and China **23%**, against 18%. Canada (2.9%), South Korea (2.6%), and Brazil (2.5%) followed.

Together the two largest countries supplied **46%** of the state's international students (U.S. 45%), 19th among the states, against 50% in 2015.

Figure 3 Is North Carolina's Mix of Countries Like the Nation's?



Share of active F-1 and M-1 students, all levels. Data from DHS, SEVIS by the Numbers, October 2026.

23.2K
Active international students, October 2026
16th of 51 · U.S. **1.19M**

23%
From India
U.S. **27%**

23%
From China
U.S. **18%**

183
Countries represented
U.S. **219**

The state's count is down 7.0% from 2025 and up 22% since 2015, when it stood at 19K. Its share of all U.S. international students is up 0.4 points since 2015, to **1.9%**.

Since 2015, the number of students from India went from 4.02K to 5.37K, China's from 5.43K to 5.25K, and Canada's from 517 to 661.

Over the same years, the number from South Korea went from 876 to 600 and Brazil's from 339 to 584. The table gives each country's count for the first, previous and latest years.

Table 1 International Students in North Carolina, 2015–2026

	2015	2025	2026	Change since 2015	U.S. 2026	Rank
ALL INTERNATIONAL STUDENTS						
Active F-1 and M-1 students	19K	24.9K	**23.2K**	+22%	1.19M	16th
Share of U.S. active F-1/M-1 students	1.6%	2.0%	**1.9%**	+0.4 pp	–	16th
Countries of citizenship represented	177	179	**183**	+3.4%	219	9th
Share from the two largest countries of origin	50%	49%	**46%**	−4.0 pp	45%	19th
LARGEST COUNTRIES OF CITIZENSHIP (STUDENTS)					**U.S. SHARE**	**STATE SHARE**
India	4.02K	6.5K	**5.37K**	+34%	27%	23%
China	5.43K	5.71K	**5.25K**	−3.3%	18%	23%
Canada	517	630	**661**	+28%	3.3%	2.9%
South Korea	876	617	**600**	−32%	3.5%	2.6%
Brazil	339	553	**584**	+72%	3.4%	2.5%

Active F-1 and M-1 students, one fall snapshot per year (the month closest to November; 2019 has none; the latest year is October 2026). Rank among the 50 states and DC. Data from DHS, SEVIS by the Numbers.

Figure 4

Which Countries Supply North Carolina's International STEM Students?

17.6K of North Carolina's international students were in STEM fields in October 2026, **76%** of the total against 63% nationally, up 89% since 2015.

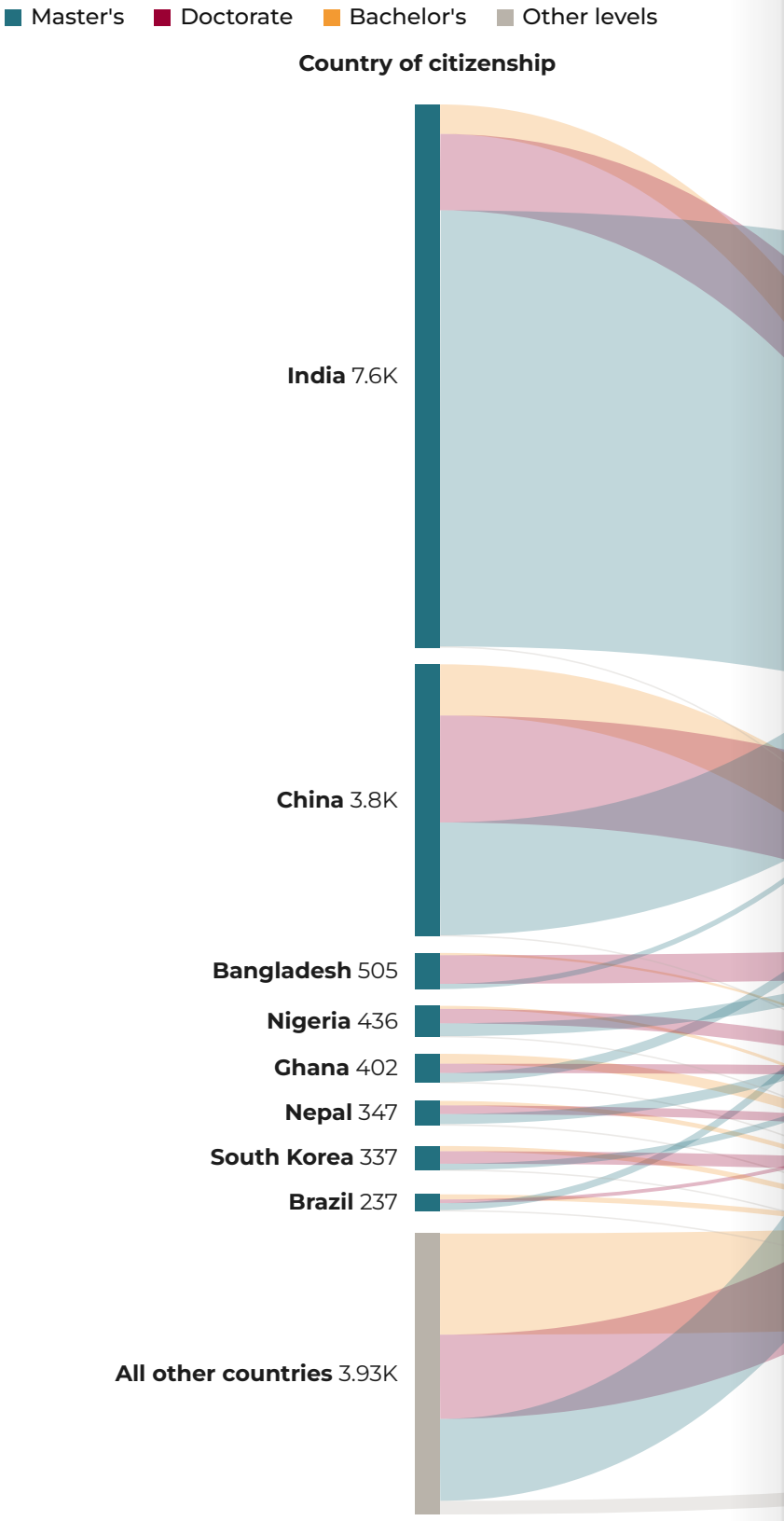
Master's students were the largest group, **9.54K**. India supplied 64% of them and China 17%.

STEM doctoral students went from 2.84K in 2015 to **4.82K**, and STEM master's students from 4.44K to 9.54K.

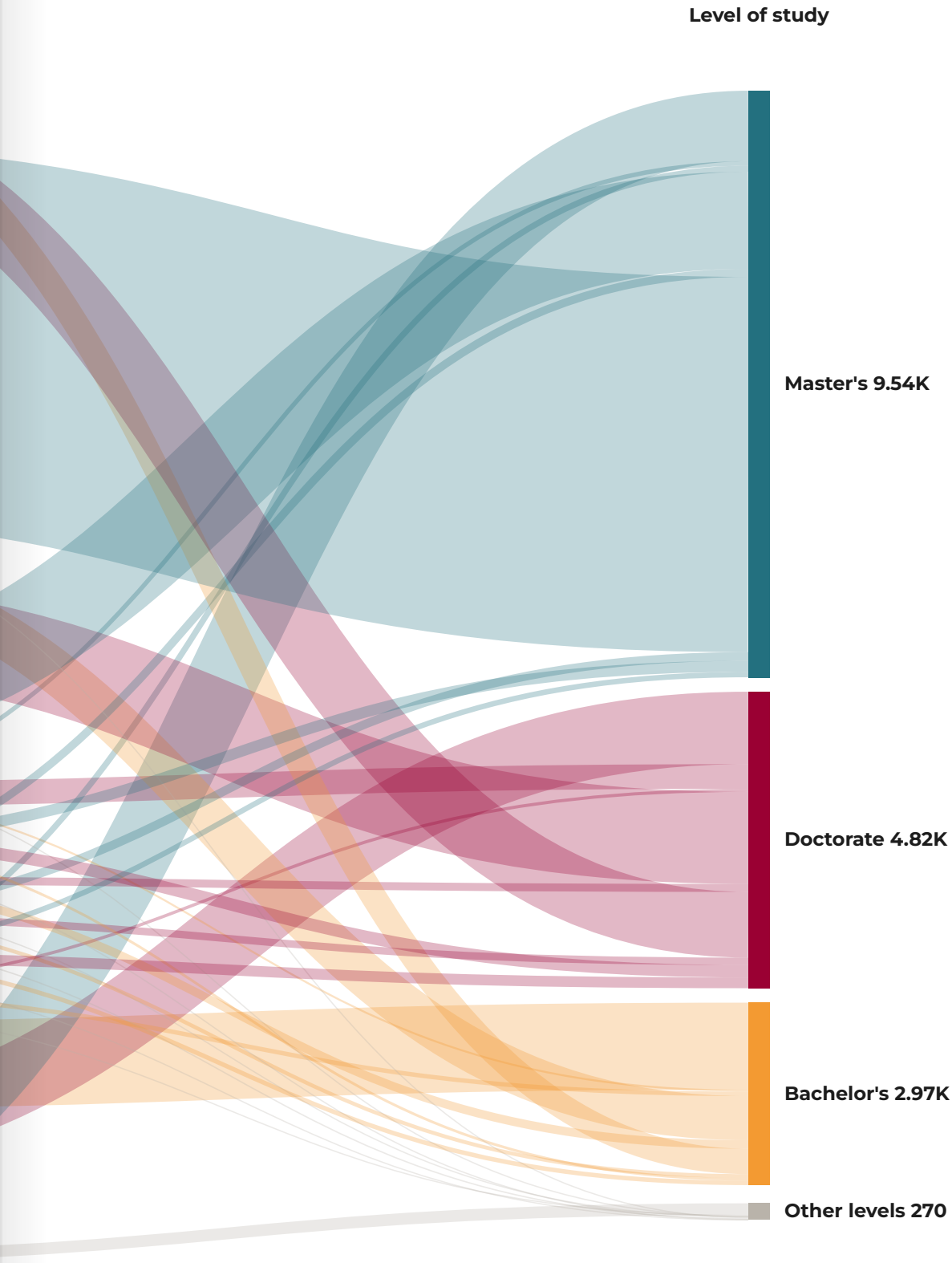
17.6K
Active STEM F-1/M-1 students, 2026
U.S. **749K** · **14th** of 51 · +89% since 2015

76%
Share in STEM fields
U.S. **63%**

9.54K
Active STEM master's students, 2026
U.S. **393K** · **14th** of 51 · 2.15× since 2015



Active F-1 and M-1 students in STEM fields by country of citizenship an
basis as its all-student pages, so the STEM count here exceeds the all-



d level of study, October 2026; the eight largest countries are shown. DHS does not draw its STEM pages on the same evel count for India (7.6K against 5.37K). Data from DHS, SEVIS by the Numbers, October 2026.

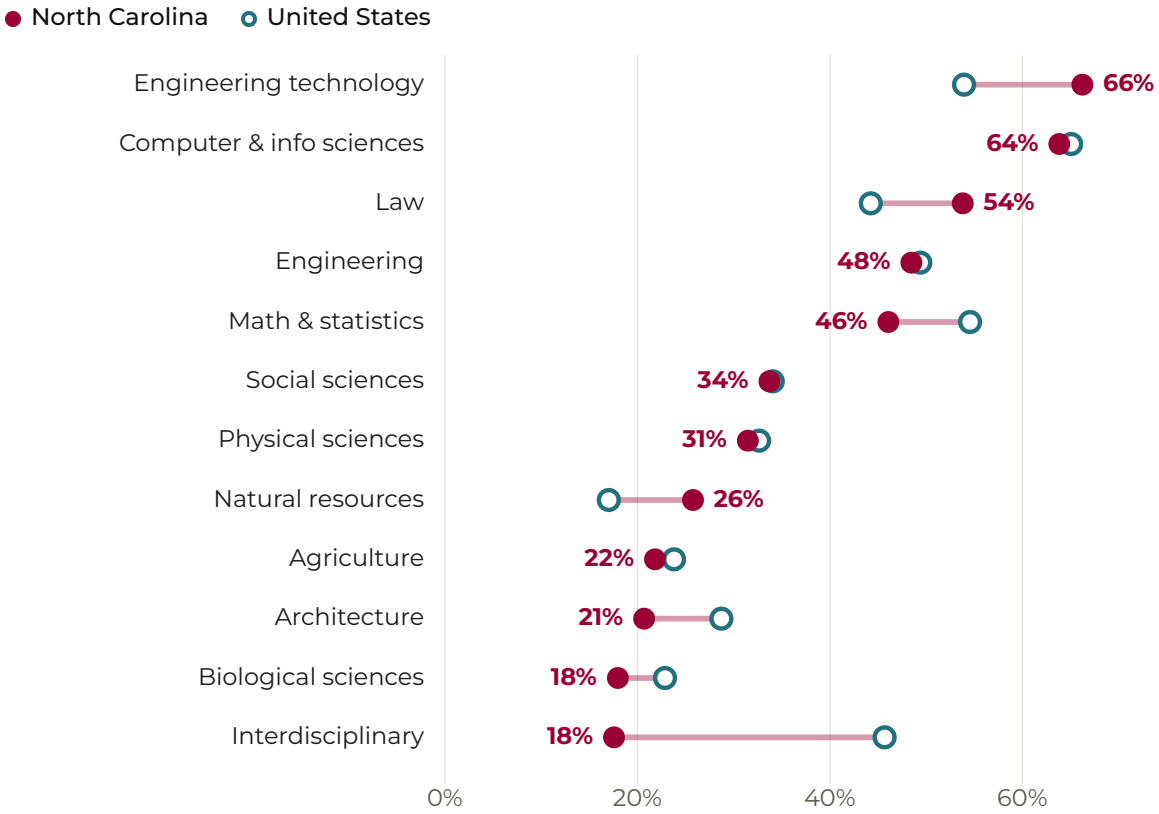
SECTION 1 · INTERNATIONAL DEPENDENCE

How International Are North Carolina's Master's Fields?

Each row is a master's field with at least 30 graduates in North Carolina in 2025, ranked by the share of degrees earned by international students (U.S. nonresidents on temporary visas). The filled dot is the state's share; the open dot is the national share for the same field.

Highest international shares. International students earned the largest shares of master's degrees in engineering technology (66%, against 54% for the U.S.), computer & info sciences (64%, against 65% for the U.S.), and law (54%, against 44% for the U.S.).

Figure 5 Which Master's Fields Depend Most on International Students?



International share of master's degrees by field, 2025; fields with 30+ graduates. Data from NCES, IPEDS Completions.

19%
Int. share of all master's, 2025
U.S. **22%** · **21st** of 51

49%
Int. share of STEM master's, 2025
U.S. **52%** · **20th** of 51

64%
Int. share of CS master's, 2025
U.S. **65%** · **25th** of 47

The table gives the international share of degrees in each STEM field at three levels: North Carolina first, the United States after the slash. The master's column is highlighted. A dash means the state awarded no degrees in the field at that level.

Highest international share among STEM master's fields. International students earned 66% of the master's degrees in engineering technology, against 54% for the U.S.

Fields with an international majority. International students earned more than half of the master's degrees in 2 STEM fields.

Table 2 How International Is Each STEM Field?

Field	Bachelor's	Master's	Doctorates
Computer & info sciences	6.4% / 8.6%	**64% / 65%**	64% / 56%
Engineering	4.2% / 7.2%	**48% / 49%**	54% / 55%
Engineering technology	1.3% / 4.5%	**66% / 54%**	– / 42%
Math & statistics	14% / 17%	**46% / 55%**	47% / 53%
Physical sciences	2.5% / 6.0%	**31% / 33%**	33% / 38%
Biological sciences	2.6% / 3.4%	**18% / 23%**	18% / 23%
Natural resources	1.3% / 2.0%	**26% / 17%**	30% / 27%
Architecture	1.2% / 7.3%	**21% / 29%**	– / 55%

International share in North Carolina / U.S., 2025. Data from NCES, IPEDS Completions.

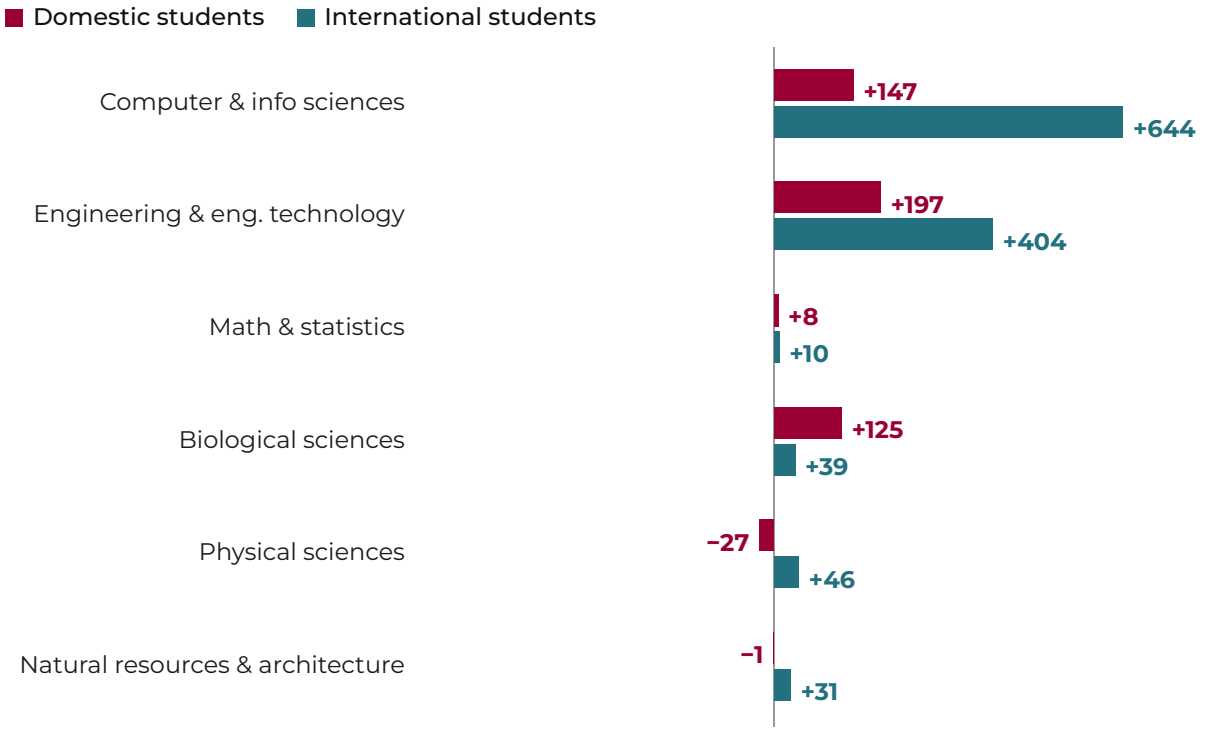
SECTION 1 · INTERNATIONAL DEPENDENCE

How Has North Carolina's STEM Master's Output Grown?

Each pair of bars is the change in the number of STEM master's degrees awarded in North Carolina between 2015 and 2025 in one field: to domestic students (U.S. citizens and permanent residents) and to international students. Bars to the left of the axis are declines.

STEM master's degrees. The state awarded 4.94K in 2025 and 3.32K in 2015, a change of +1.17K for international students and +449 for domestic students.

Figure 6 Where Did the STEM Master's Growth Come From?



Change in STEM master's degrees by field, 2015–2025. Data from NCES, IPEDS Completions.

2.42K
International STEM master's, 2025
U.S. **117K** · **17th** of 51

22%
STEM share of master's, 2025
U.S. **23%** · **24th** of 51

2.1%
Share of U.S. international STEM master's, 2025
17th of 51

4.1
STEM master's per STEM research doctorate, 2025
U.S. **6.4** · **37th** of 51

The table gives the number of STEM master's degrees awarded in North Carolina in each field in 2015 and 2025, the change between the two years, the number awarded to international students in 2025, and international students' share of the field's degrees in each year.

STEM share of master's degrees. The share was 22% in 2025, against 23% for the U.S., 24th among the states, down 0.2 points from 22% in 2024 and up 2.4 points since 2015, when it was 19%.

Table 3 How Many STEM Master's Degrees Does Each Field Award?

Field	2015	2025	Change	International 2025	Int. share 2015	Int. share 2025
Computer & info sciences	854	**1.65K**	+93%	1.05K	48%	64%
Engineering & eng. technology	1.2K	**1.8K**	+50%	968	47%	54%
Math & statistics	249	**267**	+7.2%	123	45%	46%
Biological sciences	376	**540**	+44%	97	15%	18%
Physical sciences	232	**251**	+8.2%	79	14%	31%
Natural resources & architecture	406	**436**	+7.4%	105	18%	24%

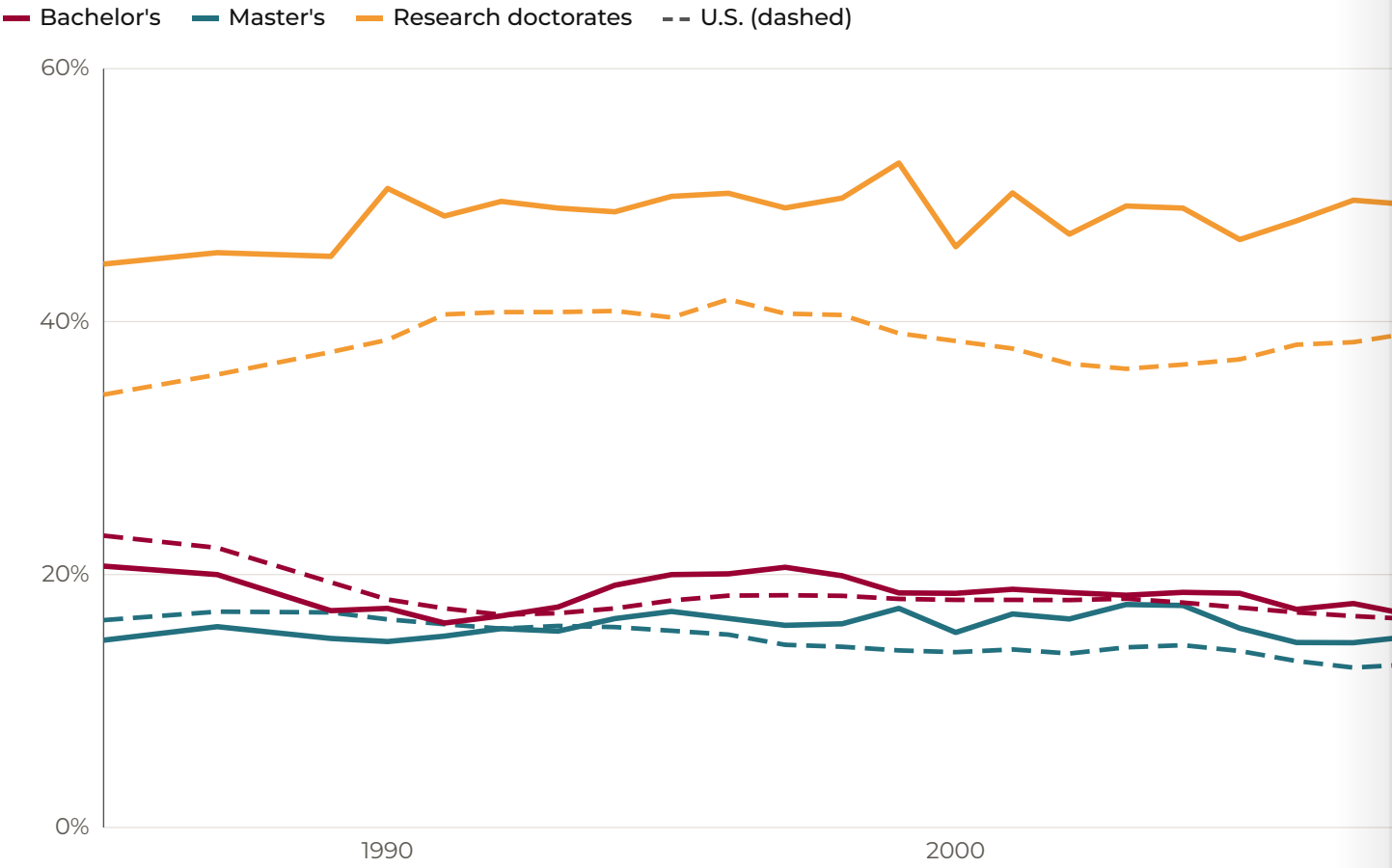
STEM master's degrees awarded in North Carolina, all students and international students, 2015 and 2025. Data from NCES, IPEDS Completions.

SECTION 2

STEM Degrees

North Carolina ranks 13th of 51 on the STEM share of bachelor's, at 26% in 2025 against 24% for the U.S.

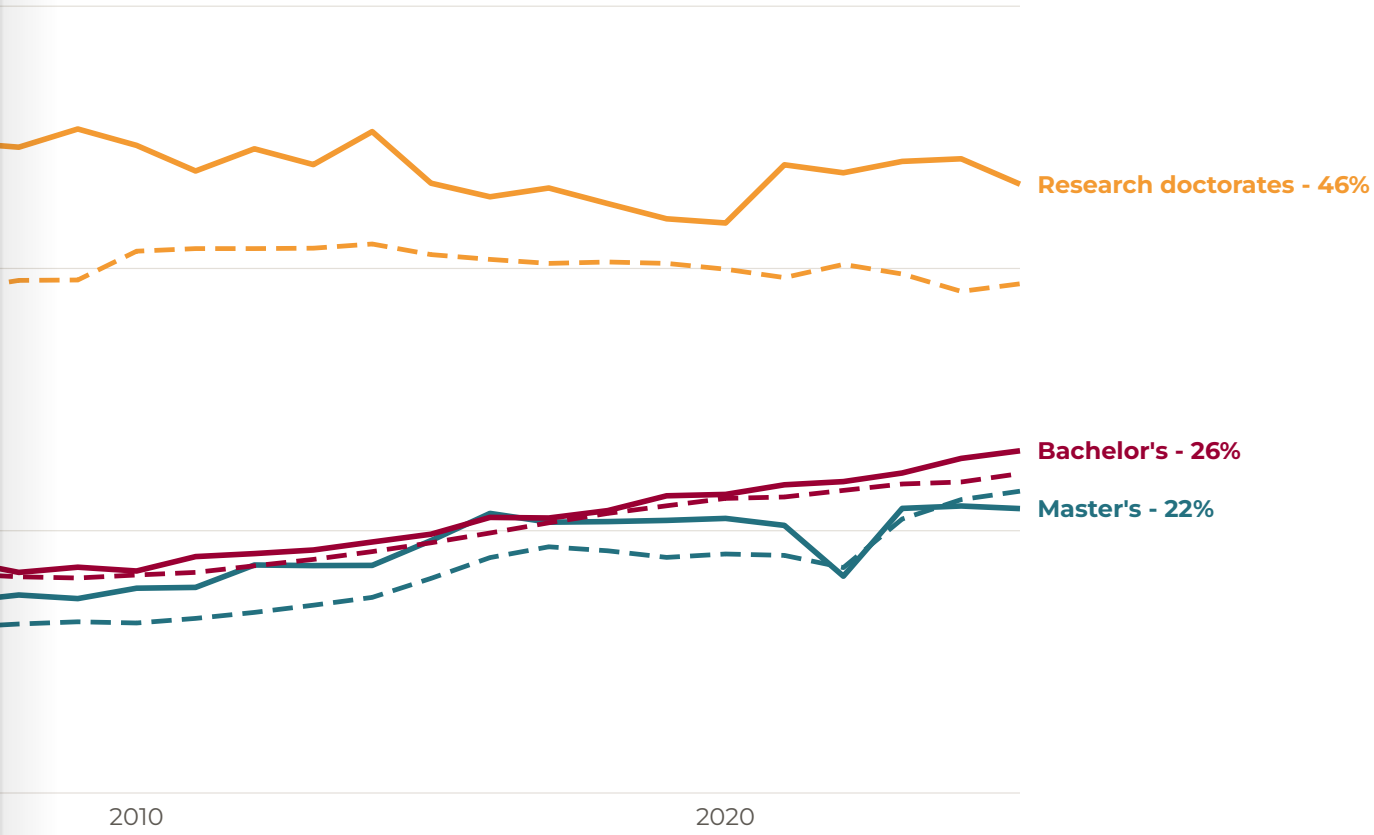
Figure 7 How Much of North Carolina's Degree Output Is STEM?



STEM share of degrees by level; state solid, U.S. dashed. Data from NCES, IPEDS Completions.

STEM fields account for 26% of North Carolina's bachelor's degrees, 22% of its master's and 46% of its research doctorates (U.S. 24%, 23% and 39%).

The STEM share of master's degrees is up 2.4 points since 2015, and the state ranks 24th of 51 on it. On the STEM share of bachelor's degrees it ranks 13th of 51.



26%

STEM share of bachelor's, 2025

U.S. **24%** · **13th** of 51 · +6.4 pp since 2015

22%

STEM share of master's, 2025

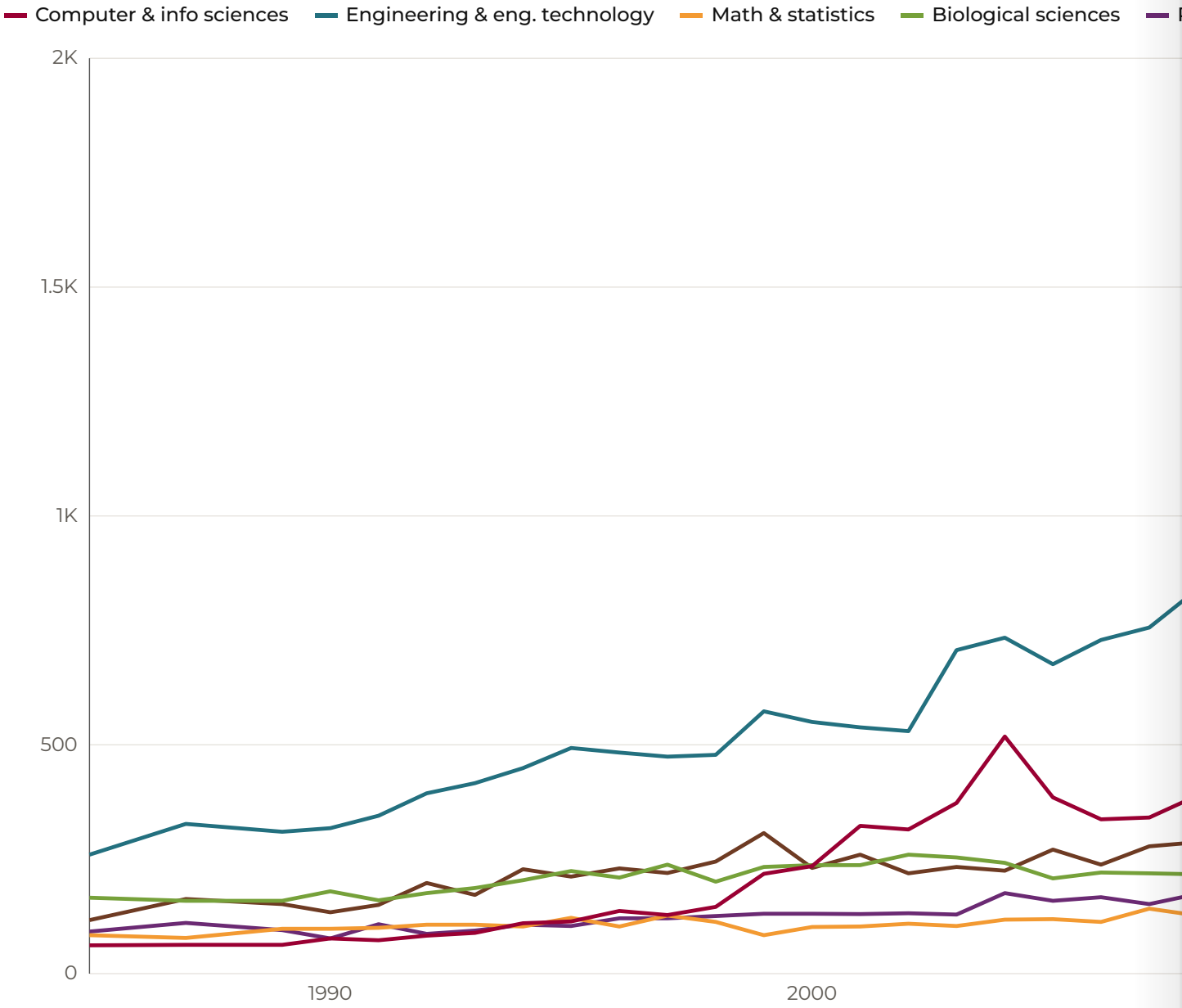
U.S. **23%** · **24th** of 51 · +2.4 pp since 2015

46%

STEM share of research doctorates, 2025

U.S. **39%** · **14th** of 51 · −0.1 pp since 2015

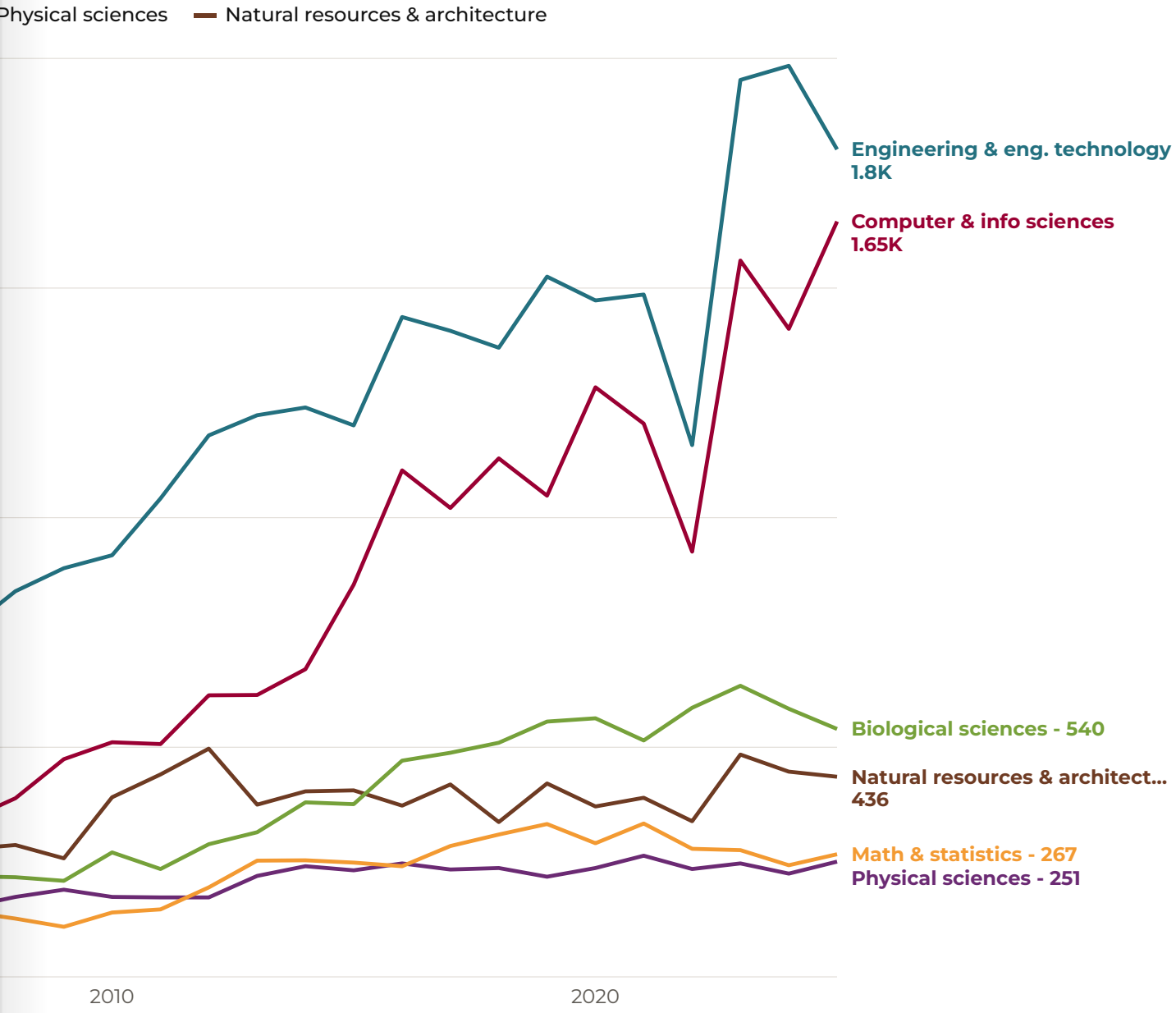
Figure 8 Which STEM Fields Produce North Carolina's Master's Degrees?



STEM master's degrees by field, all students. Data from NCES, IPEDS Completions.

Engineering & eng. technology produced the most STEM master's degrees in 2025, 1.8K, followed by computer & info sciences (1.65K) and biological sciences (540). Master's degrees in engineering & eng. technology are up 50% since 2015.

International students earned 2.42K of the state's STEM master's degrees in 2025, 2.1% of the national total (17th of 51).



2.42K

International STEM master's, 2025

U.S. **117K** · **17th** of 51 · +94% since 2015

2.1%

Share of U.S. international STEM master's, 2025

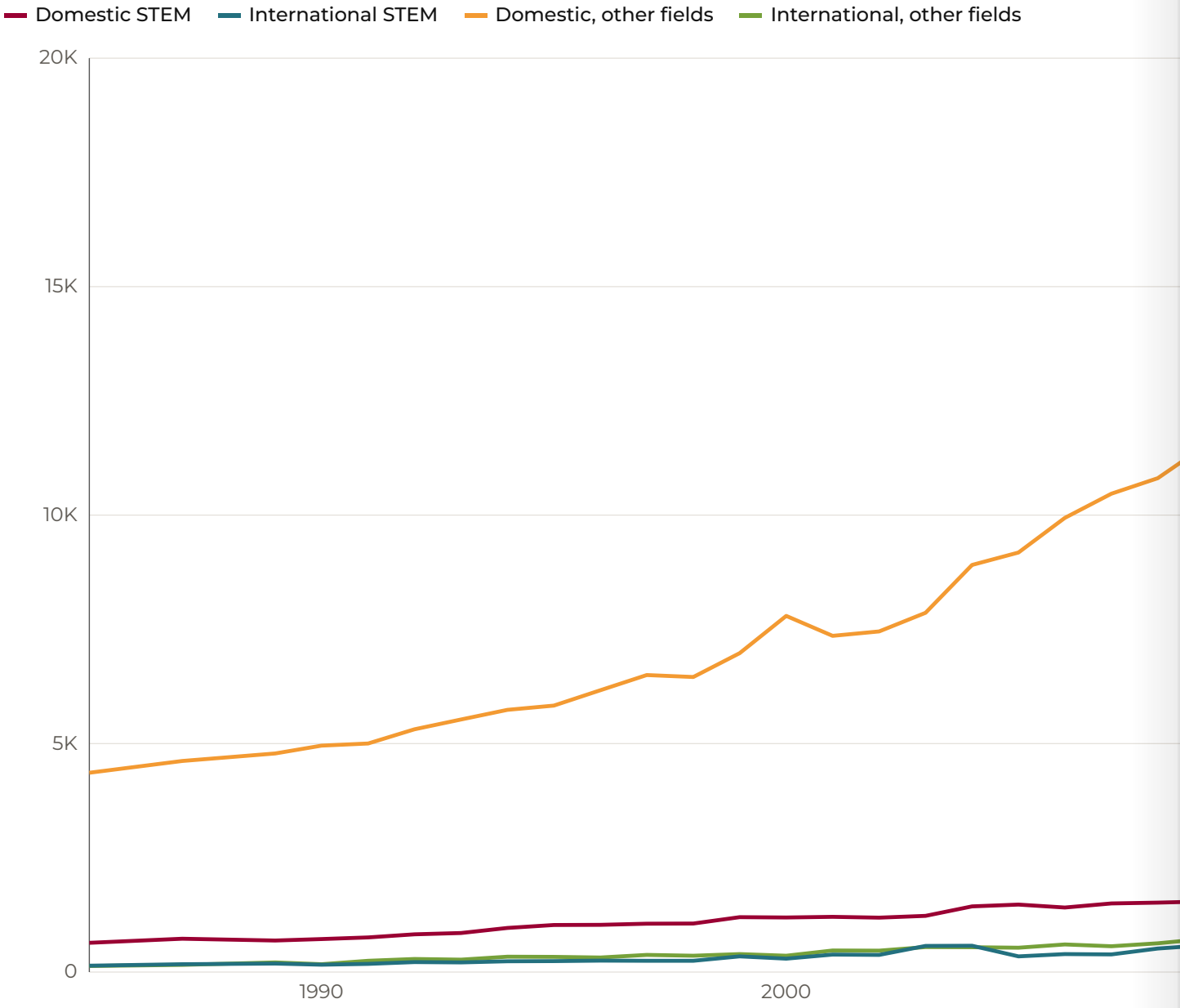
· **17th** of 51 · −0.3 pp since 2015

64%

Int. share of CS master's, 2025

U.S. **65%** · **25th** of 47 · +16.3 pp since 2015

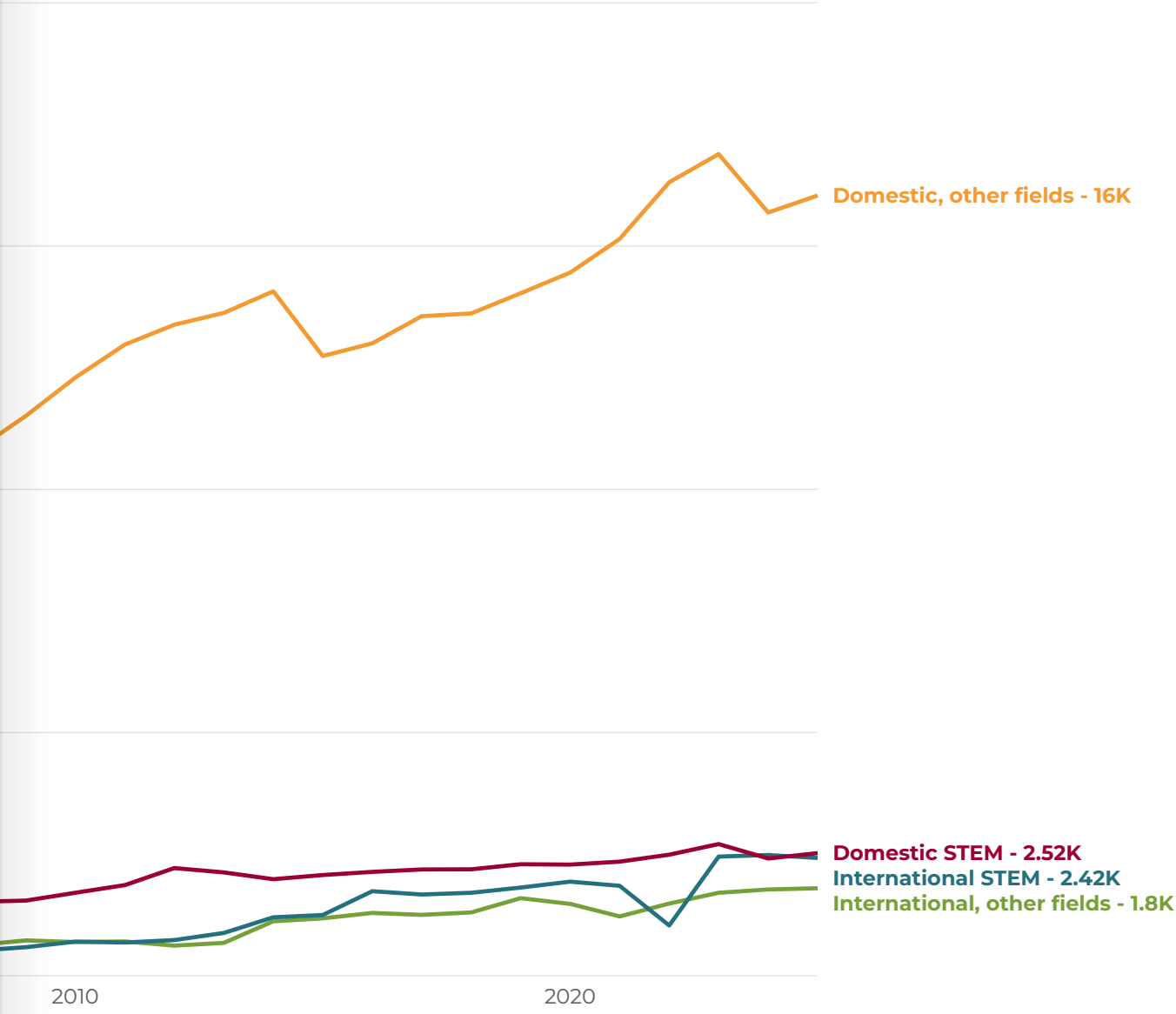
Figure 9 Who Earns North Carolina's Master's Degrees?



Master's degrees by student residency and field group. Data from NCES, IPEDS Completions.

Between 2015 and 2025, 1.17K more STEM master's degrees went to international students and 449 more to domestic students.

International students earn 49% of North Carolina's STEM master's but 10% of master's in other fields (U.S. 52% and 12%).



19%

Int. share of all master's, 2025

U.S. **22%** · **21st** of 51 · +4.4 pp since 2015

49%

Int. share of STEM master's, 2025

U.S. **52%** · **20th** of 51 · +11.4 pp since 2015

4.1

STEM master's per STEM research doctorate (IPEDS), 2025

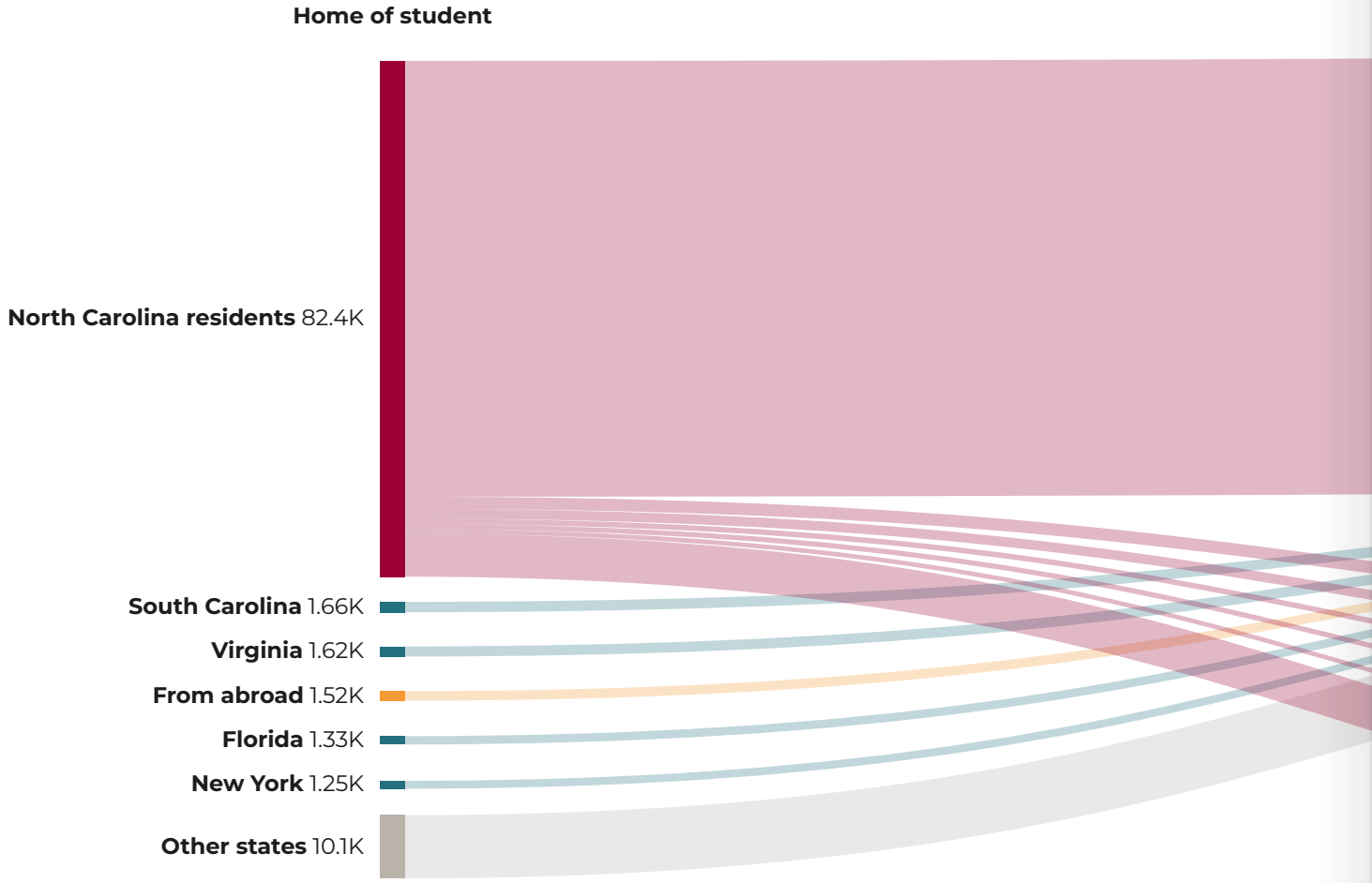
U.S. **6.4** · **37th** of 51 · +0.8 since 2015

SECTION 3

Where Students Come From

North Carolina ranks 11th of 51 on the share of first-time students who are in-state residents, at 79% in 2024.

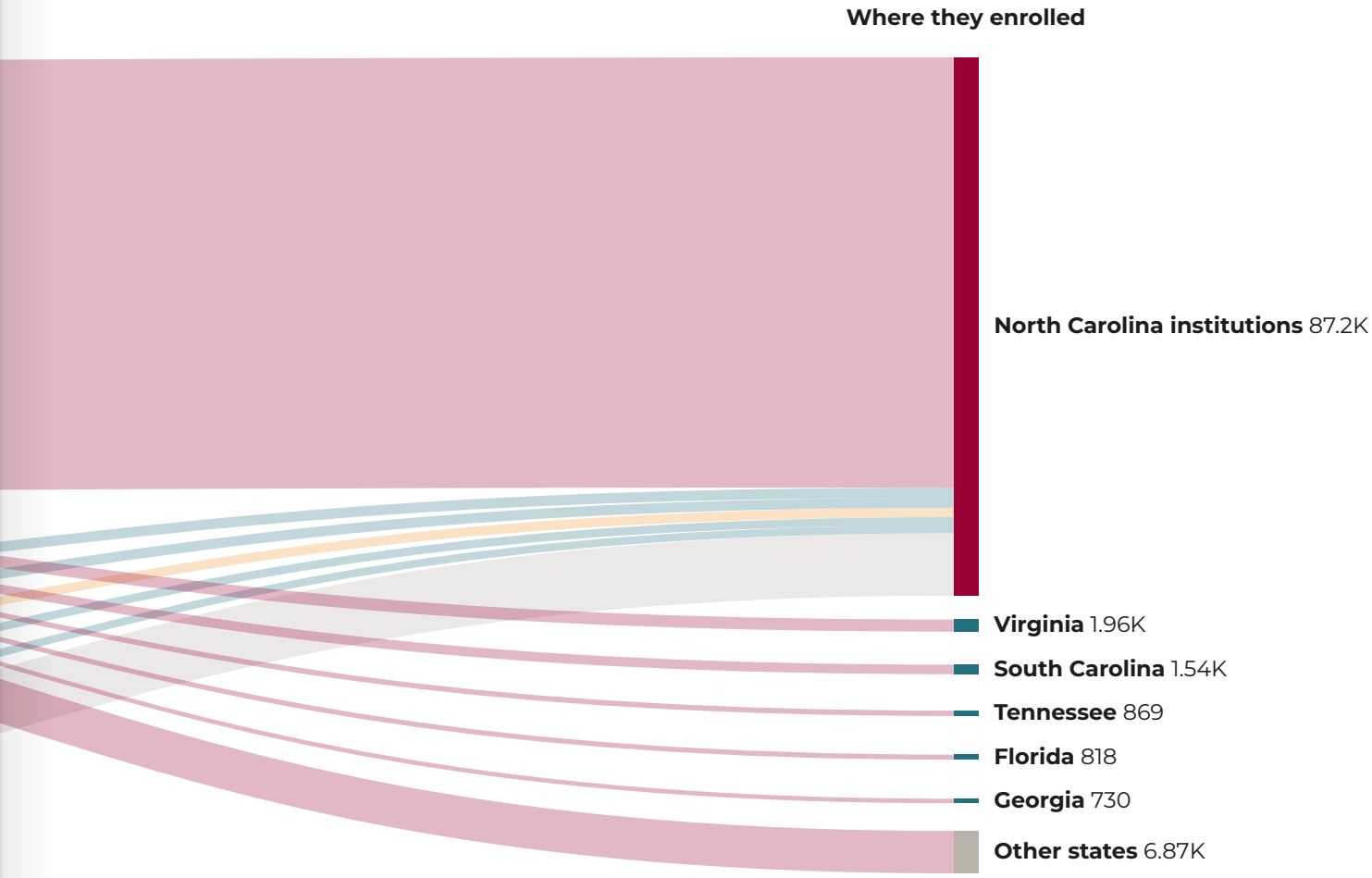
Figure 10 First-Time Students by Home and Destination, Fall 2024



Home of first-time degree-seeking undergraduates (left) and the state of the institution entered (right); the five largest s

69.7K residents stayed in North Carolina for college in fall 2024; 16K students arrived from other states and 1.52K from abroad, while 12.8K residents left.

Each band runs from a student's home on the left to the state where they enrolled on the right. 79% of the state's first-time students are residents (U.S. –).



ources and destinations, the rest grouped. Data from NCES, IPEDS Fall Enrollment.

79%

First-time students from in state, 2024

· **11th** of 51 · −3.0 pp since 2016

1.7%

First-time students from abroad, 2024

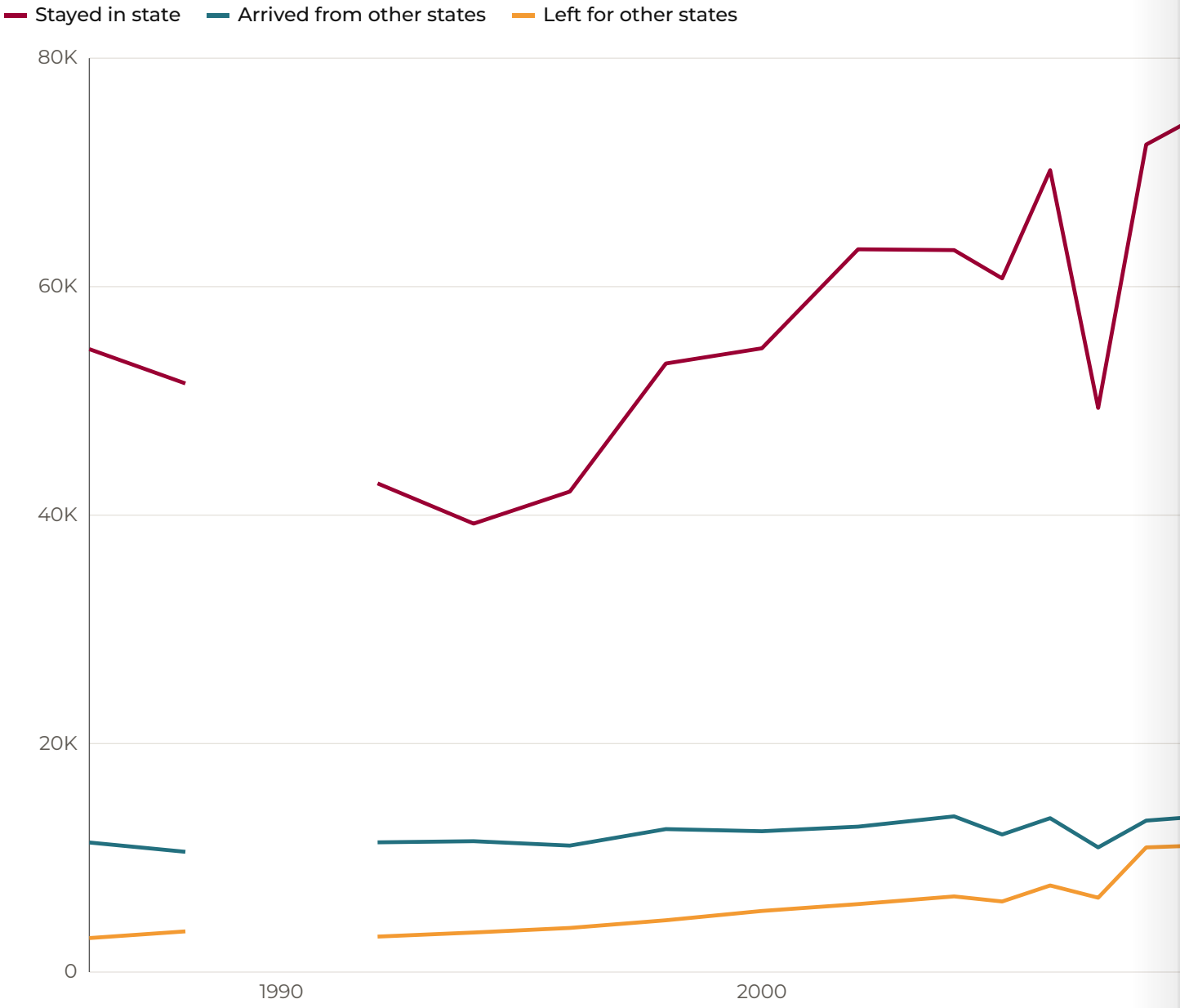
U.S. **3.1%** · **39th** of 51 · +0.2 pp since 2016

+4

Net migration per 100 residents, 2024

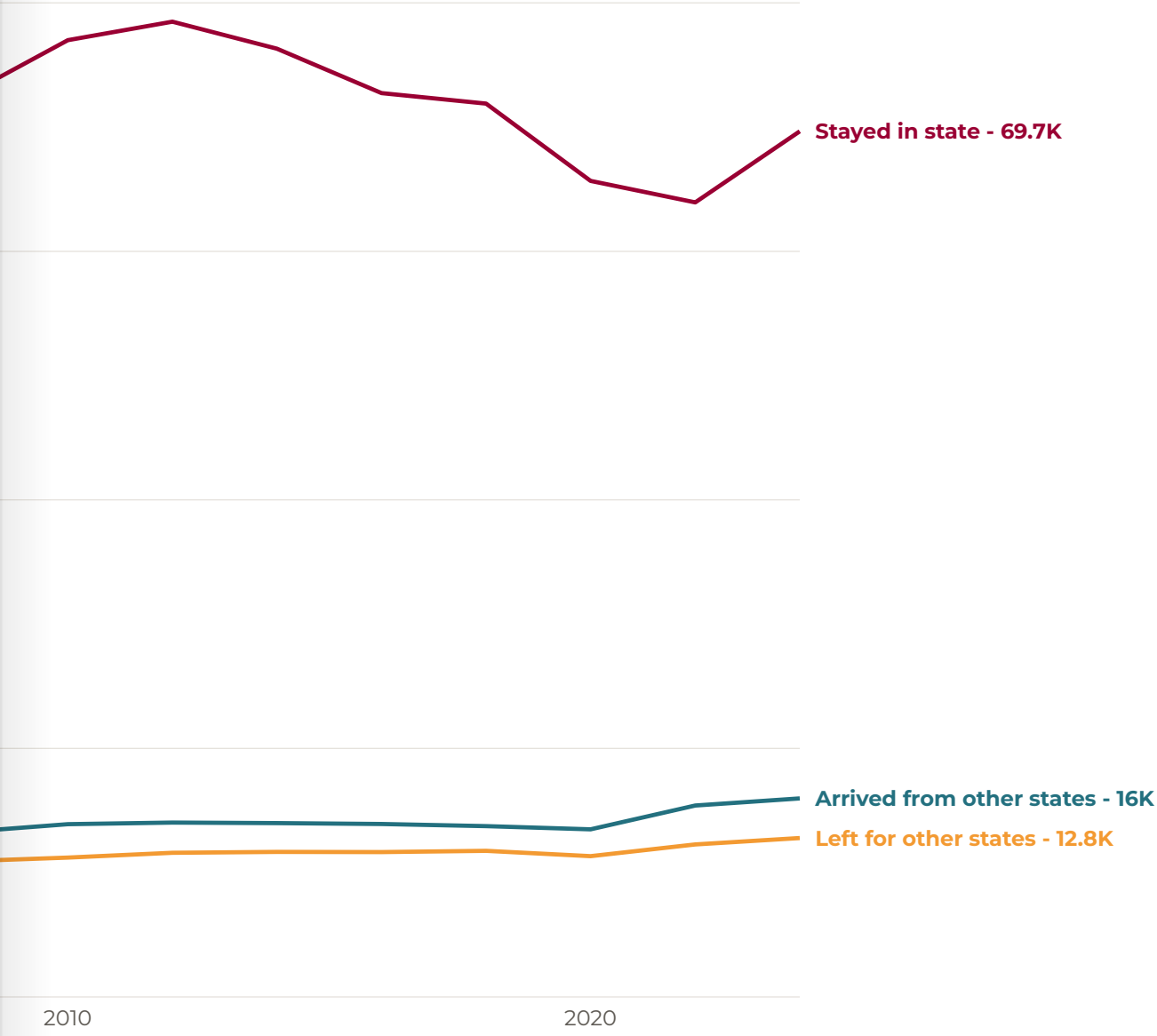
· **29th** of 51 · +1.2 since 2016

Figure 11 How Have the Flows Changed?



First-time degree-seeking undergraduates by flow; even years before 2001. Data from NCES, IPEDS Fall Enrollment.

Since 2015, the number of residents staying in state for college is down 4%, arrivals from other states are up 15%, and residents leaving are up 10%.



Net migration was +4 per 100 resident first-time students (29th of 51). The top destination for residents who left was Virginia (1.96K).

Figure 12

Where Do North Carolina's Newcomers Come From?

The bands show where first-time degree-seeking undergraduates who enrolled at North Carolina's institutions in fall 2024 but did not live in the state came from: the 12 largest home states and students from abroad, with all other states grouped. Band widths are proportional to the number of students. Home is the state where the student lived when admitted.

17.5K

First-time students from other states and abroad, fall 2024

Abroad **1.52K**

1.7%

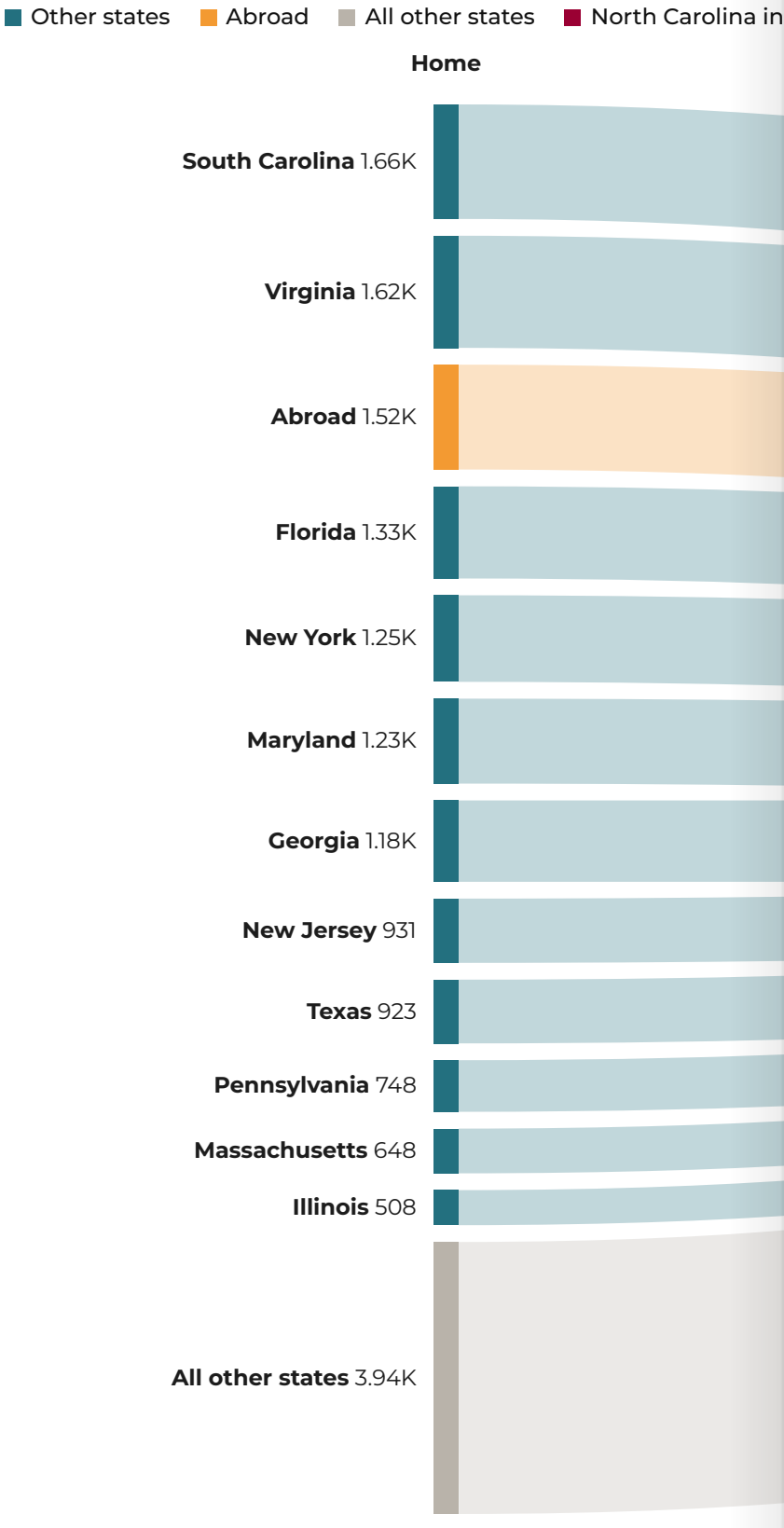
First-time students from abroad, 2024

U.S. **3.1%** · **39th** of 51 · +0.2 pp since 2016

79%

First-time students from in state, 2024

11th of 51 · −3.0 pp since 2016



First-time degree-seeking undergraduates, fall 2024; the 12 largest flow

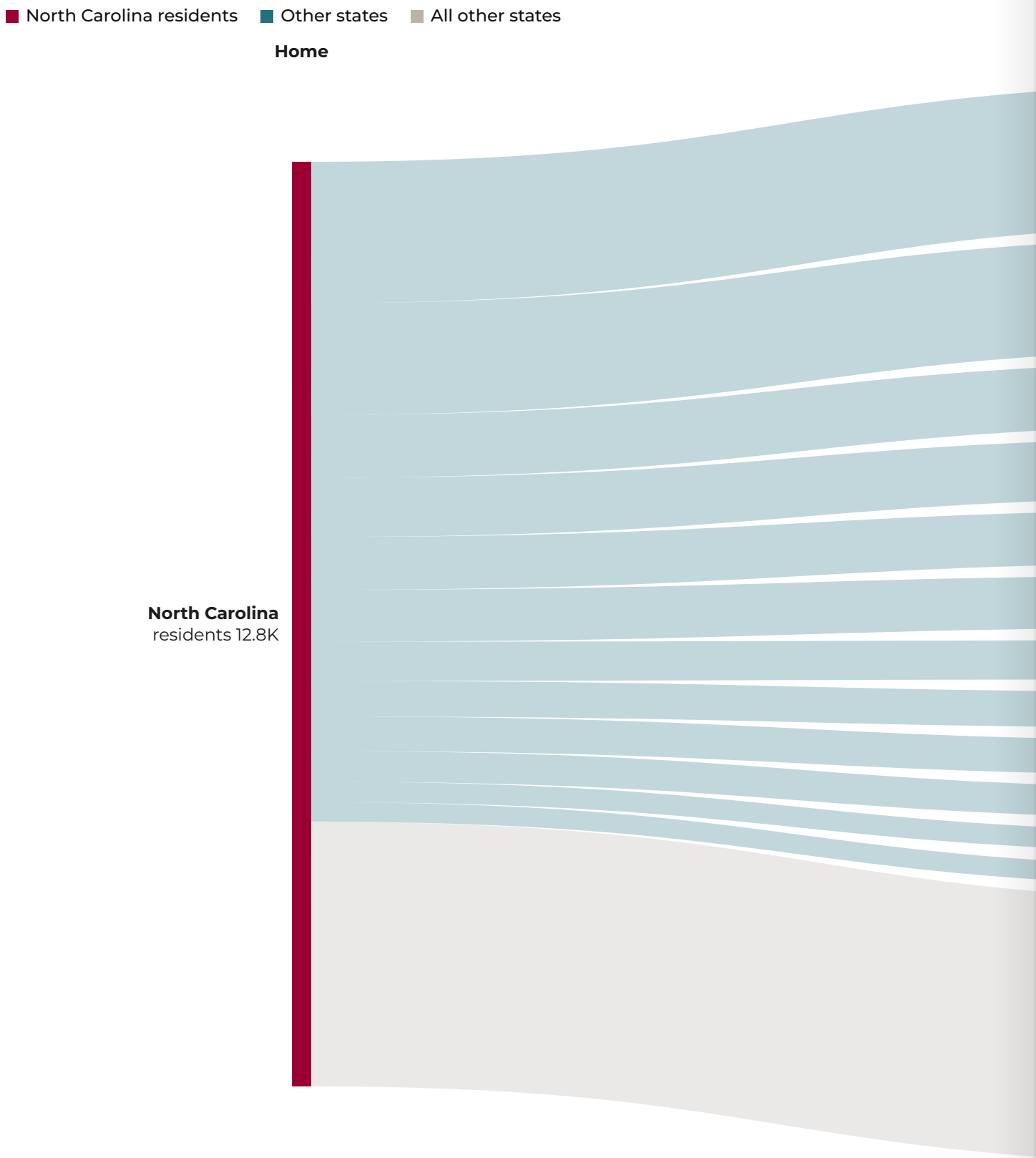
stitutions

Enrolled in



North Carolina
institutions 17.5K

ws. Data from NCES, IPEDS Fall Enrollment.



First-time degree-seeking undergraduates, fall 2024; the 12 largest flows. Data from NCES, IPEDS Fall Enrollment.

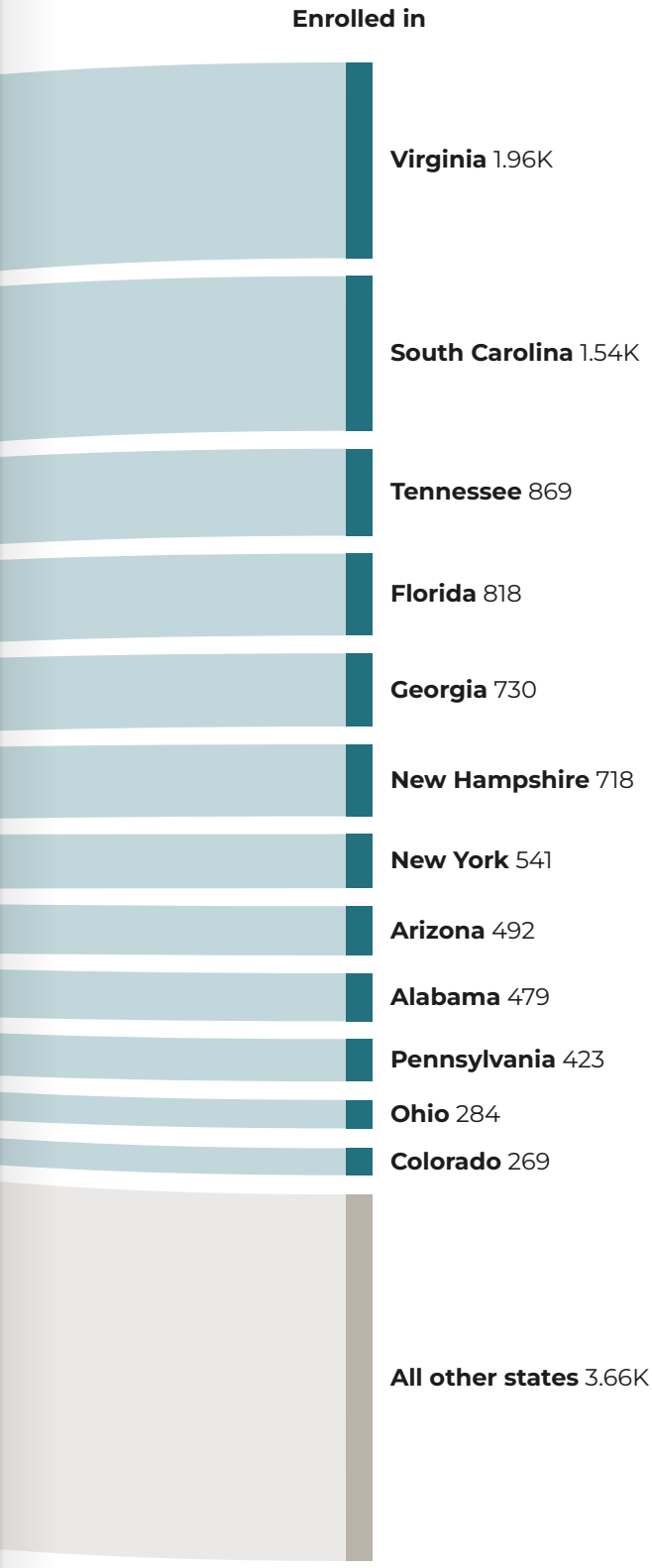


Figure 13

Where Do North Carolina's Residents Go When They Leave?

The bands show where North Carolina residents who left the state to start college in fall 2024 enrolled: the 12 largest destination states, with all other states grouped. Band widths are proportional to the number of students. Residents who enrolled in the state are not shown.

Residents leaving for other states. The number was 12.8K in fall 2024, up 4.1% from 12.3K in fall 2022 and up 9.7% since fall 2016, when it was 11.7K.

12.8K
Residents who started college in another state, fall 2024
Arrived from other states **16K** · **17.5K** with abroad

+4
Net migration per 100 residents, 2024
29th of 51 · +1.2 since 2016

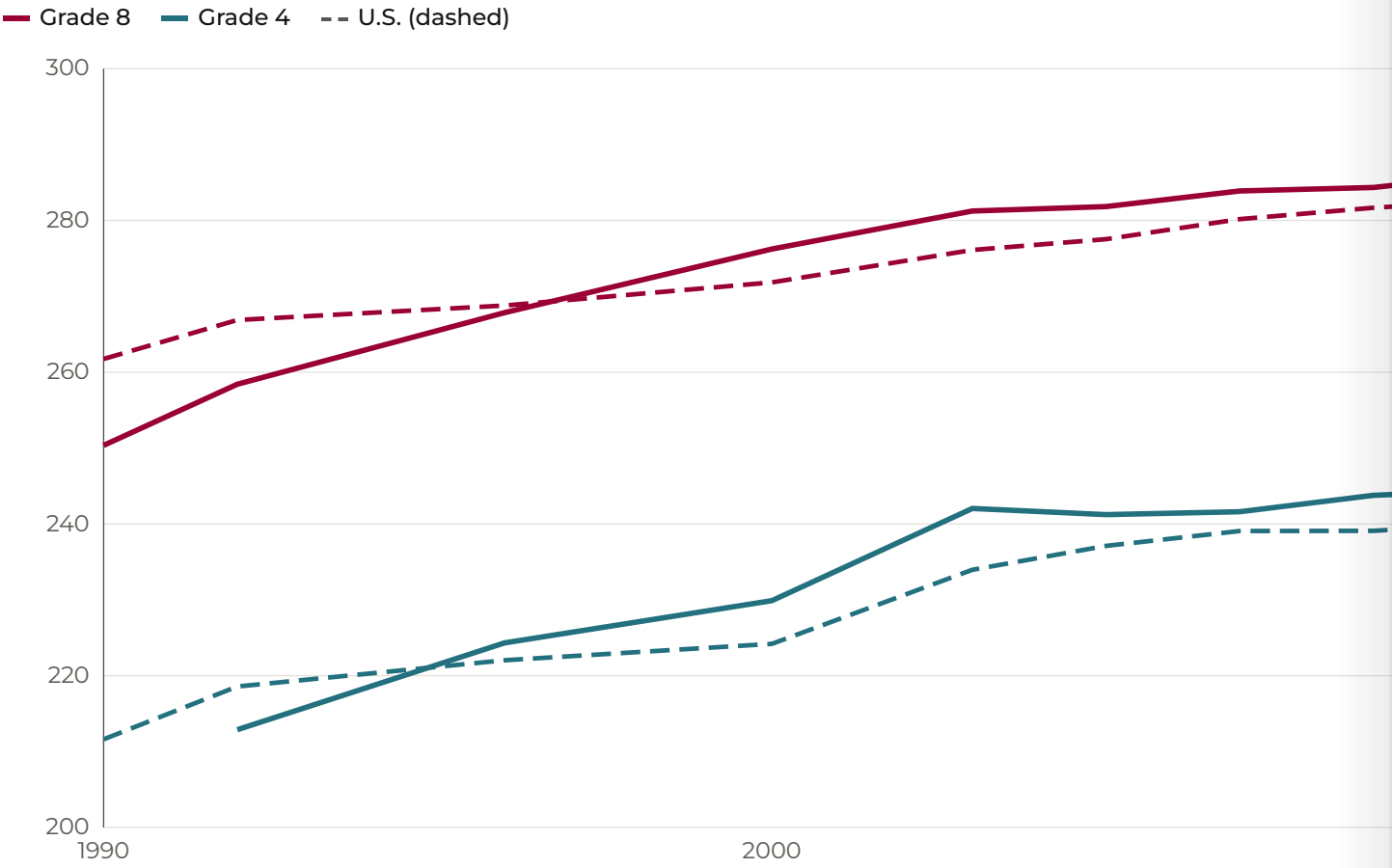
1.96K
Went to Virginia
The largest destination

SECTION 4

K-12 Preparation

North Carolina ranks 14th of 51 on the share of grade 8 mathematics: at or above Proficient, at 31% in 2024 against 27% for the U.S.

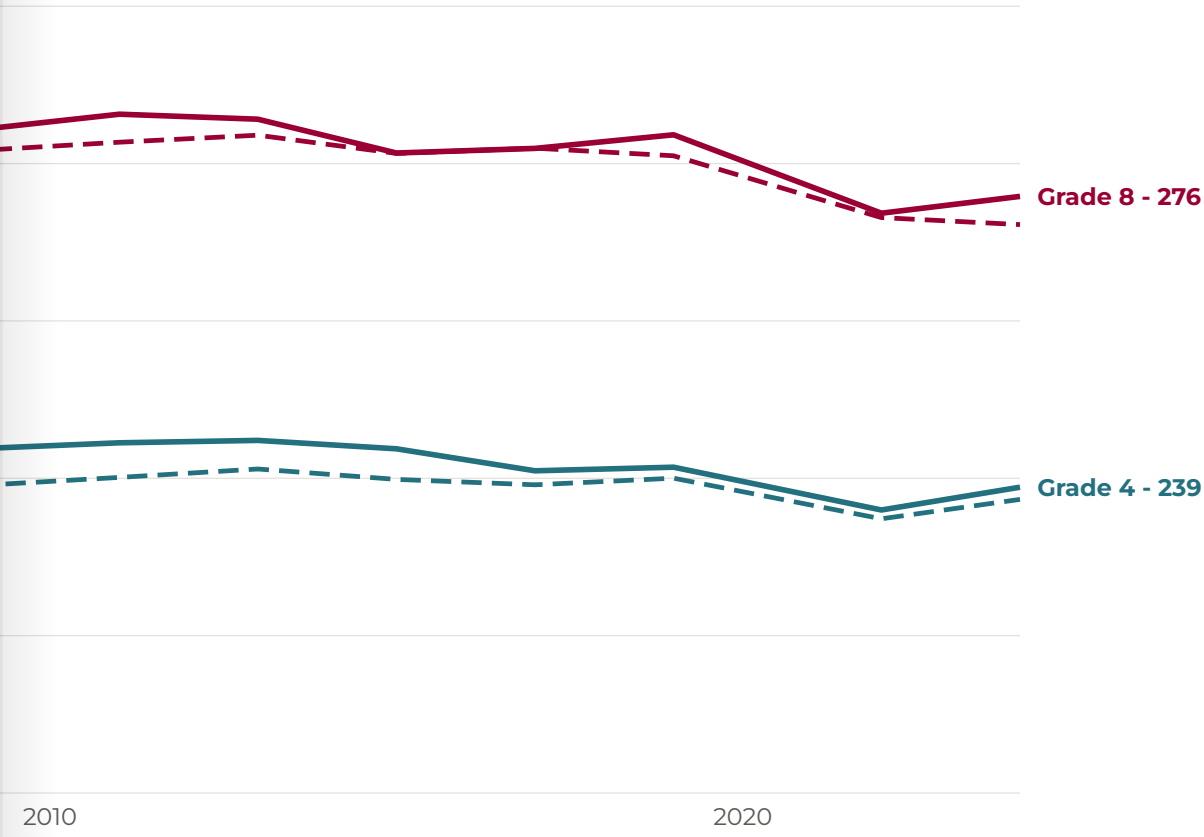
Figure 14 How Well Prepared in Mathematics Are North Carolina's Students?



Average NAEP mathematics score, public schools, 0–500 scale; state solid, U.S. public schools dashed. Data from NCES, N

The talent pipeline starts in school. North Carolina's eighth graders averaged 276 in mathematics on the 2024 NAEP, against 272 for U.S. public schools, 20th of 51 among the states.

31% scored at or above the Proficient level (U.S. 27%). Differences of a point or two are often within the margin of error.



lational Assessment of Educational Progress.

276

Grade 8 math score (NAEP), 2024

U.S. **272** · **20th** of 51 · −5.5 since 2015

31%

Grade 8 math, at or above Proficient, 2024

U.S. **27%** · **14th** of 51 · −1.1 pp since 2015

255

Grade 8 reading score (NAEP), 2024

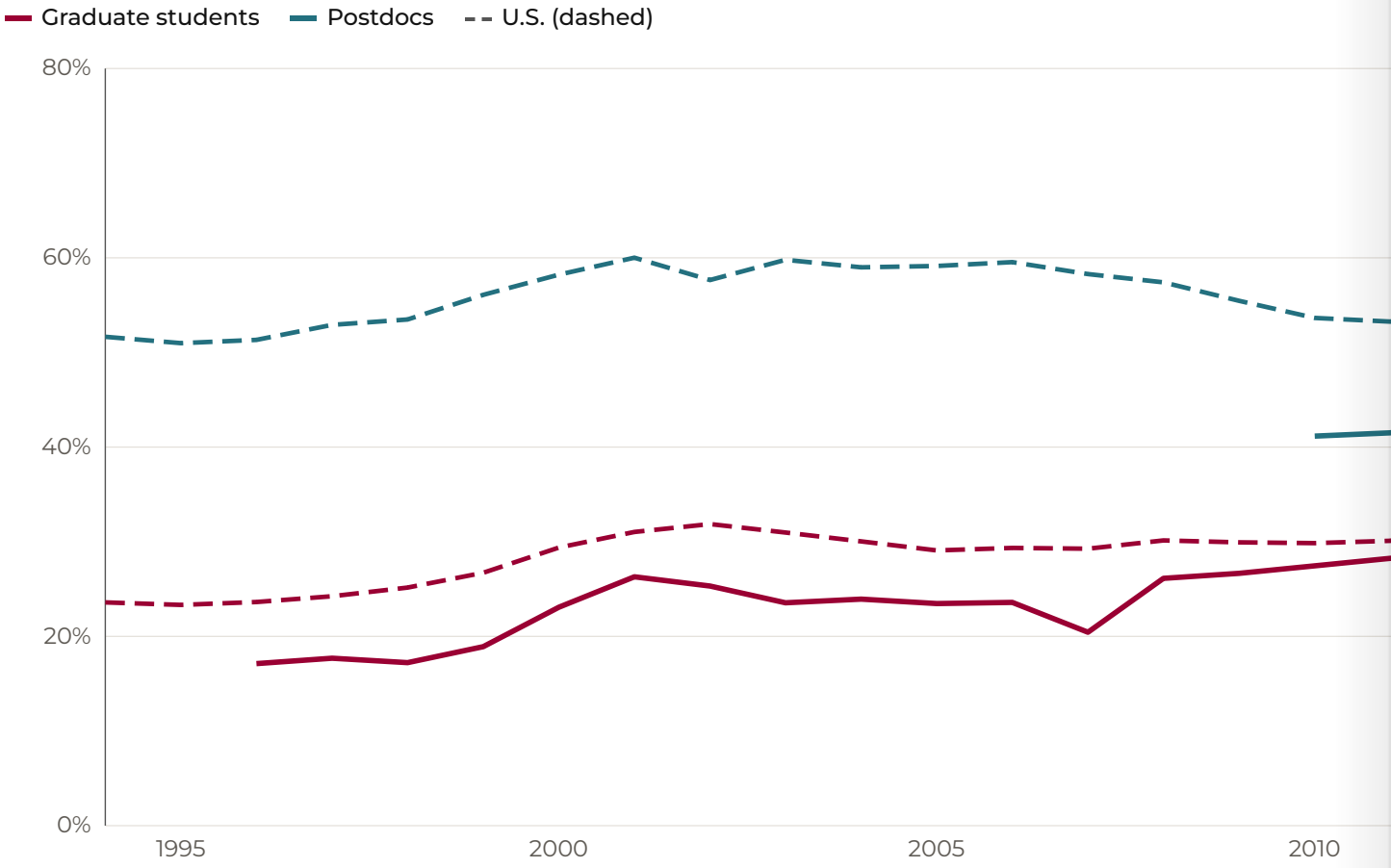
U.S. **257** · **35th** of 51 · −5.7 since 2015

SECTION 5

Graduate Research

North Carolina ranks 8th of 51 on research doctorates awarded, at 1.97K in 2025 against 59K for the U.S.

Figure 15 How Many of North Carolina's Science and Engineering Graduate Students H



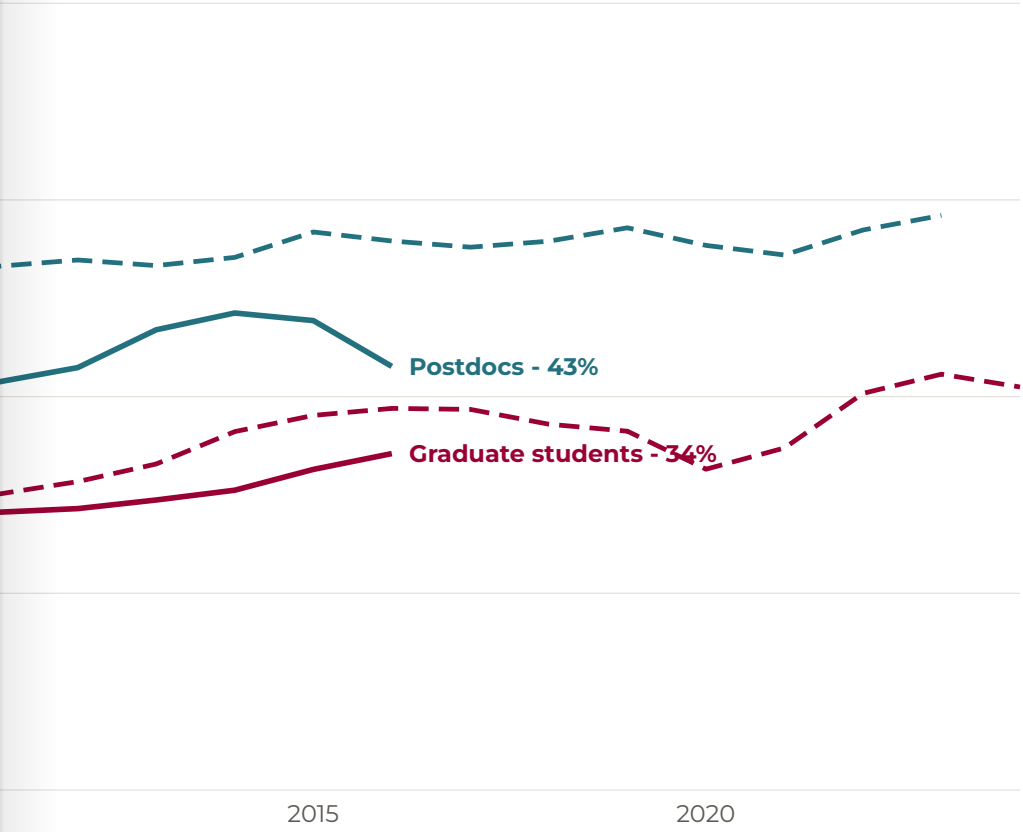
Temporary-visa share of science and engineering graduate students and postdoctoral appointees, by the institution's sta
Students and Postdoctorates.

In 2016, the last year NSF published state tables, 34% of North Carolina's science and engineering graduate students held temporary visas (U.S. 39%; 25th of 51).

Among postdoctoral researchers, the share was 43% (U.S. 56%). Postdocs are the stage of research training most dependent on international talent, and IPEDS does not count them.

The state's institutions awarded 1.97K research doctorates in 2025, 3.3% of the U.S. total (Data from NSF NCSES, Survey of Earned Doctorates; NSF publishes doctorates by citizenship only for the nation).

old Temporary Visas?



te; state solid, U.S. dashed. NSF stopped publishing state tables after 2016. Data from NSF NCSES, Survey of Graduate

34%

S&E grad students on temporary visas, 2016

U.S. **39%** · **25th** of 51 · +1.6 pp since 2015

43%

S&E postdocs on temporary visas, 2016

U.S. **56%** · **43rd** of 51 · −4.7 pp since 2015

1.97K

Research doctorates awarded (NSF SED), 2025

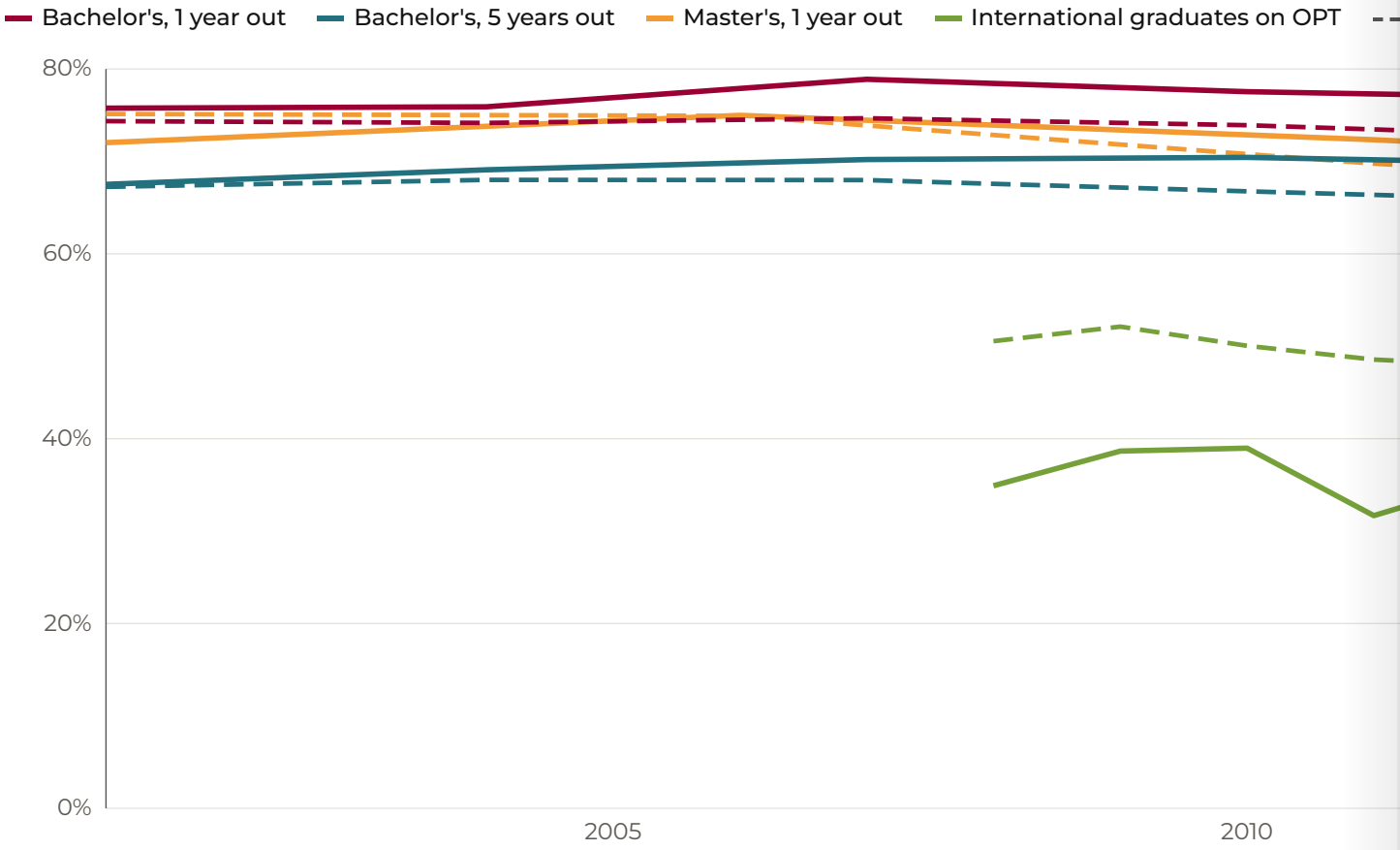
U.S. **59K** · **8th** of 51 · +15% since 2015

SECTION 6

Do Graduates Stay?

North Carolina ranks 4th of 33 on the share of bachelor's graduates working in the state 1 year after graduating, at 79% in 2019 against 72% for the U.S.

Figure 16 How Many of North Carolina's Graduates Go On to Work in the State?

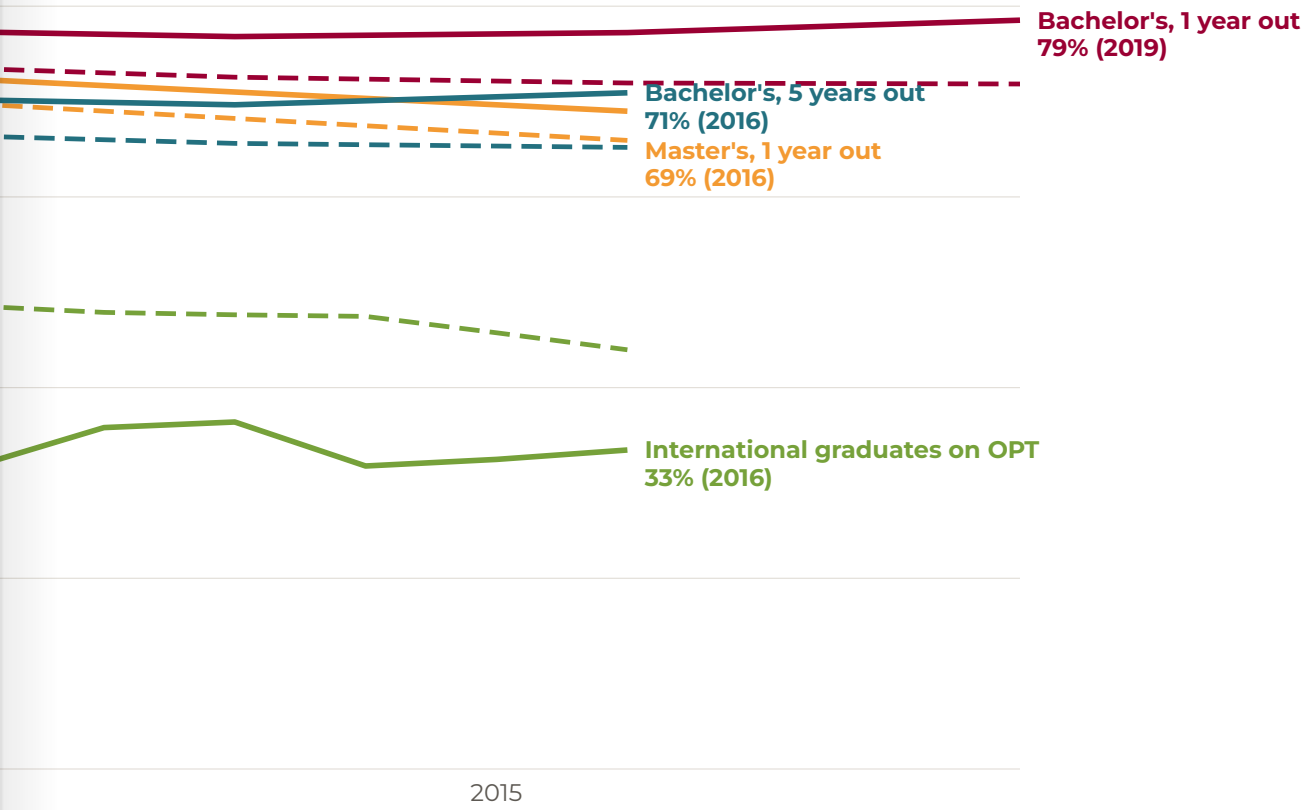


Share of employed graduates working in the state of their institution, by graduating cohort (Data from U.S. Census Burea state where they studied (Data from ICE, SEVIS FOIA releases; shares only, the file is truncated). State solid, U.S. or particip

Of North Carolina's bachelor's graduates who were working a year after graduating, 79% worked in the state (participating states 72%). Five years out the share was 71%.

Among international graduates on Optional Practical Training in 2016, 33% worked in the state where they studied (U.S. 44%). That is the last year the released records cover, before most of the growth in international STEM master's degrees since 2015. Through migration between states, the state gained 5.4 bachelor's-plus adults per 1,000 such residents in 2025 (Data from U.S. Census Bureau, American Community Survey).

U.S. (dashed)



au, Post-Secondary Employment Outcomes; participating institutions only); international graduates' OPT jobs in the
pating states dashed.

79%

Bachelor's grads working in-state, 1 year out, 2019

U.S. **72%** · **4th** of 33 · +1.3 pp since 2016

33%

OPT graduates working in the state, 2016

U.S. **44%** · **30th** of 51 · +1.0 pp since 2015

+5.4

Net migration, bachelor's-plus, per 1,000, 2025

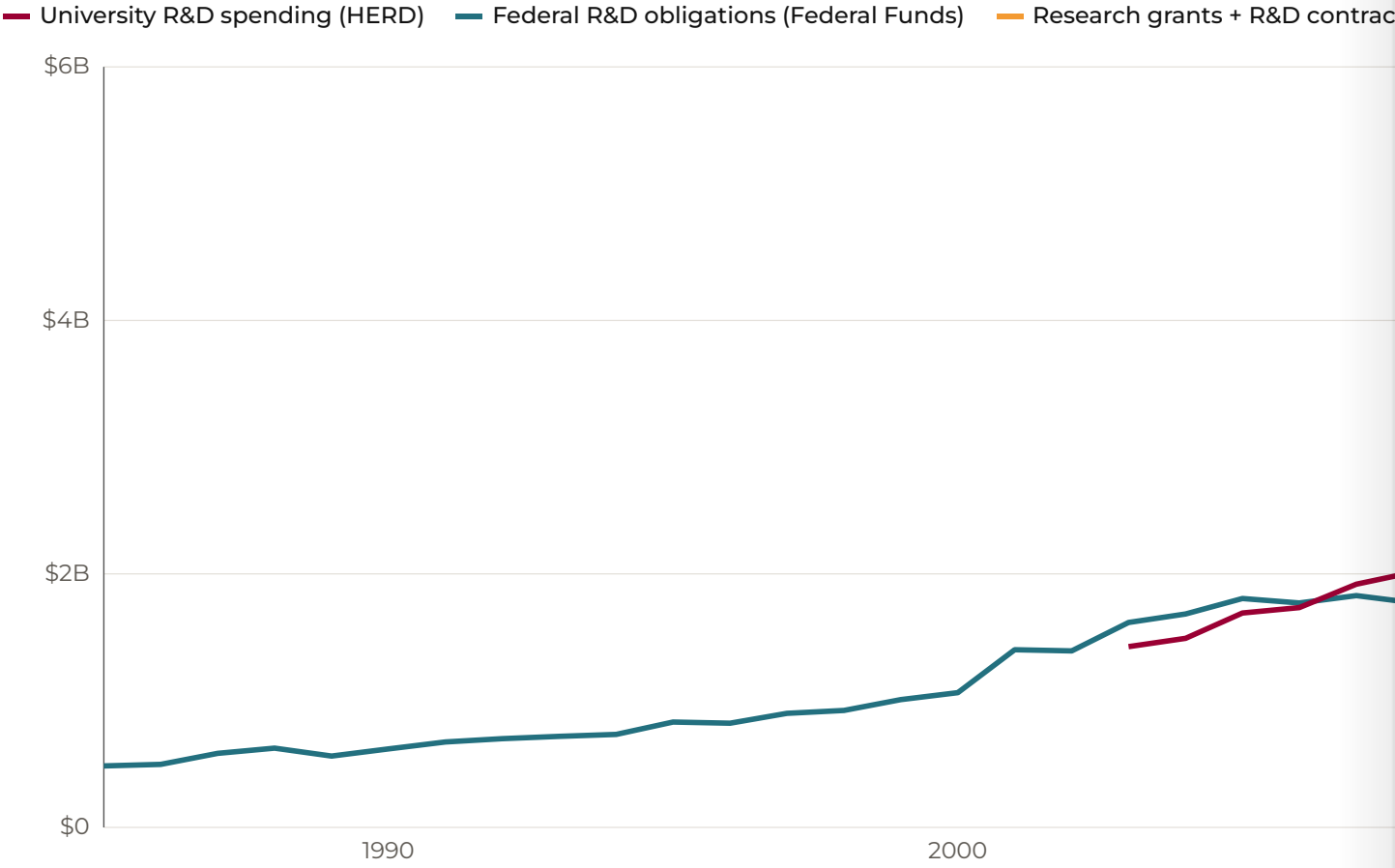
· **11th** of 51 · −5.9 since 2015

SECTION 7

Research Funding

North Carolina ranks 7th of 51 on university R&D spending, at $4.65B in 2024 against $118B for the U.S.

Figure 17 How Much Research Money Flows Into North Carolina?

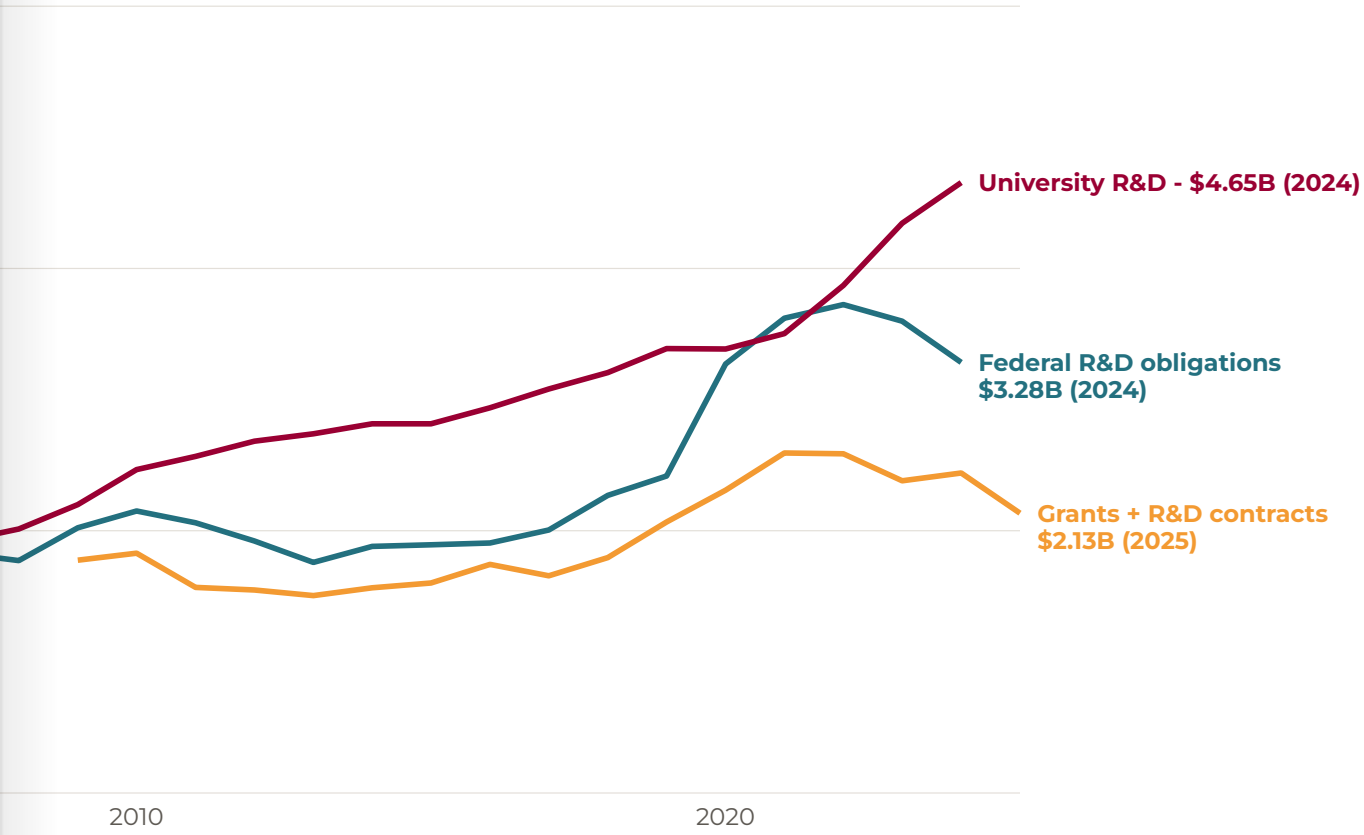


Nominal dollars (not adjusted for inflation), fiscal years; a line ending before the others is labeled with its last year. Univers
Federal R&D obligations by state of performance: Data from NSF NCSES, Survey of Federal Funds for R&D (coverage wide

North Carolina's universities spent $4.65B on research and development in FY2024, 7th of 51 among the states; 59% of it was federally funded (U.S. 55%).

Federal agencies obligated $3.28B for R&D performed in the state in FY2024. Research funding drives demand for graduate students and postdocs, the stage of training most dependent on international talent.

ts (USAspending)



sity R&D: Data from NSF NCSES, Higher Education Research and Development Survey (FY2003-09 predecessor survey).
ens in FY2021). Grants from NSF, NIH, NASA and DOE Science plus R&D contracts: Data from USAspending.gov.

$4.65B

University R&D spending, 2024

U.S. **$118B** · **7th** of 51 · +65% since 2015

59%

Federal share of university R&D, 2024

U.S. **55%** · **14th** of 51 · +1.9 pp since 2015

$3.28B

Federal R&D obligations, 2024

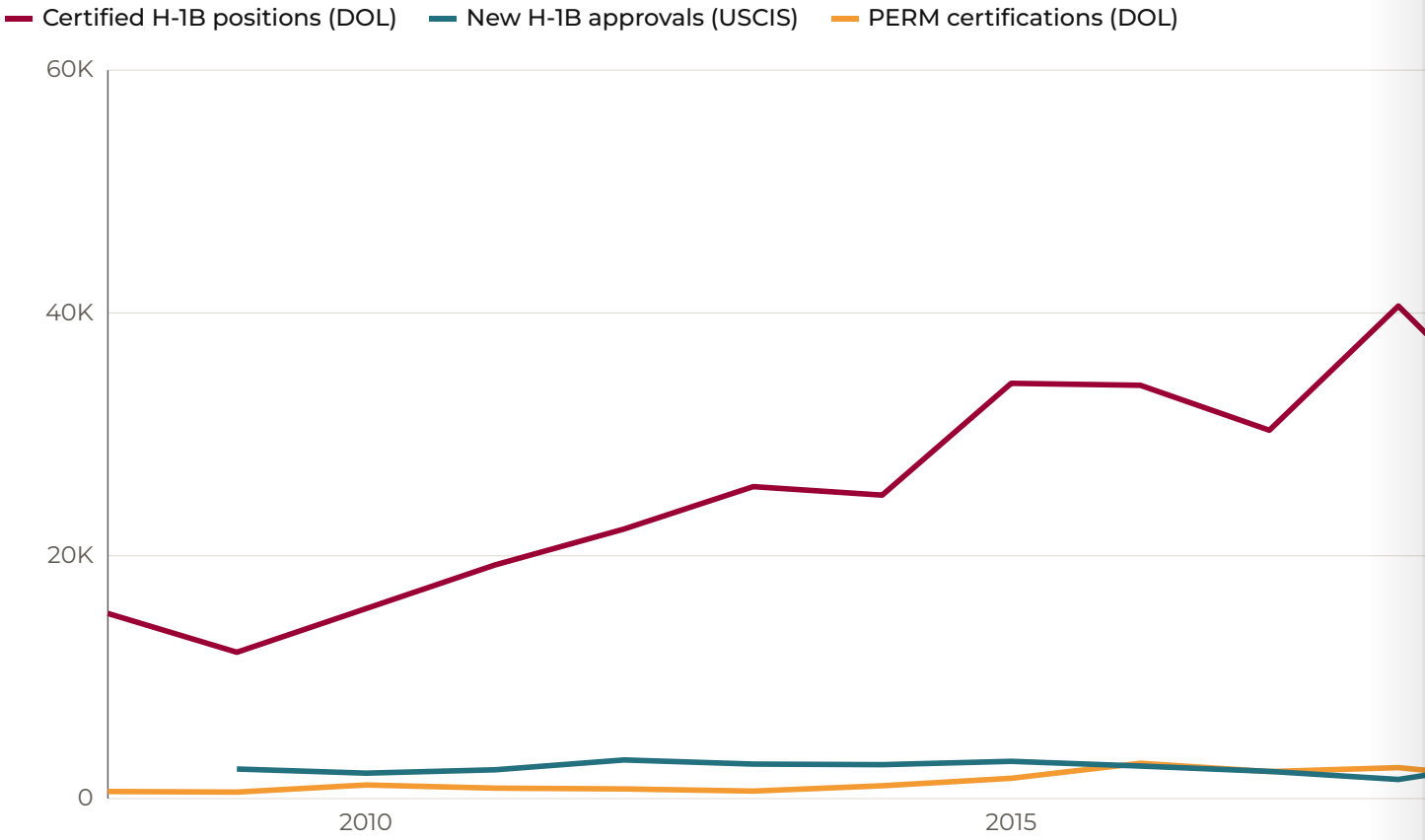
U.S. **$193B** · **17th** of 51 · +74% since 2015

SECTION 8

H-1B Jobs and Green Cards

North Carolina ranks 7th of 51 on certified H-1B positions, by worksite, at 29.3K in 2025 against 914K for the U.S.

Figure 18 How Many Skilled-Visa Jobs and Green-Card Certifications Does North Caroli

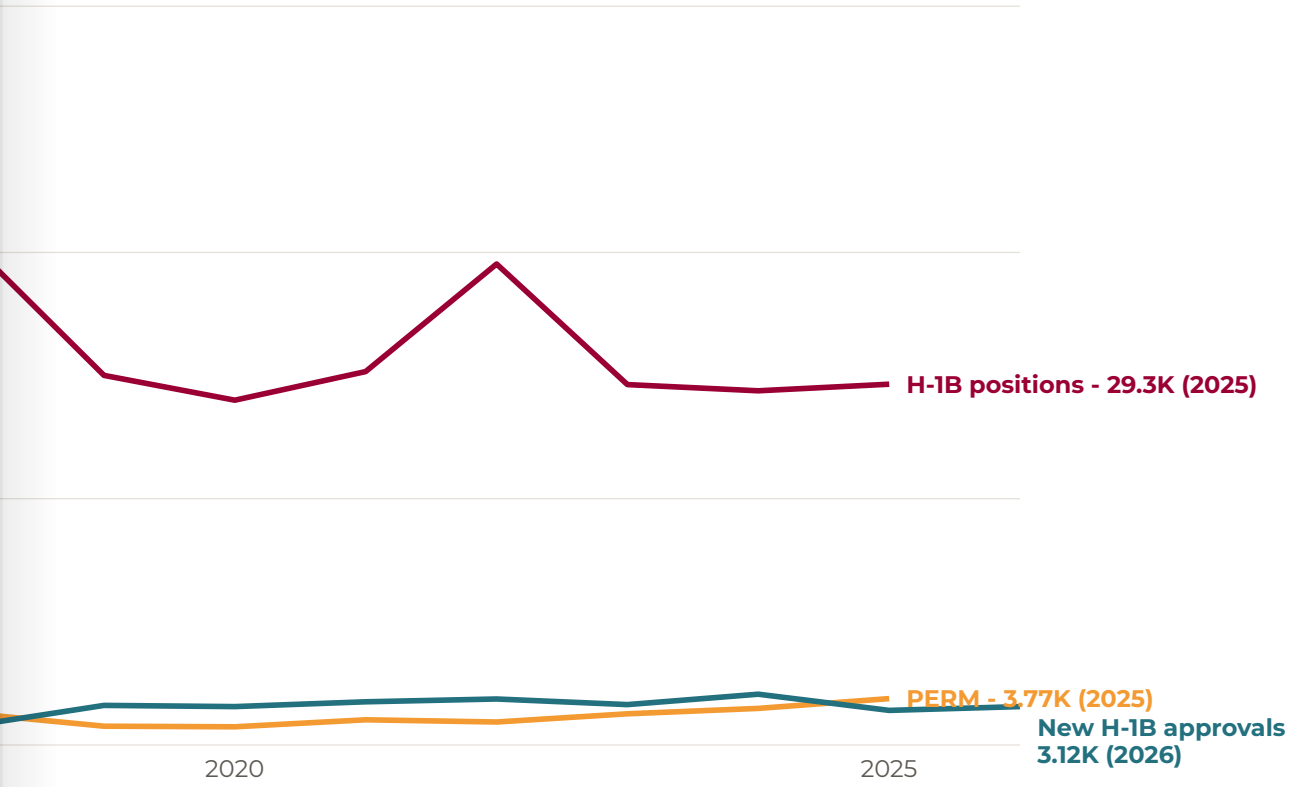


Certified H-1B positions on labor condition applications by worksite and certified PERM green-card applications, by fiscal
from USCIS, H-1B Employer Data Hub).

Employers were certified for 29.3K H-1B positions at North Carolina worksites in FY2025, 7th of 51 among the states. A labor condition application comes before the petition and covers renewals and job changes as well as new hires, by the worksite's state, so it runs well above approvals. USCIS approved 3.12K new H-1B petitions in FY2026 and 6.18K continuing ones, counted by the employer's address.

DOL certified 3.77K PERM applications, the employer's step toward an employment-based green card, for jobs in the state in FY2025. In FY2024, 6.6% were for workers still on F-1 student visas (U.S. 8.7%): the most direct link from classroom to permanent residence.

na Generate?



year (Data from DOL, Office of Foreign Labor Certification); new H-1B petition approvals by the employer's address (Data

29.3K

Certified H-1B positions (worksite), 2025

U.S. **914K** · **7th** of 51 · −14% since 2015

3.77K

PERM green-card certifications, 2025

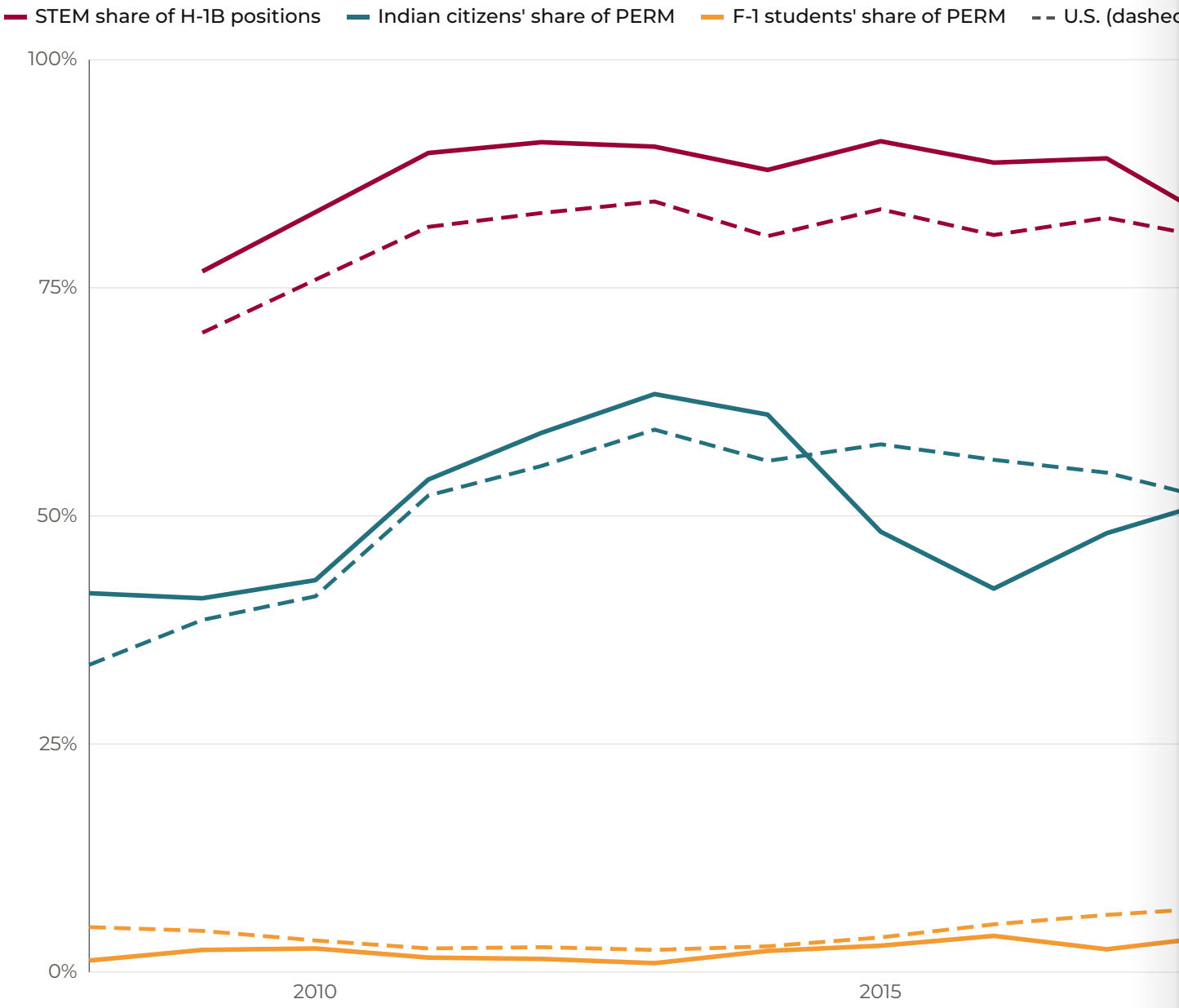
U.S. **138K** · **10th** of 51 · 2.27× since 2015

6.6%

PERM workers on F-1 student visas, 2024

U.S. **8.7%** · **39th** of 51 · +3.7 pp since 2015

Figure 19 Who Holds North Carolina's Skilled-Visa Jobs?

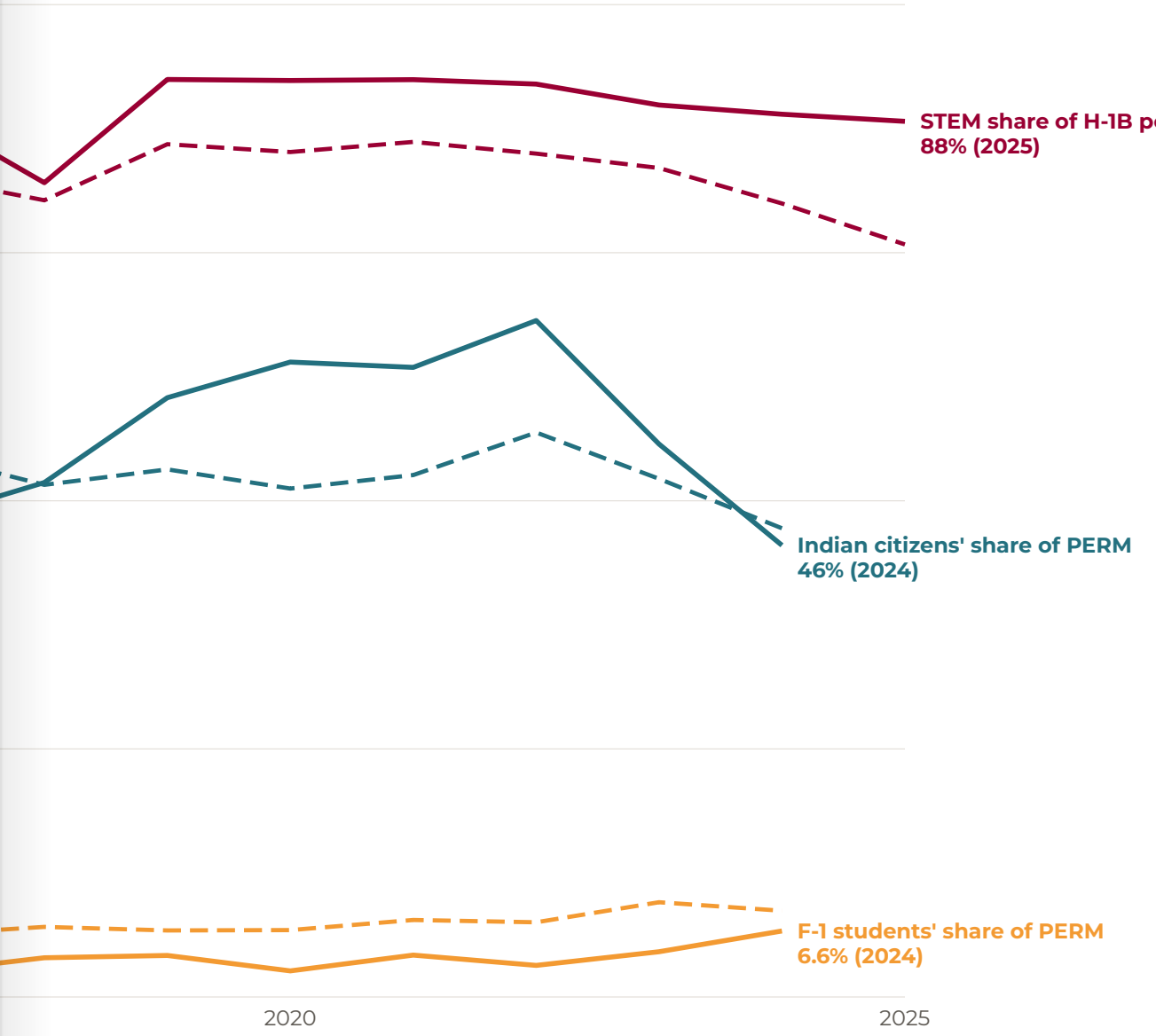


State solid, U.S. dashed. PERM citizenship and class of admission are not on the form DOL uses from FY2025. Data from D

88% of the state's certified H-1B positions were in STEM occupations in FY2025 (U.S. 76%).

Citizens of India held 46% of the state's PERM certifications in FY2024 (U.S. 47%). Under per-country caps they face the longest waits for a green card, so this share measures the state's exposure to the backlog.

d)



OL, Office of Foreign Labor Certification.

88%

STEM share of H-1B positions, 2025

U.S. **76%** · **3rd** of 51 · −2.8 pp since 2015

46%

PERM certifications for Indian citizens, 2024

U.S. **47%** · **17th** of 51 · −2.7 pp since 2015

$113K

Median offered H-1B wage (nominal), 2025

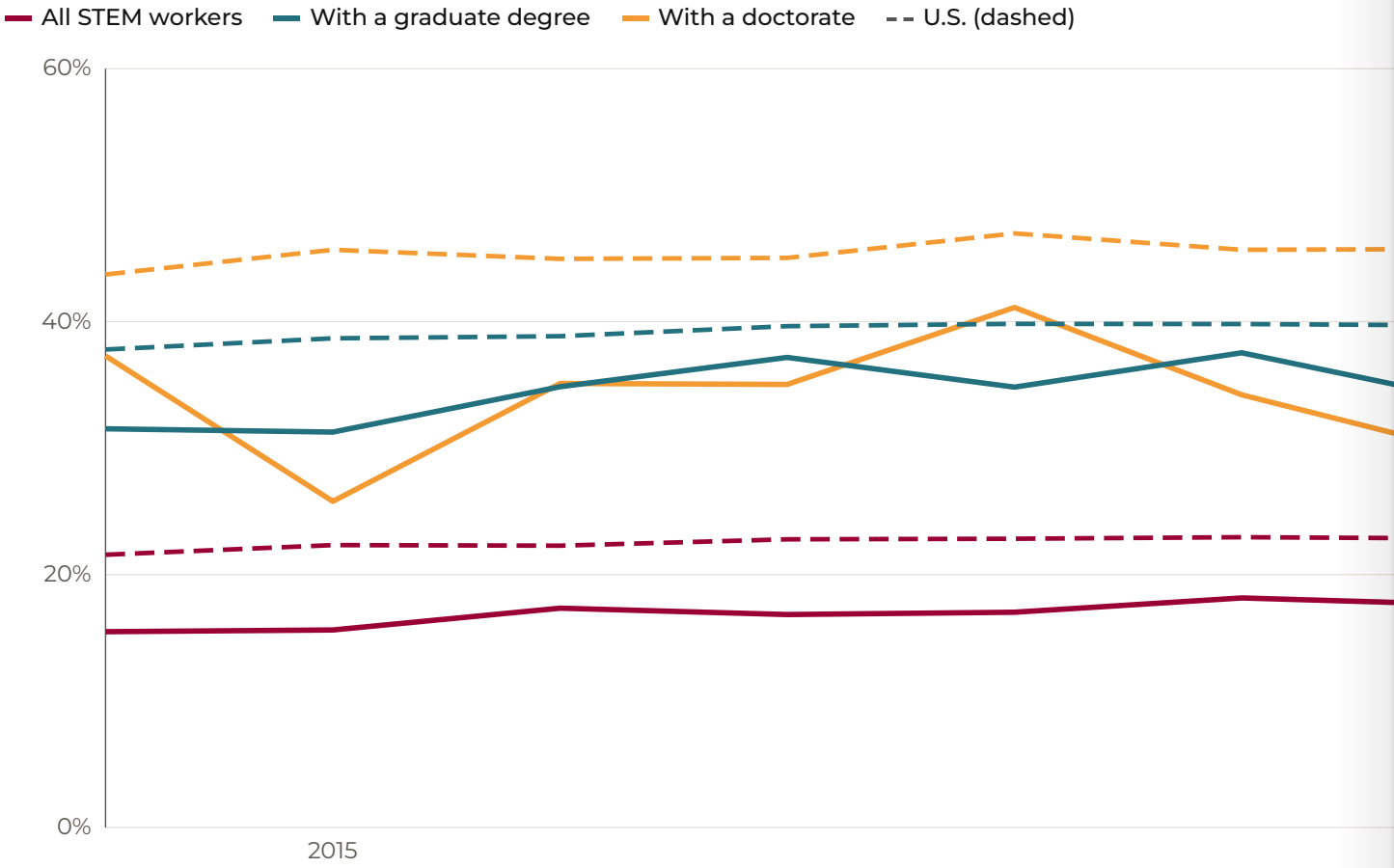
U.S. **$119K** · **12th** of 51 · +59% since 2015

SECTION 9

STEM Workforce

North Carolina ranks 11th of 51 on the noncitizen share of STEM workers, at 11% in 2024 against 11% for the U.S.

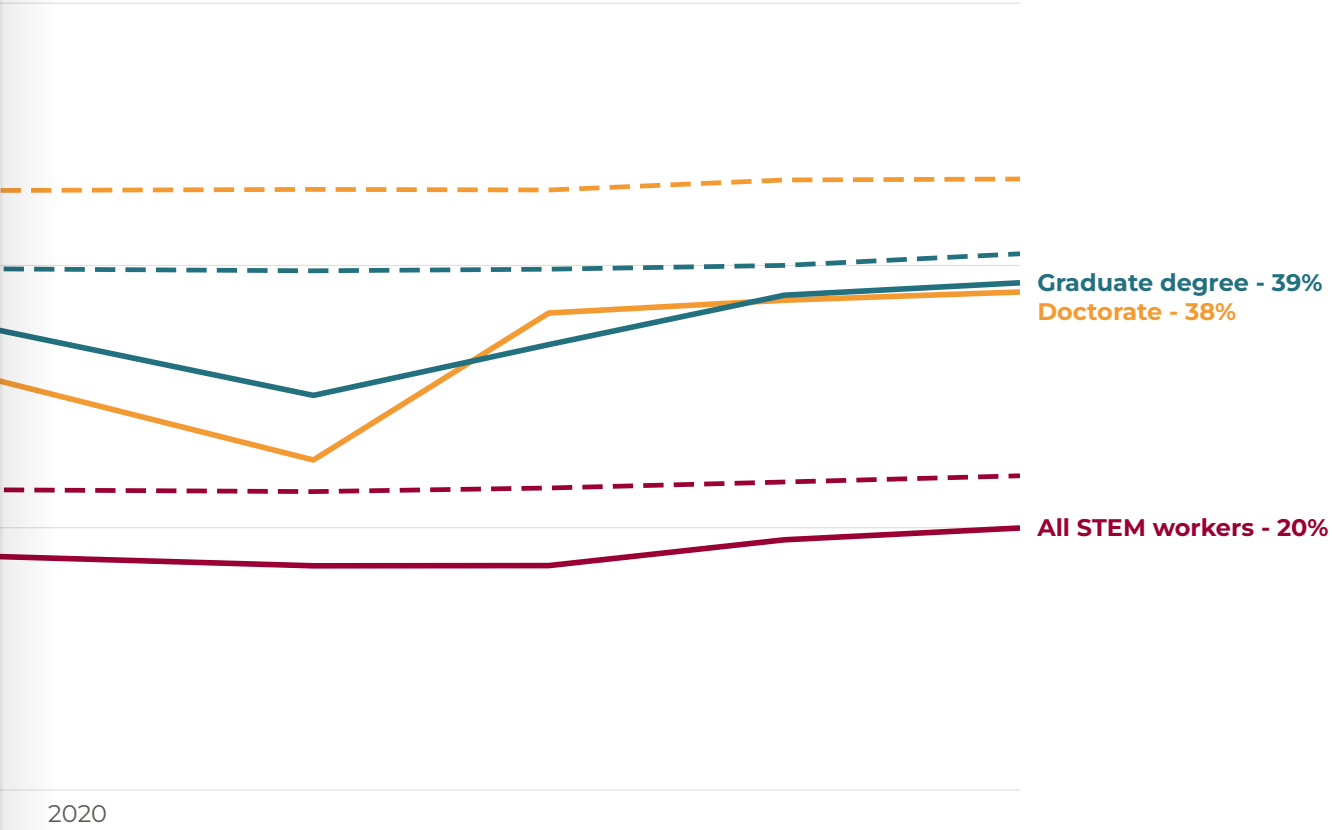
Figure 20 How Much of North Carolina's STEM Workforce Was Born Abroad?



Foreign-born share of employed workers in STEM occupations, by state of residence; state solid, U.S. dashed. Shares resti
Community Survey.

In 2024, 20% of the people working in STEM occupations in North Carolina were born abroad (U.S. 24%; 15th of 51).

The share rises with education: 39% of STEM workers with a graduate degree were foreign-born (U.S. 41%), and 38% of those with a doctorate (U.S. 47%; 15th of 20). 11% of all STEM workers were not U.S. citizens.



ng on fewer than 100 survey records are left out. No 2020 (experimental data). Data from U.S. Census Bureau, American

20%

Foreign-born share of STEM workers, 2024

U.S. **24%** · **15th** of 51 · +4.4 pp since 2015

39%

Foreign-born, STEM workers with grad degrees, 2024

U.S. **41%** · **12th** of 44 · +7.4 pp since 2015

11%

Noncitizen share of STEM workers, 2024

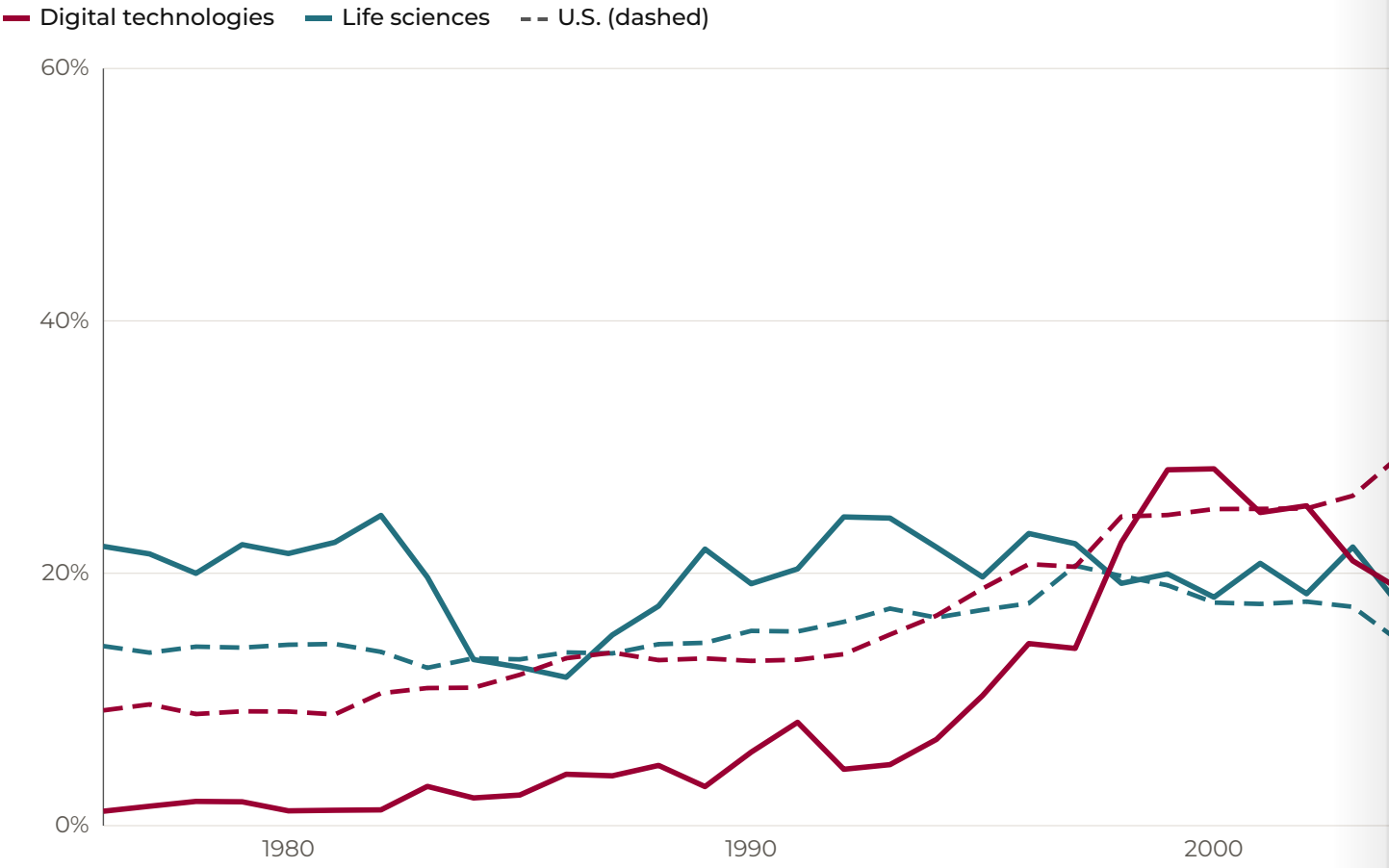
U.S. **11%** · **11th** of 51 · +1.8 pp since 2015

SECTION 10

Patents

North Carolina ranks 8th of 51 on patents granted to assignees in the state, at 4.33K in 2025 against 135K for the U.S.

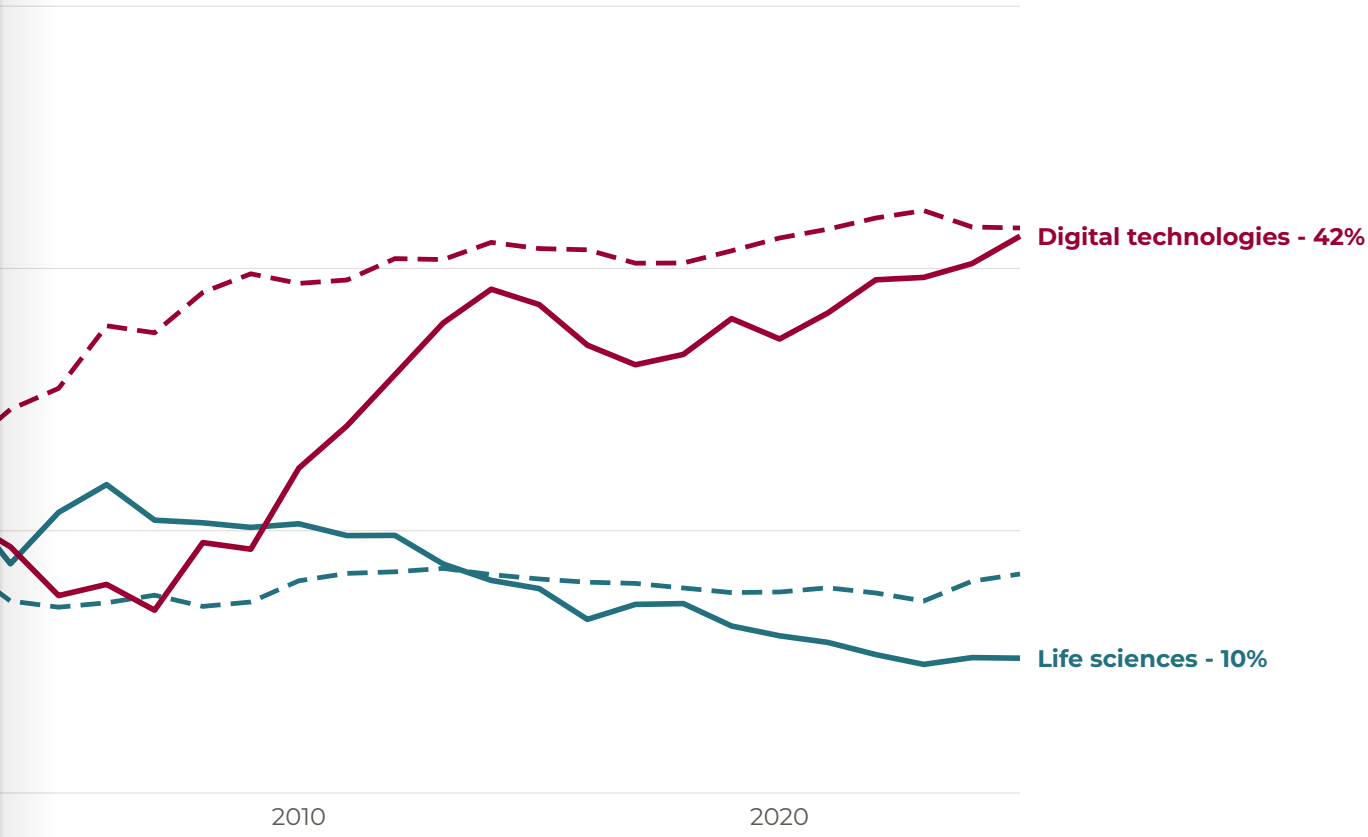
Figure 21 What Kinds of Technology Does North Carolina Patent?



Share of patents granted to owners in the state, by WIPO technology field of the first classification; state solid, U.S. dashed

Organizations and people in North Carolina were granted 4.33K U.S. patents in 2025, 3.2% of all patents with a U.S. owner (8th of 51).

42% were in digital technologies (U.S. 43%) and 10% in the life sciences (U.S. 17%).



d. Fractional counts by the owner's (assignee's) location, not where inventors live. Data from USPTO, PatentsView.

4.33K

Patents granted to owners in the state, 2025

U.S. **135K** · **8th** of 51 · +70% since 2015

42%

Patents in digital technologies, 2025

U.S. **43%** · **11th** of 51 · +5.2 pp since 2015

10%

Patents in life sciences, 2025

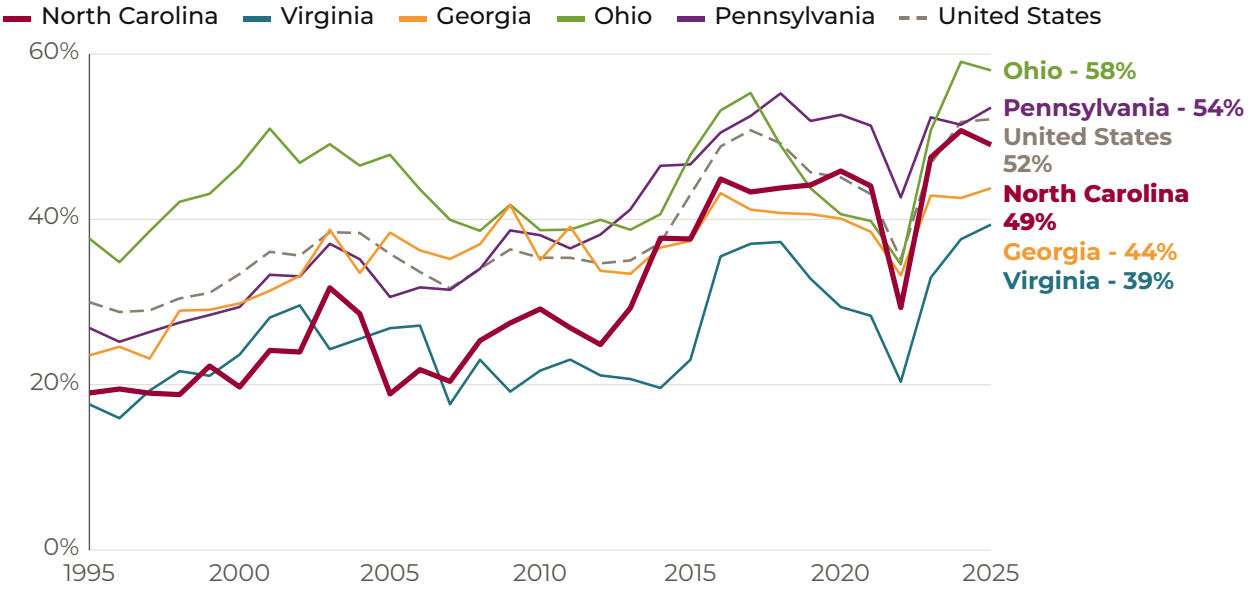
U.S. **17%** · **42nd** of 51 · −5.3 pp since 2015

SECTION 11

Competitors

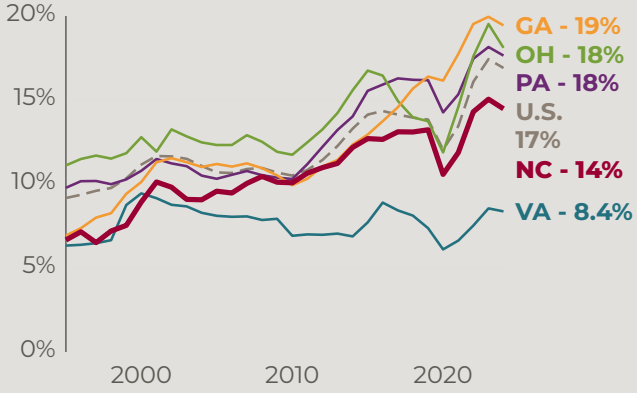
The four states closest in total fall enrollment, measured proportionally (a state 1.5 times as large counts as near as one two-thirds the size), are Virginia, Georgia, Ohio, and Pennsylvania. North Carolina ranks 3rd of the five on the international share of STEM master's; Ohio is the most dependent at 58%.

Figure 22 How Dependent Are North Carolina and Its Peers on International STEM Master's?



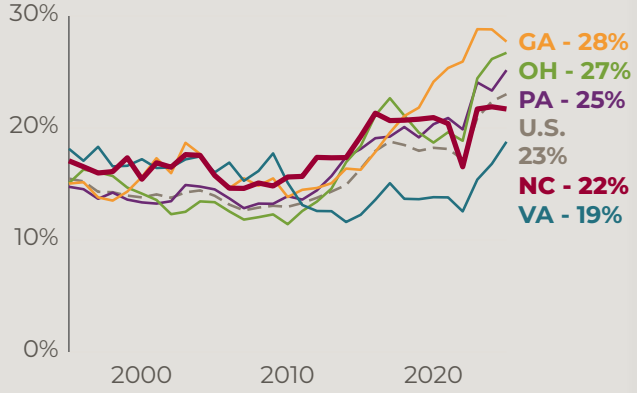
Data from NCES, IPEDS Completions.

Figure 23 Int. share of grad students



Data from NCES, IPEDS.

Figure 24 STEM share of master's



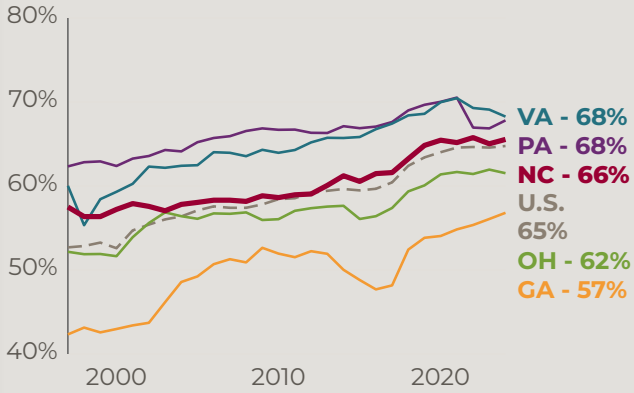
Data from NCES, IPEDS.

Table 4 How Does North Carolina Compare With States of Similar Size?

Measure	NC	VA	GA	OH	PA	U.S.
Students enrolled, 2024	**582K**	575K	594K	611K	637K	19.7M
Graduate share of students, 2024	**14%**	22%	18%	14%	23%	17%
STEM share of bachelor's, 2025	**26%**	24%	27%	24%	27%	24%
STEM share of master's, 2025	**22%**	19%	28%	27%	25%	23%
International STEM master's, 2025	**2.42K**	2.07K	3.78K	3.84K	5.51K	117K
Share of U.S. international STEM master's, 2025	**2.1%**	1.8%	3.2%	3.3%	4.7%	–
Int. share of STEM master's, 2025	**49%**	39%	44%	58%	54%	52%
Int. share of STEM research doctorates, 2025	**39%**	41%	44%	45%	45%	43%
Int. share of grad students, 2024	**14%**	8.4%	19%	18%	18%	17%
Int. share of undergraduate enrollment, 2024	**2.1%**	2.7%	2.6%	3.3%	3.9%	3.0%
Six-year graduation rate, 2024	**66%**	68%	57%	62%	68%	65%
In-state tuition, public 4-year, 2023	**$7.54K**	$15.7K	$7.14K	$11K	$17K	$9.71K
Appropriations per student, 2024	**$15.6K**	$11.3K	$10.8K	$7.76K	$7.14K	$11.4K
Average faculty salary, 2024	**$95.1K**	$106K	$95.3K	$93.3K	$109K	$106K
First-time students from in state, 2024	**79%**	75%	82%	76%	68%	–
Net migration per 100 residents, 2024	**+4**	+2	−6	+5	+8	–

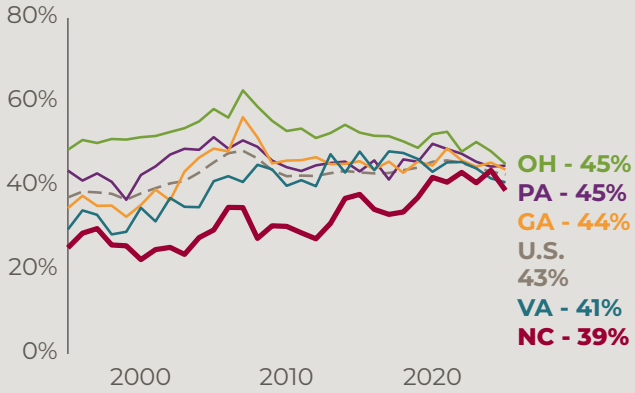
Latest year for each measure, given after its name. Dollars are 2024 dollars (CPI-U). Data from NCES, IPEDS.

Figure 25 Six-year graduation rate



Data from NCES, IPEDS.

Figure 26 Int. share of STEM doctorates

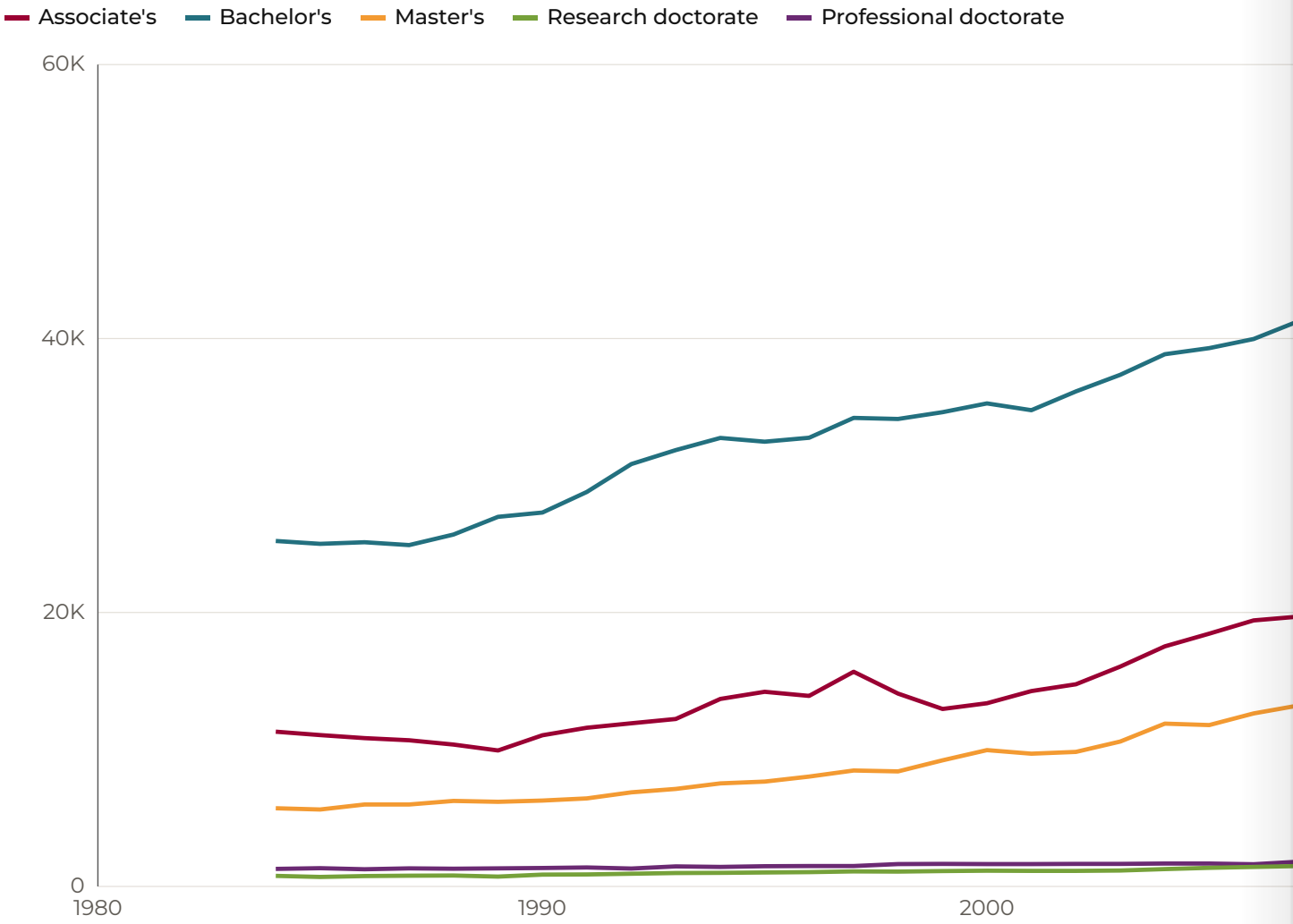


Data from NCES, IPEDS.

SECTION 12

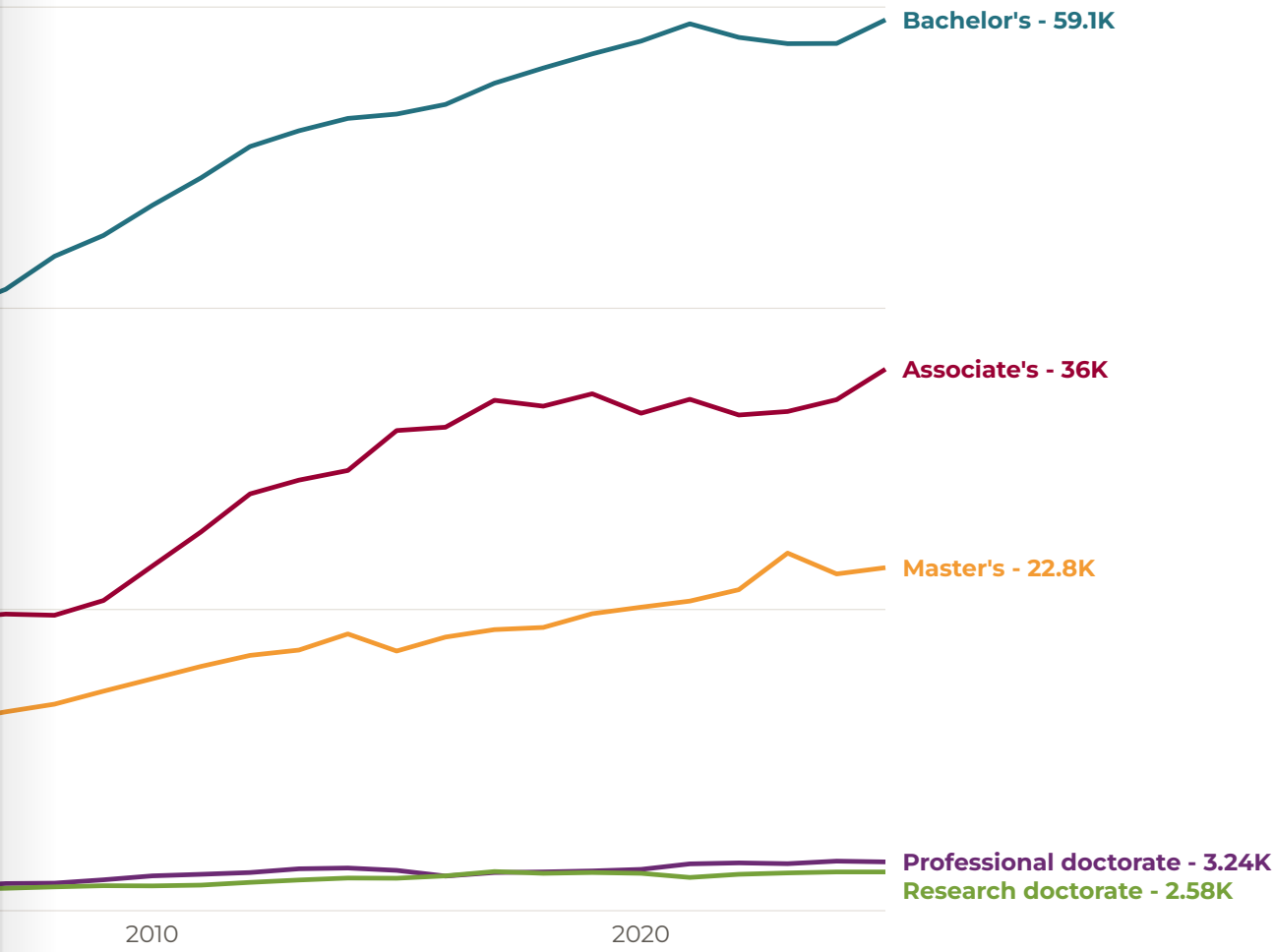
Fields of Study

Figure 27 How Many Degrees Does North Carolina Award?



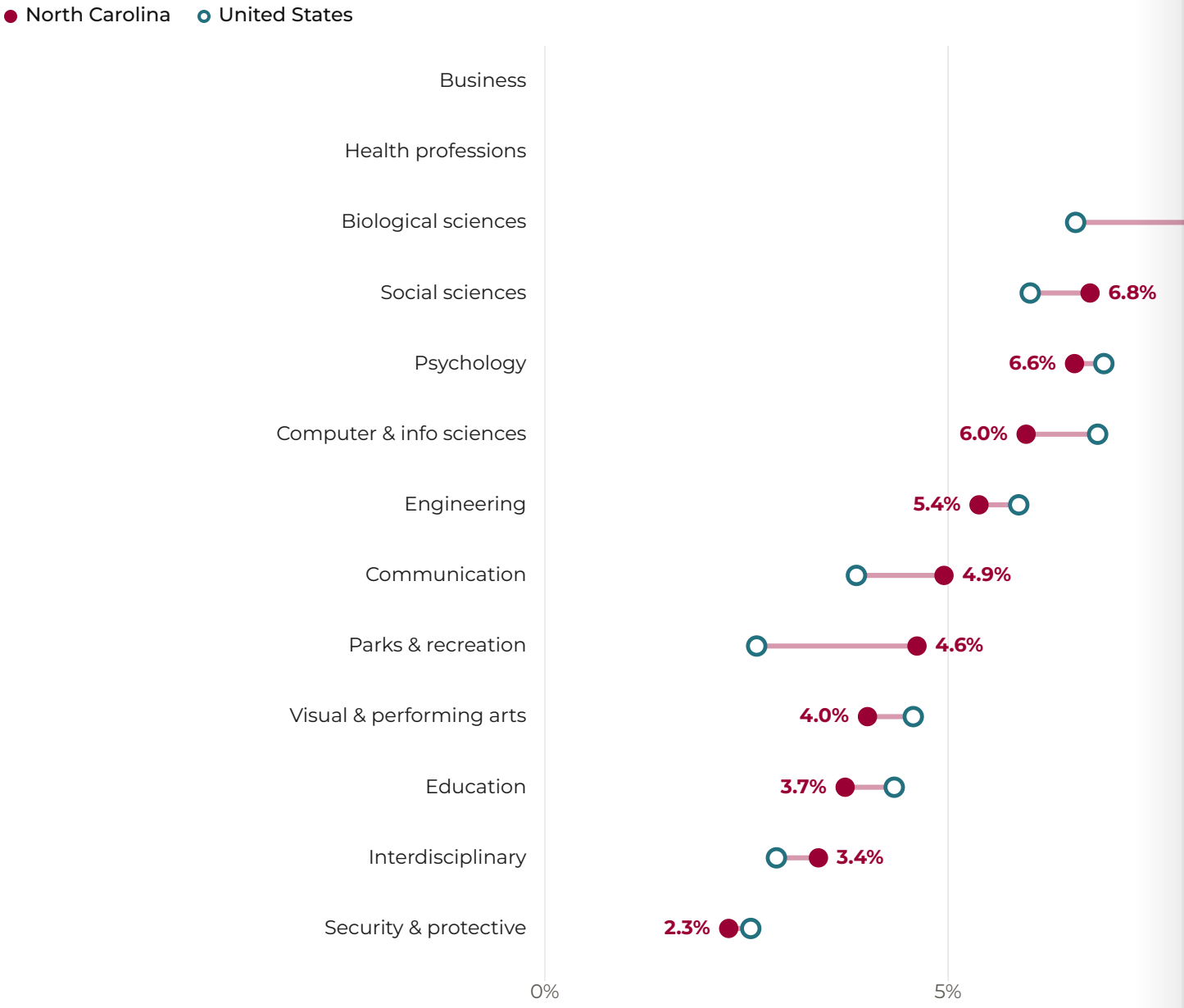
Degrees awarded by level, first majors. Data from NCES, IPEDS Completions.

In 2025, North Carolina's institutions awarded 36K associate's degrees, 59.1K bachelor's degrees, 22.8K master's degrees, 2.58K research doctorate degrees, 3.24K professional doctorate degrees.



Since 2015: associate's up 13%; bachelor's up 12%; master's up 32%; research doctorate up 19%; professional doctorate up 21%. The state awards 11.8 bachelor's degrees for every 100 undergraduates (U.S. 12.3).

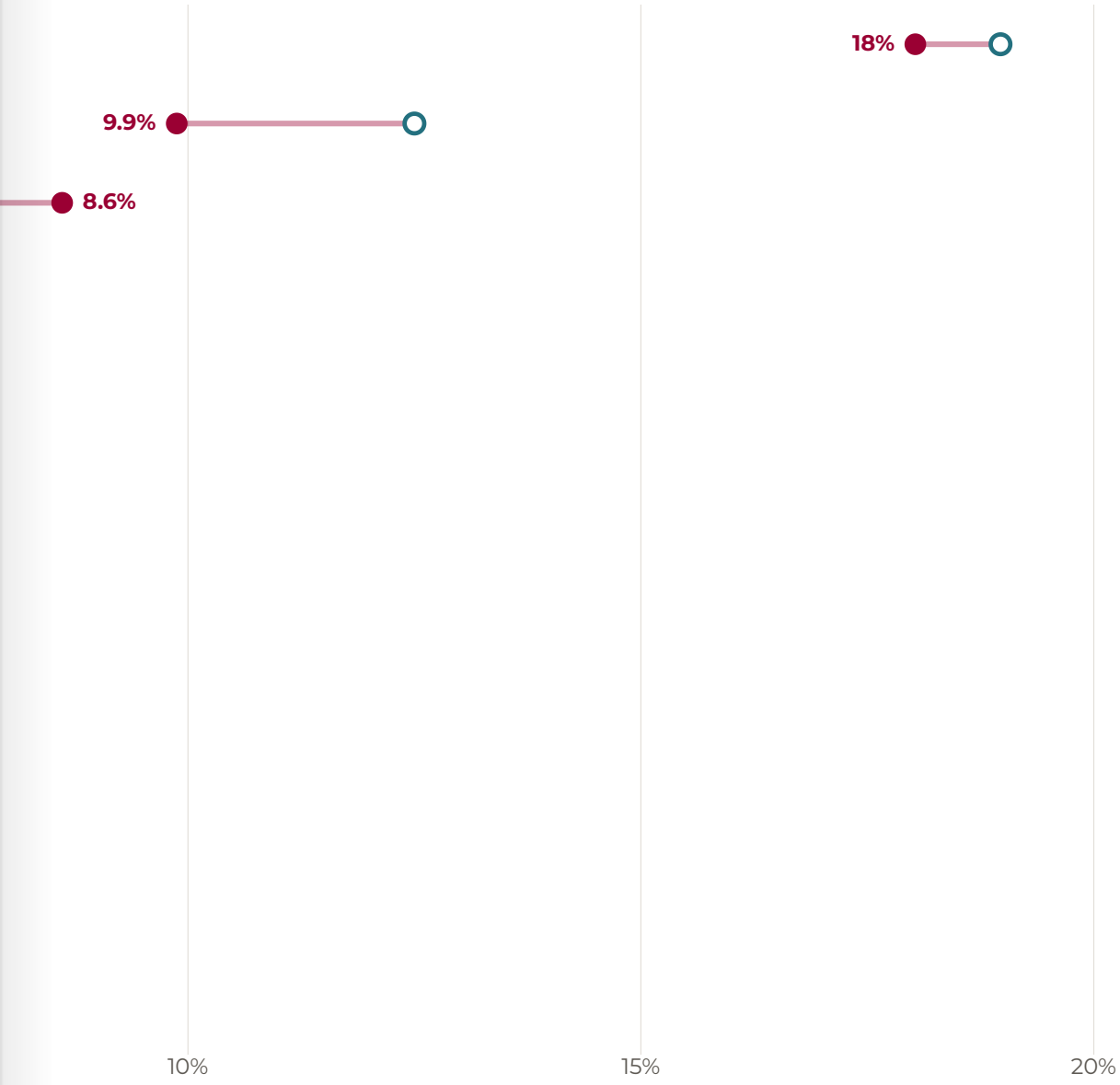
Figure 28 What Do North Carolina's Bachelor's Graduates Study?



Share of bachelor's degrees by field, 2025. Data from NCES, IPEDS.

North Carolina's largest bachelor's fields are business (18%; U.S. 19%), health professions (9.9%; U.S. 13%), biological sciences (8.6%; U.S. 6.6%).

Relative to the nation, North Carolina's bachelor's degrees are most concentrated in parks & recreation (1.76× the U.S. share), engineering technology (1.55× the U.S. share), public administration (1.36× the U.S. share).



6.0%

Bachelor's in computer science, 2025

U.S. **6.9%** · **24th** of 51 · +3.4 pp since 2015

6.8%

Bachelor's in engineering & eng. technology, 2025

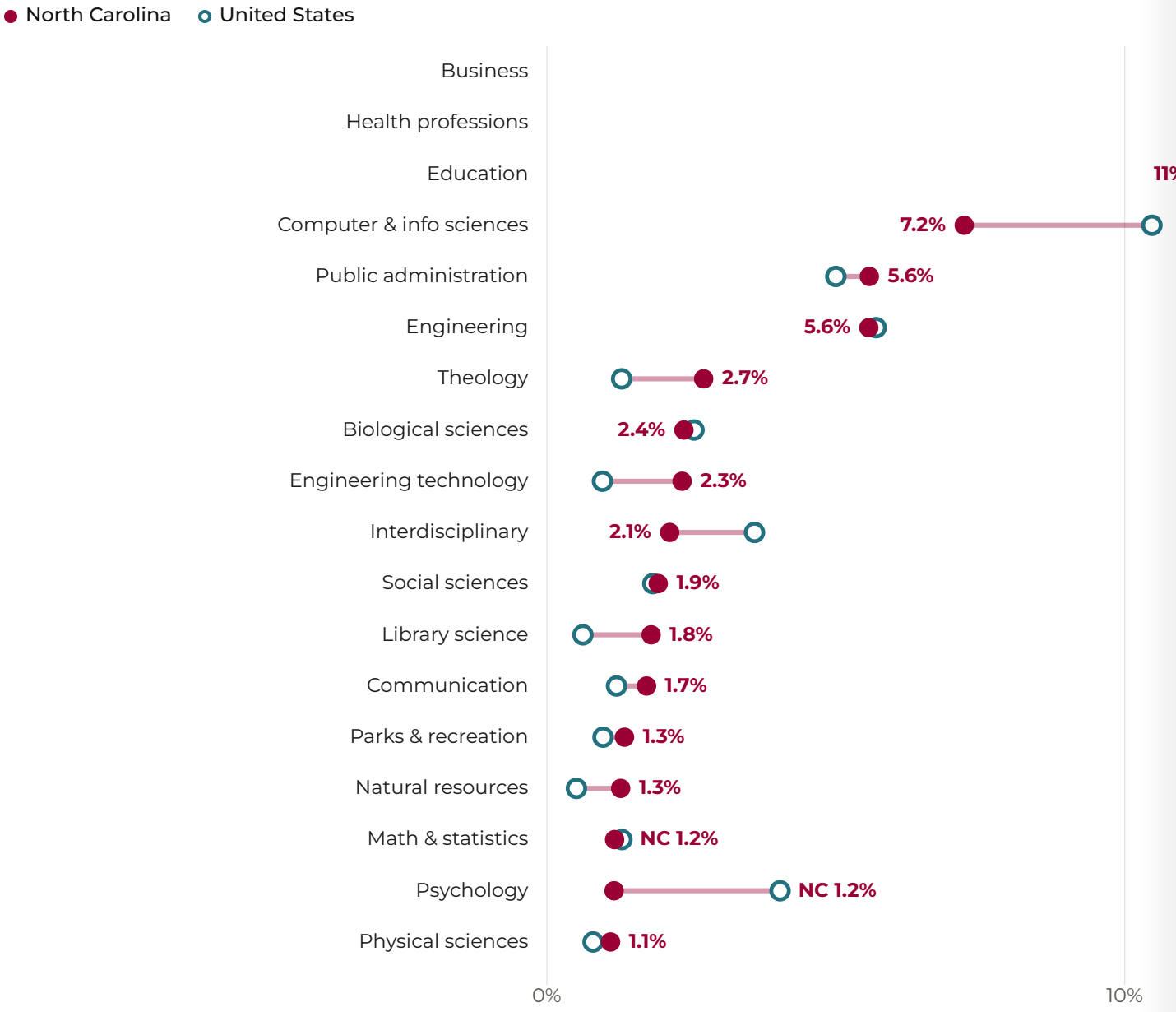
U.S. **6.8%** · **23rd** of 51 · +1.3 pp since 2015

18%

Bachelor's in business, 2025

U.S. **19%** · **33rd** of 51 · +1.2 pp since 2015

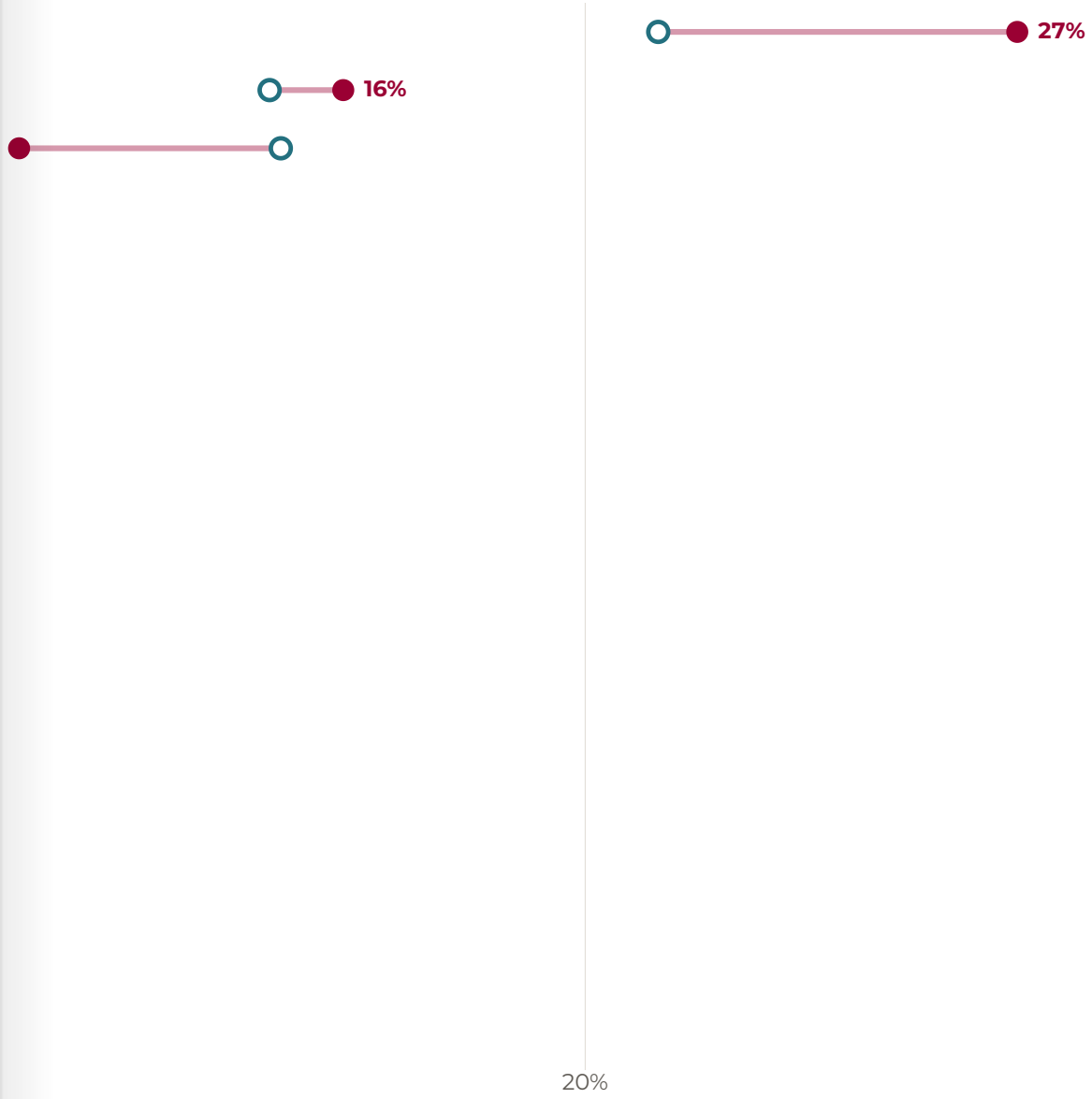
Figure 29 What Do North Carolina's Master's Graduates Study?



Share of master's degrees by field, 2025; largest fields. Data from NCES, IPEDS.

North Carolina's largest master's fields are business (27%; U.S. 21%), health professions (16%; U.S. 15%), education (11%; U.S. 15%).

International students earn 19% of all master's degrees in the state (U.S. 22%), and STEM fields account for 22% of them.



22%

STEM share of master's, 2025

U.S. **23%** · **24th** of 51 · +2.4 pp since 2015

19%

Int. share of all master's, 2025

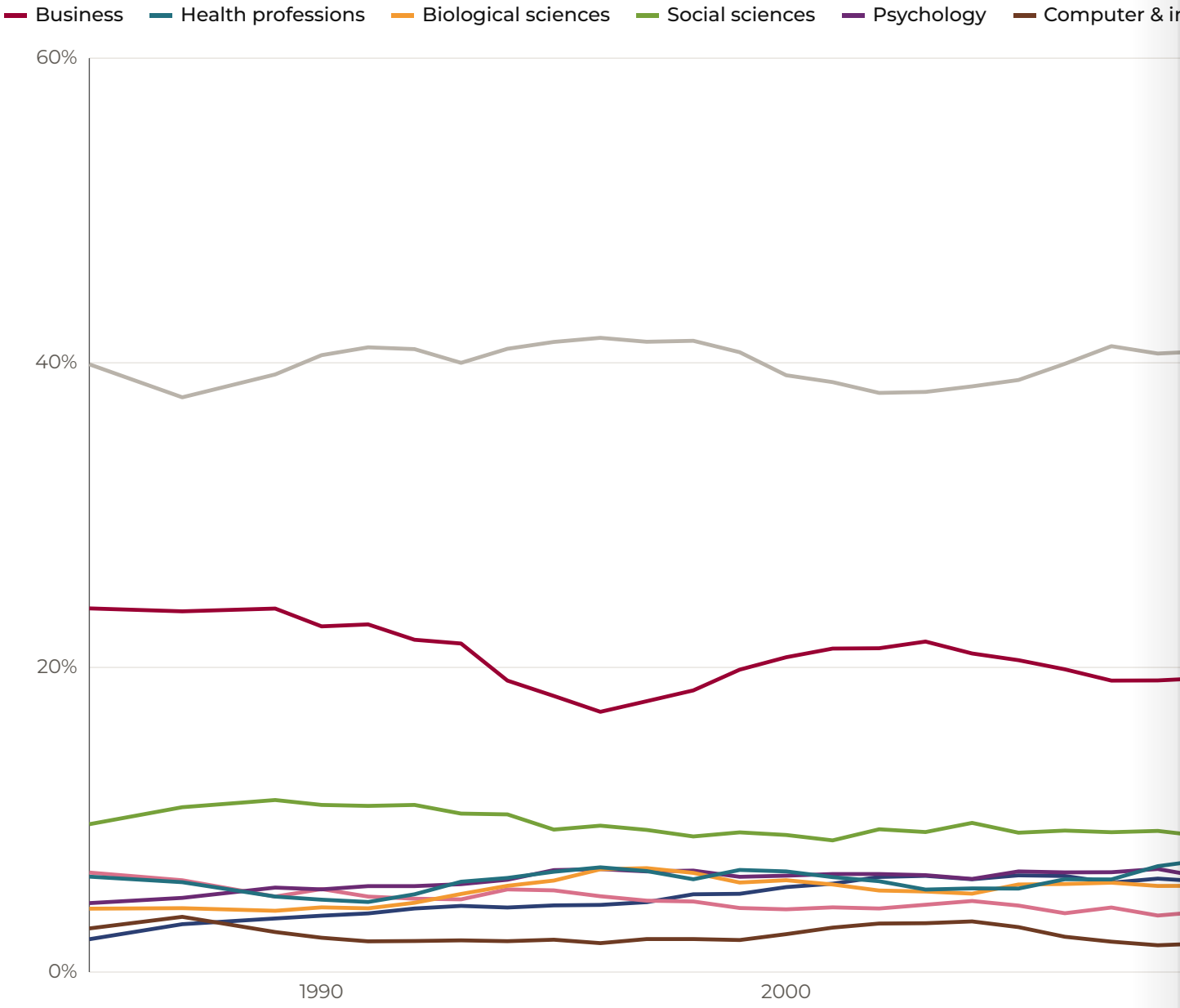
U.S. **22%** · **21st** of 51 · +4.4 pp since 2015

64%

Int. share of CS master's, 2025

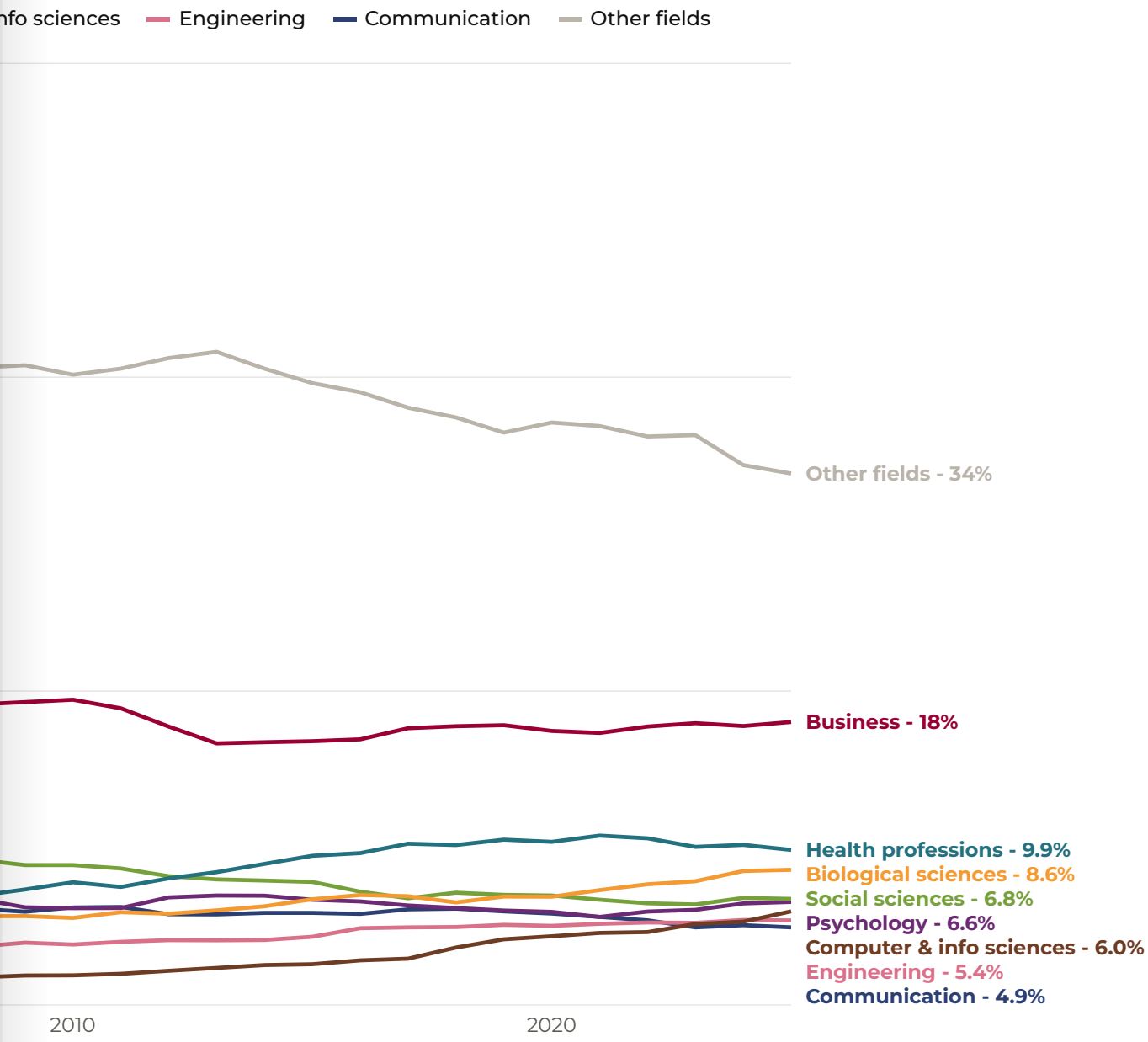
U.S. **65%** · **25th** of 47 · +16.3 pp since 2015

Figure 30 How Has North Carolina's Bachelor's Field Mix Changed?



Share of bachelor's degrees in the largest fields. Data from NCES, IPEDS Completions.

Since 2015, computer & info sciences has gained the most ground, 3.4 points of the state's bachelor's degrees, while social sciences lost the most (1.1 points).



The chart follows the share of bachelor's degrees in each of the state's largest fields; smaller fields are grouped as other fields.

SECTION 12 · FIELDS OF STUDY

Where Is North Carolina Specialized, and What Do Its Doctoral Graduates Study?

The location quotient divides a field's share of North Carolina's bachelor's degrees by its share of all U.S. bachelor's degrees in 2025. A quotient of 1.00 means the same share as the nation; 2.00 means twice the national share. The table lists the 10 fields with the highest quotients among fields with at least 50 degrees.

Highest location quotients. Parks & recreation (1.76, with 4.6% of the state's bachelor's degrees against 2.6% for the U.S.), engineering technology (1.55, with 1.4% of the state's bachelor's degrees against 0.9% for the U.S.), and public administration (1.36, with 2.0% of the state's bachelor's degrees against 1.5% for the U.S.) had the highest quotients.

Table 5 Where Is North Carolina Most Specialized?

Field	Location quotient	State share	U.S. share	Degrees
Parks & recreation	**1.76**	4.6%	2.6%	2.73K
Engineering technology	**1.55**	1.4%	0.9%	819
Public administration	**1.36**	2.0%	1.5%	1.18K
Biological sciences	**1.31**	8.6%	6.6%	5.09K
Physical sciences	**1.29**	1.7%	1.3%	981
Natural resources	**1.28**	1.4%	1.1%	847
Communication	**1.28**	4.9%	3.9%	2.93K
Interdisciplinary	**1.18**	3.4%	2.9%	2.01K
Math & statistics	**1.18**	1.4%	1.2%	817
Agriculture	**1.13**	1.2%	1.0%	685

Bachelor's degrees; location quotient = state share ÷ U.S. share; fields with 50+ degrees. Data from NCES, IPEDS.

26%
STEM share of bachelor's, 2025
U.S. **24%** · **13th** of 51

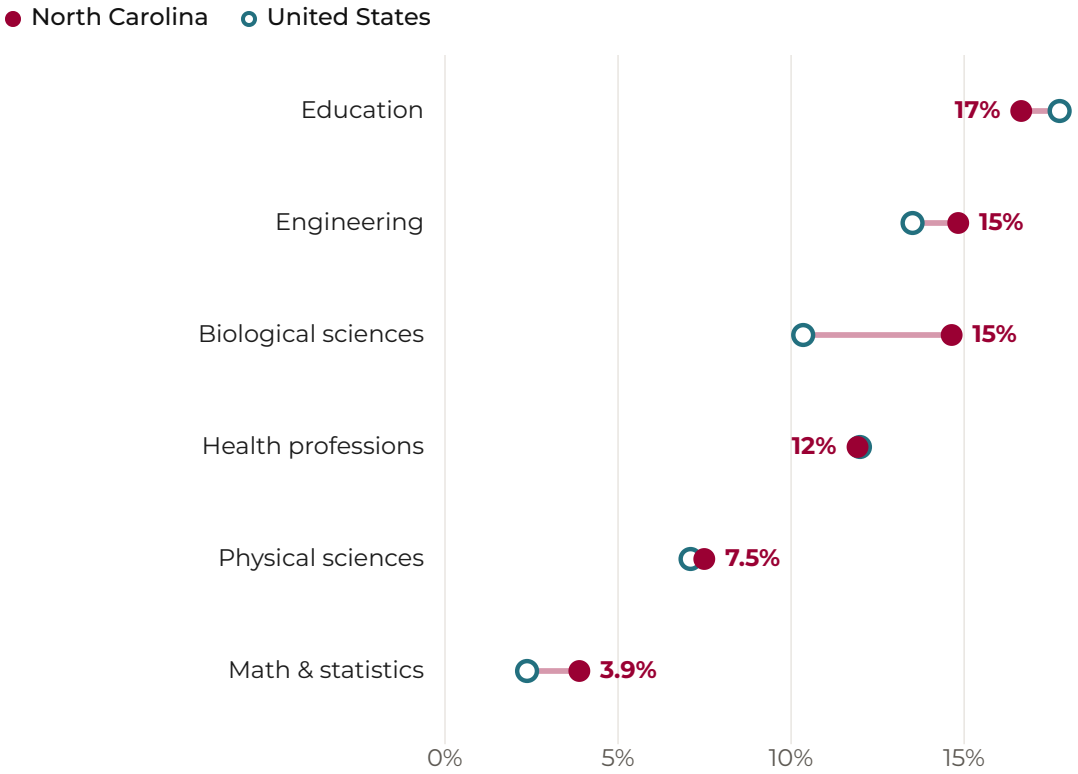
6.0%
Bachelor's in computer science, 2025
U.S. **6.9%** · **24th** of 51

6.8%
Bachelor's in engineering & eng. technology, 2025
U.S. **6.8%** · **23rd** of 51

Each row is a field's share of the research doctorates awarded in North Carolina in 2025 (filled dot), against its share nationally (open dot), for the ten largest fields with at least 20 doctorates. Research doctorates exclude professional doctorates such as the M.D. and J.D.

Largest doctoral fields. The state's largest doctoral fields were education (17% of research doctorates, against 18% for the U.S.), engineering (15% of research doctorates, against 14% for the U.S.), and biological sciences (15% of research doctorates, against 10% for the U.S.).

Figure 31 What Do North Carolina's Doctoral Graduates Study?



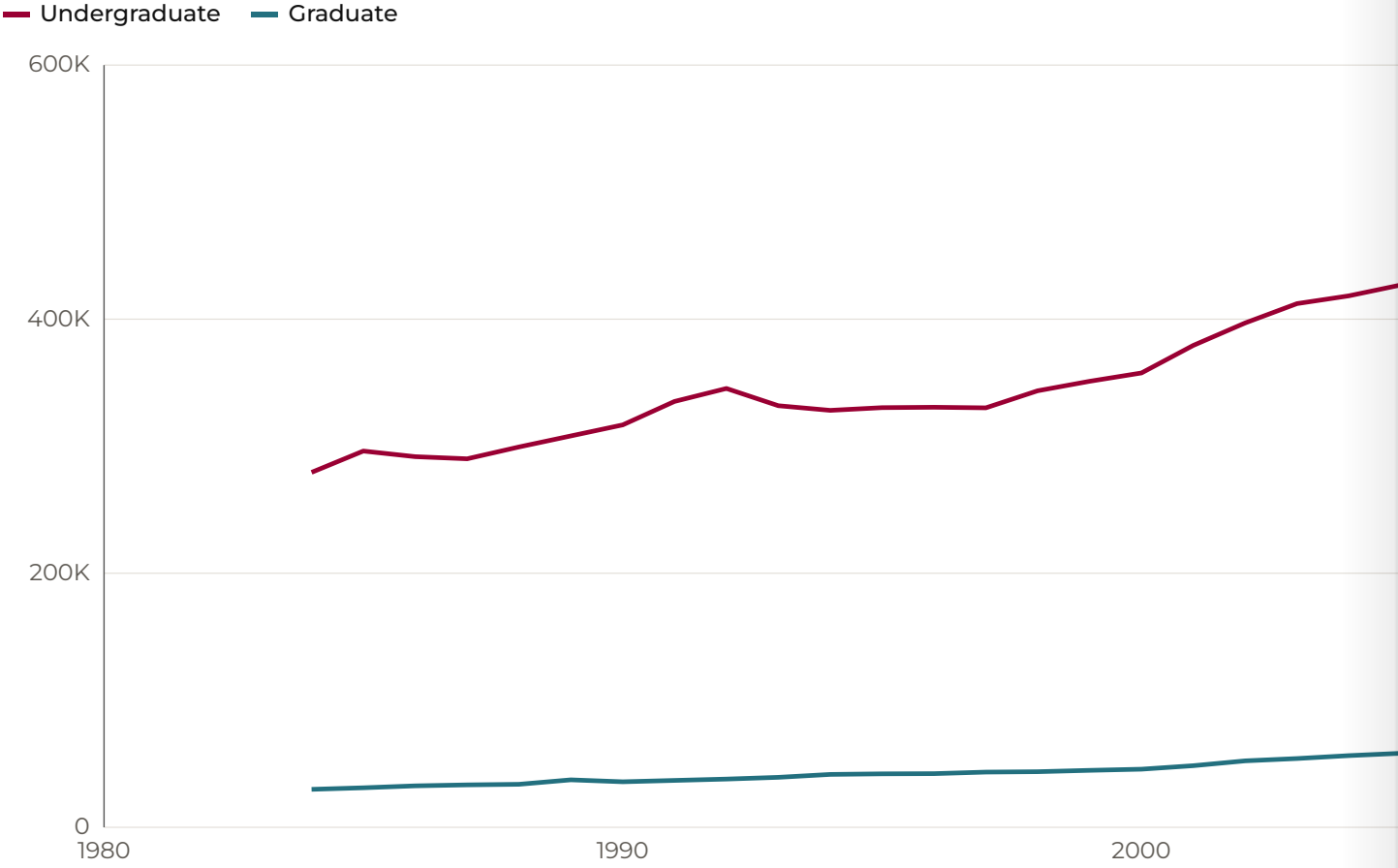
Share of research doctorates by field, 2025; fields with 20+ doctorates. Data from NCES, IPEDS.

SECTION 13

Students and Money

North Carolina ranks 10th of 51 on fall enrollment, at 582K in 2024 against 19.7M for the U.S.

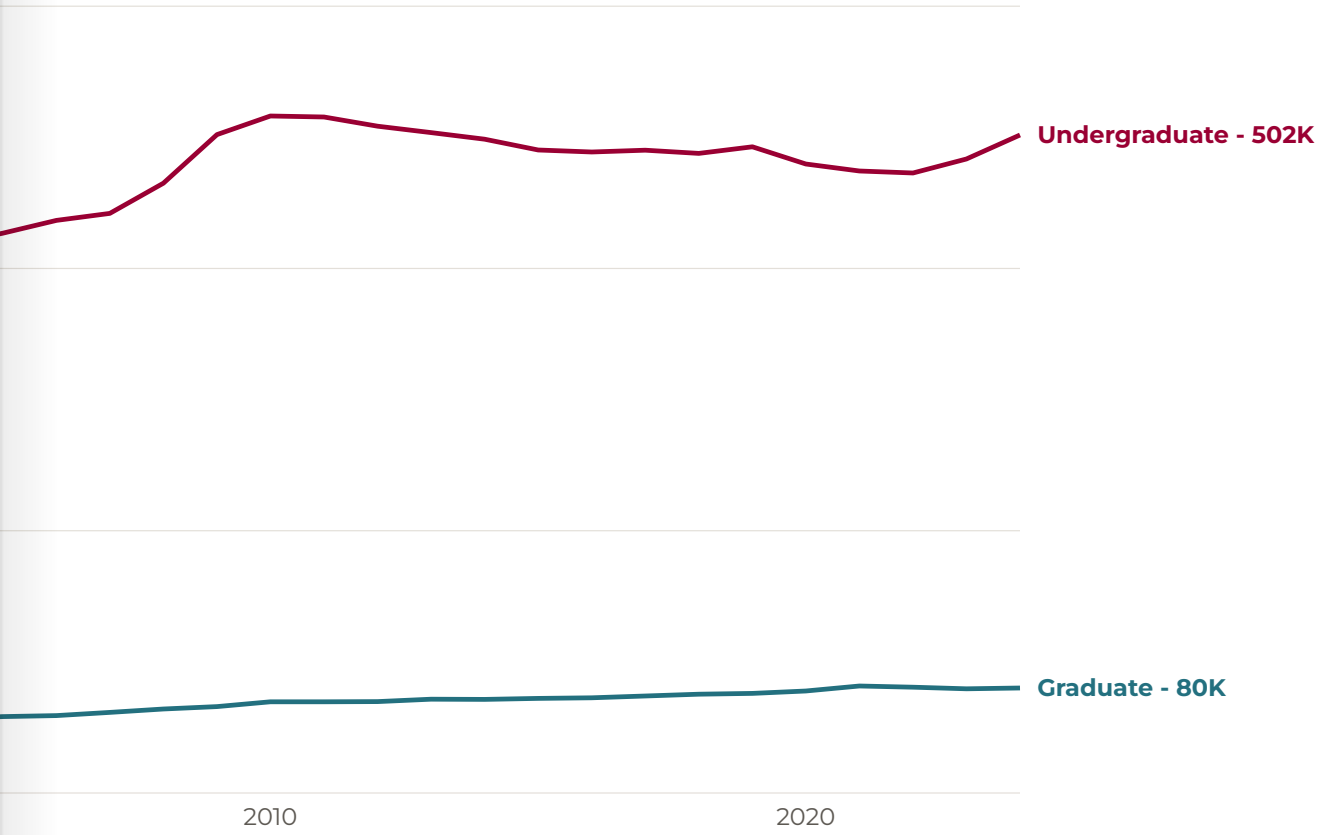
Figure 32 How Has Enrollment Changed?



Fall headcount enrollment. Data from NCES, IPEDS Fall Enrollment.

In fall 2024, North Carolina enrolled 502K undergraduates and 80K graduate students, 582K in all (10th of 51).

Undergraduate enrollment is up 2% since 2015 and graduate enrollment up 11% since 2015. Graduate students are 14% of all students (U.S. 17%).



582K

Students enrolled, 2024

U.S. **19.7M** · **10th** of 51 · +3.4% since 2015

14%

Graduate share of students, 2024

U.S. **17%** · **34th** of 51 · +0.9 pp since 2015

39%

Students attending part-time, 2024

U.S. **39%** · **29th** of 51 · +2.7 pp since 2015

SECTION 13 · STUDENTS AND MONEY

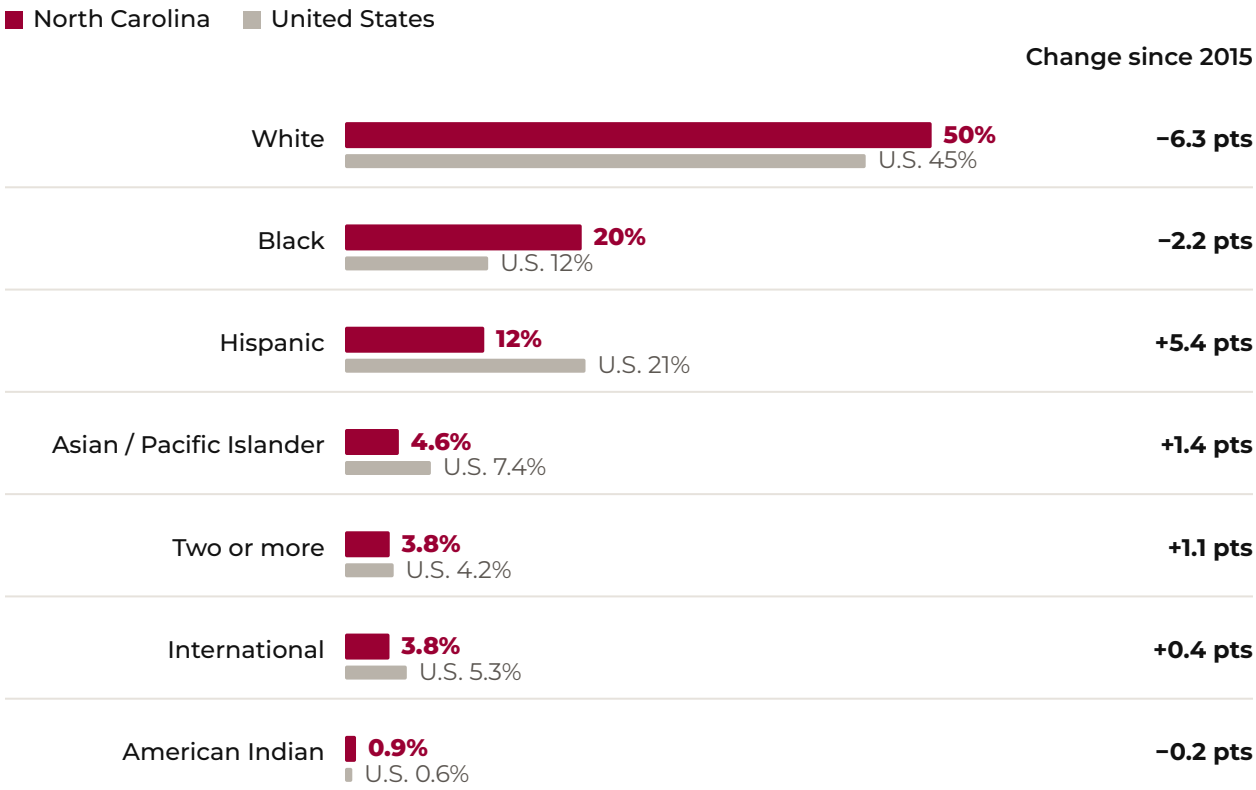
Who Enrolls in North Carolina, and Where?

In fall 2024, White students were the largest group at North Carolina's degree-granting institutions, **50%** of enrollment against 45% nationally, down 6.3 points since 2015.

Black students were **20%** (U.S. 12%), down 2.2 points since 2015, and Hispanic students were **12%** (U.S. 21%), up 5.4 points since 2015.

Asian and Pacific Islander students were **4.6%** (U.S. 7.4%), up 1.4 points since 2015, and students of two or more races were **3.8%** (U.S. 4.2%), up 1.1 points since 2015. "International" counts U.S. nonresidents on temporary visas, whatever their race.

Figure 33 Who Enrolls in North Carolina?



Share of fall 2024 headcount enrollment by race and ethnicity; students of unknown race are left out of the chart. Data from NCES, IPEDS.

50%
White students
U.S. **45%** · −6.3 pp since 2015

20%
Black students
U.S. **12%** · −2.2 pp since 2015

43%
At public 4-year institutions
+2.7 pp since 2015

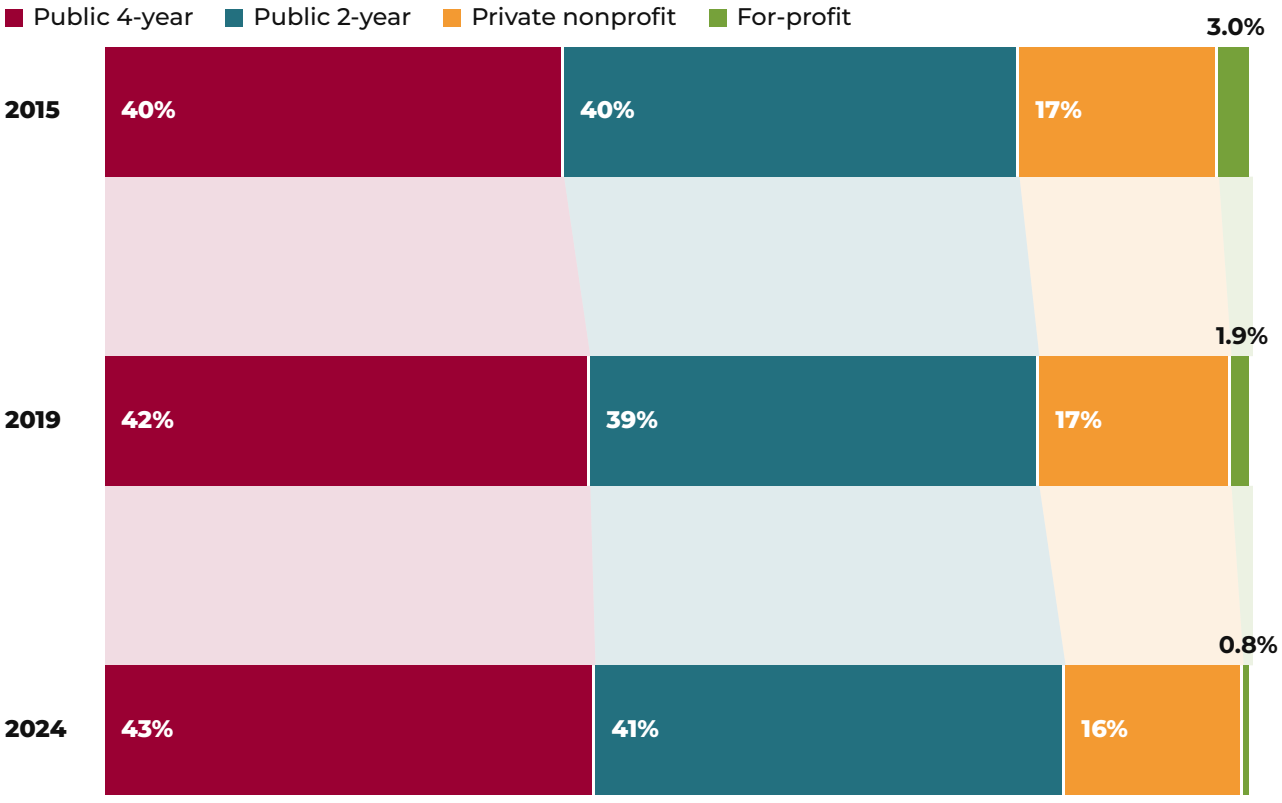
41%
At public 2-year institutions
+1.3 pp since 2015

In fall 2024, public 4-year institutions enrolled **43%** of students, up 2.7 points from 40% in 2015. Public 2-year institutions enrolled **41%** of students, up 1.3 points from 40% in 2015.

Private nonprofit institutions enrolled **16%** of students, down 1.8 points from 17% in 2015, and for-profit institutions enrolled **0.8%** of students, down 2.2 points from 3.0% in 2015.

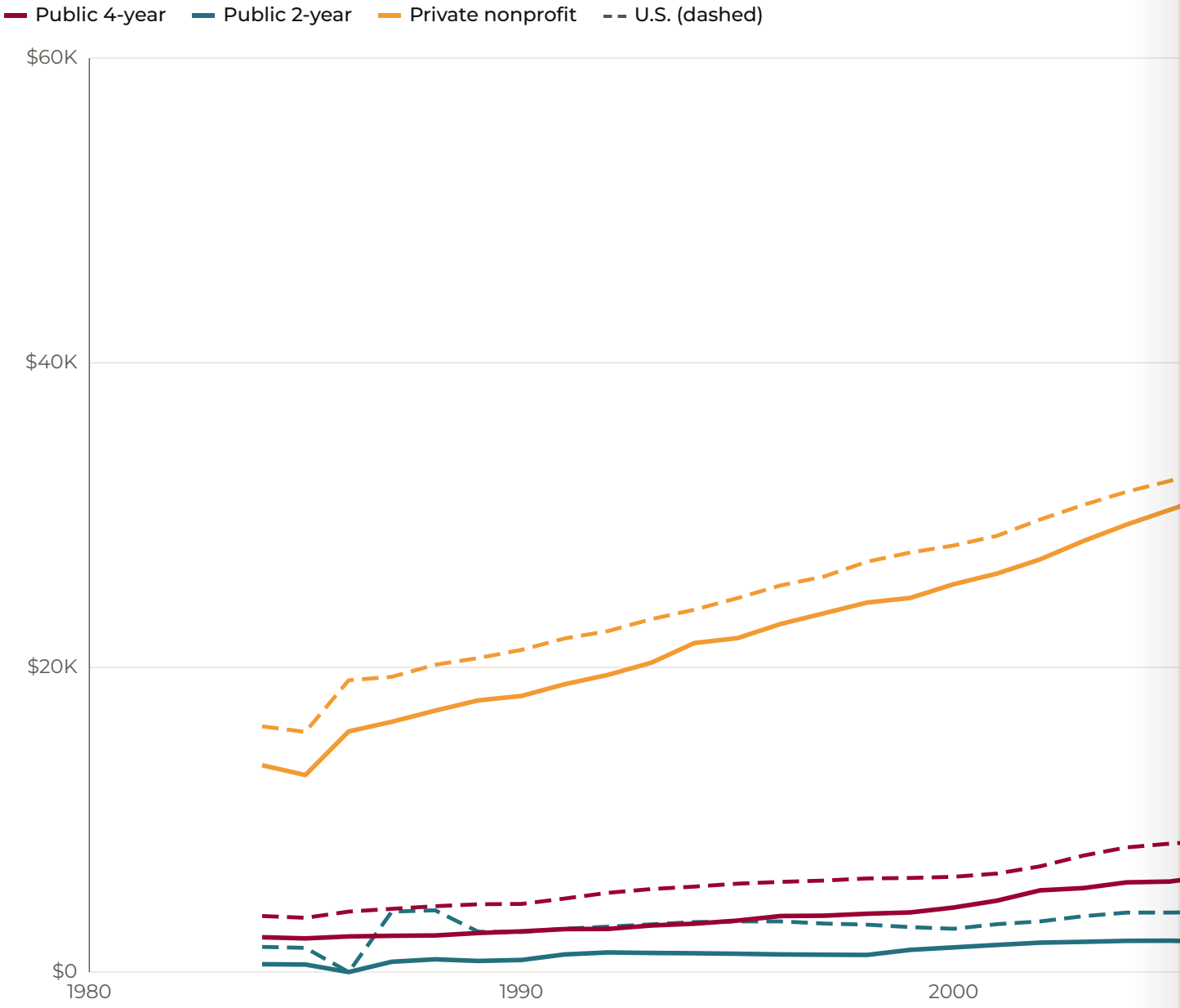
Public institutions together enrolled **84%** of students, against 73% nationally, 18th among the states.

Figure 34 Where Do Students Enroll?



Share of fall headcount enrollment by sector, 2015, 2019, 2024. Sectors as IPEDS classifies them each year. Data from NCES, IPEDS.

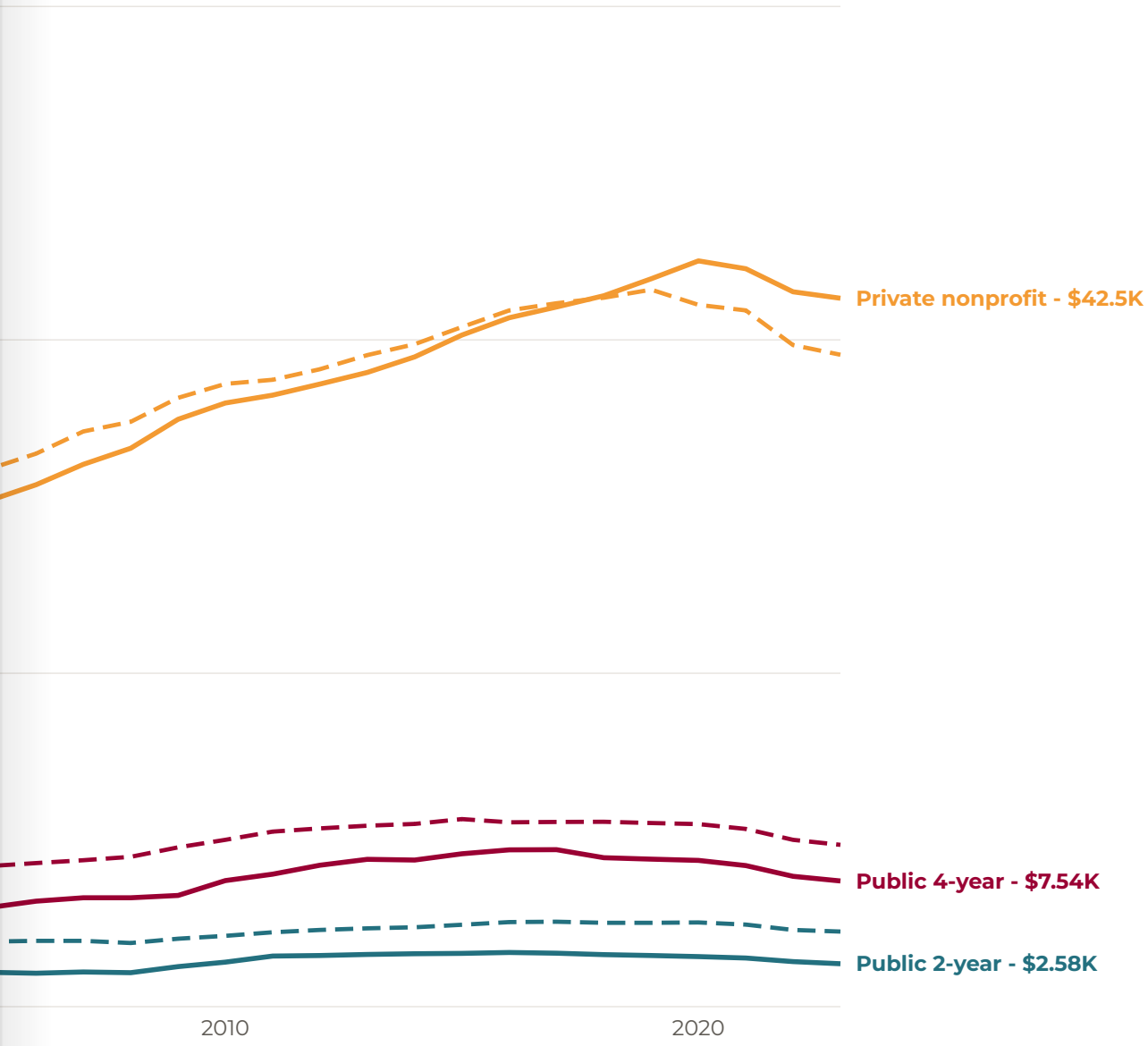
Figure 35 What Does In-State Tuition Cost?



Published in-state tuition and fees, 2024 dollars (CPI-U); state solid, U.S. dashed. Data from NCES, IPEDS.

In-state tuition and fees at North Carolina's public four-year institutions are $7.54K (2023, 2024 dollars), 22% below the U.S. ($9.71K); the state ranks 45th of 51.

State and local appropriations come to $15.6K per full-time-equivalent student at public institutions (9th of 51), and net tuition makes up 28% of net tuition plus state and local appropriations (U.S. 43%).



$7.54K

In-state tuition, public 4-year, 2023

U.S. **$9.71K** · **45th** of 51 · −18% since 2015

$15.6K

Appropriations per student, 2024

U.S. **$11.4K** · **9th** of 51 · +12% since 2015

28%

Tuition share of tuition plus appropriations, 2024

U.S. **43%** · **46th** of 51 · −4.3 pp since 2015

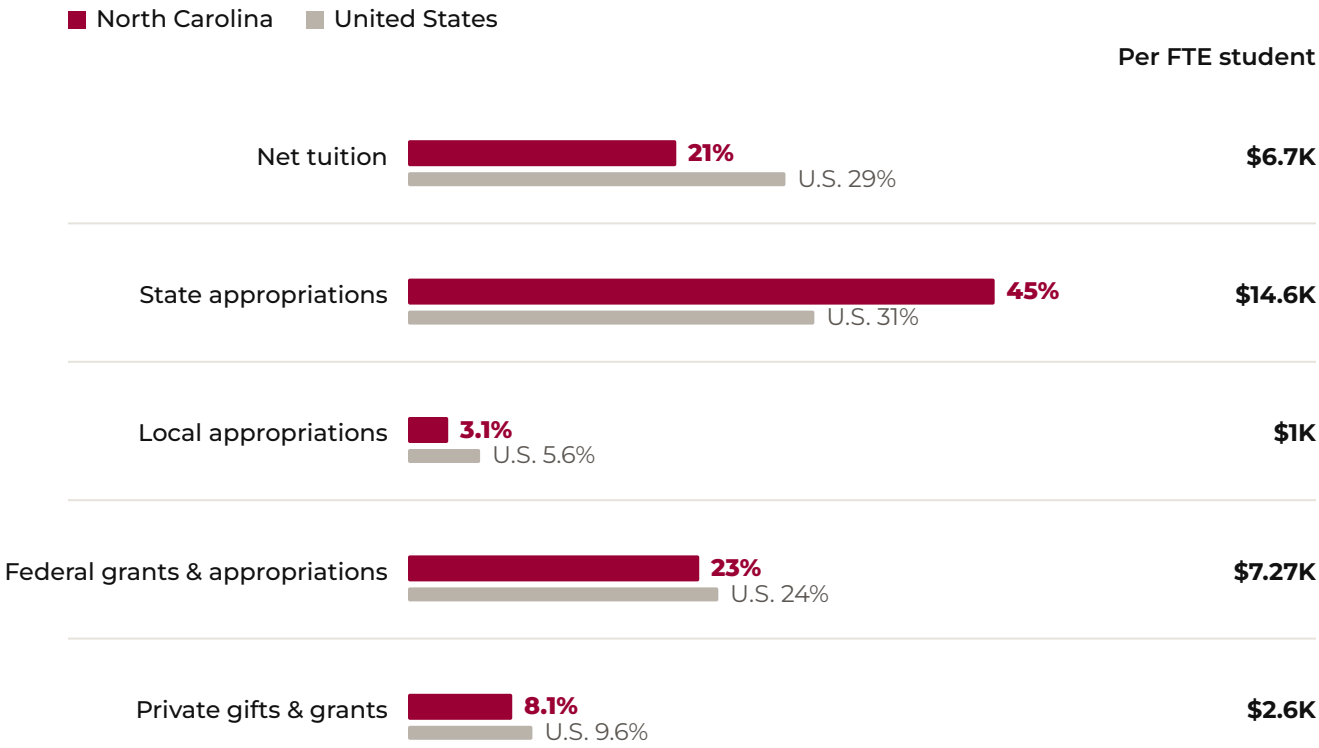
SECTION 13 · STUDENTS AND MONEY

How Are North Carolina's Public Institutions Funded, and What Do They Produce?

The bars give each source's share of the core revenue of North Carolina's public institutions, with the nation for comparison, and at right the dollars per full-time-equivalent (FTE) student it provides. Values are in 2024 dollars (CPI-U). Net tuition is tuition and fees after institutional discounts and allowances.

Net tuition per FTE student. The figure was $6.7K in FY2024, against $8.96K for the U.S., down 1.1% from $6.77K in FY2023 and down 5.6% since FY2015, when it was $7.09K.

Figure 36 How Are Public Institutions Funded?



Share of core revenue per FTE student by source, public institutions, FY2024; dollars per FTE at right, 2024 dollars (CPI-U). Shares are of all five sources; the tuition share elsewhere in this profile divides net tuition by net tuition plus state and local appropriations only. Data from NCES, IPEDS Finance.

$7.54K
In-state tuition, public 4-year, 2023
U.S. **$9.71K** · **45th** of 51

$15.6K
Appropriations per student, 2024
U.S. **$11.4K** · **9th** of 51

28%
Tuition share of tuition plus appropriations, 2024
U.S. **43%** · **46th** of 51

The table brings together five measures of what public higher education costs and what it produces in North Carolina: in-state tuition at public four-year institutions, state and local appropriations per full-time-equivalent student, tuition's share of public institutions' core revenue, the six-year bachelor's graduation rate, and the average full-time faculty salary.

In-state tuition, public 4-year. The figure was $7.54K in 2023, against $9.71K for the U.S., 45th among the states, down 3.5% from $7.81K in 2022 and down 18% since 2015, when it was $9.17K.

Table 6 Money and Results

Measure	Year	North Carolina	U.S.	Rank	Change since 2015
In-state tuition, public 4-year	2023	**$7.54K**	$9.71K	45th of 51	−18%
Appropriations per student	2024	**$15.6K**	$11.4K	9th of 51	+12%
Tuition share of tuition plus appropriations	2024	**28%**	43%	46th of 51	−4.3 pp
Six-year graduation rate	2024	**66%**	65%	22nd of 51	+5.0 pp
Average faculty salary	2024	**$95.1K**	$106K	28th of 51	−6.3%

Rank among the 50 states and DC (1 = highest). Dollars in 2024 dollars (CPI-U). Data from NCES, IPEDS.

SECTION 14

Student Outcomes

North Carolina ranks 37th of 51 on bachelor's degrees per 100 undergraduates, at 11.8 in 2025 against 12.3 for the U.S.

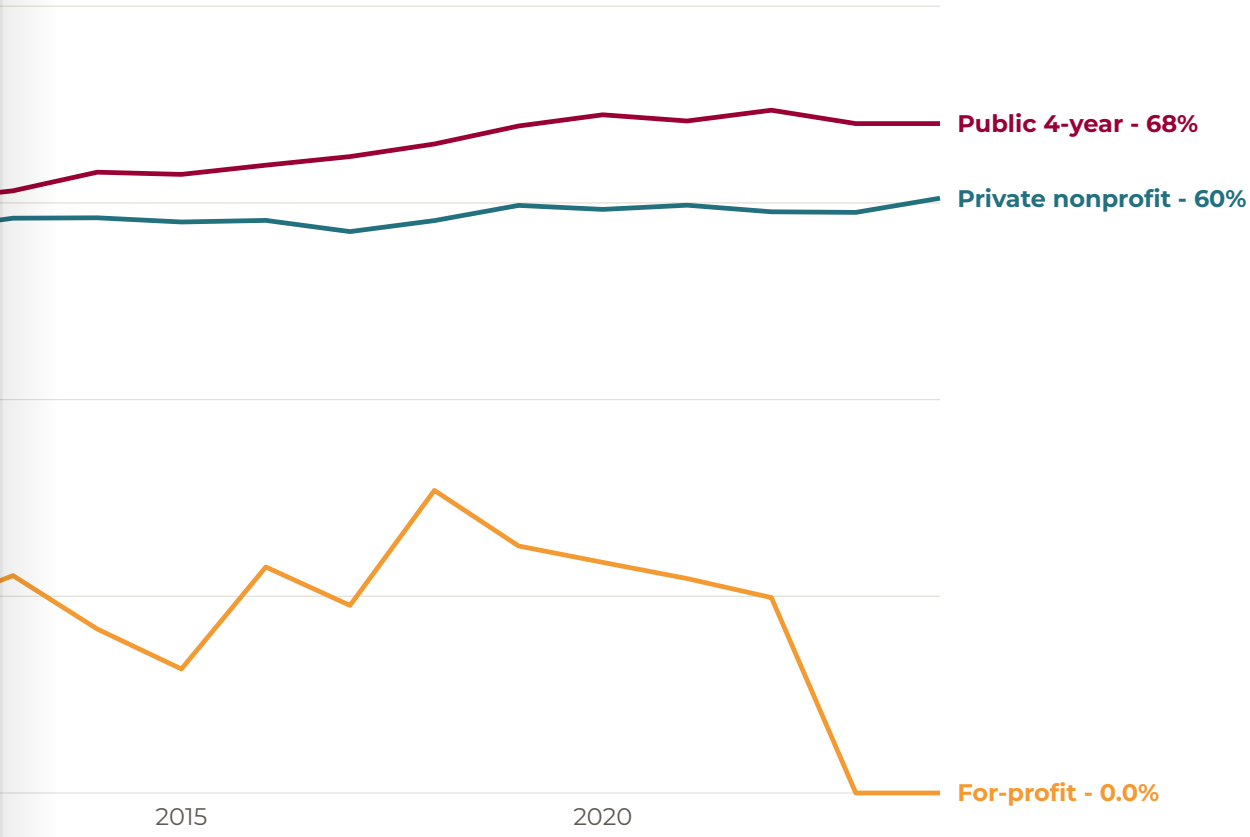
Figure 37 How Do Graduation Rates Differ by Sector?



Six-year bachelor's graduation rate by sector of institution. Data from NCES, IPEDS Graduation Rates.

66% of students who started a bachelor's degree at a four-year institution in North Carolina graduated within six years (2024), 0.8 points above the U.S. (65%); the state ranks 22nd of 51.

By sector: public 4-year 68%, private nonprofit 60%, for-profit 0.0%. At two-year institutions, 41% complete within 150% of normal time (U.S. 41%).



66%

Six-year graduation rate, 2024

U.S. **65%** · **22nd** of 51 · +5.0 pp since 2015

41%

Two-year completion rate, 2024

U.S. **41%** · **26th** of 50 · +22.2 pp since 2015

11.8

Bachelor's per 100 undergraduates, 2025

U.S. **12.3** · **37th** of 51 · +1.2 since 2015

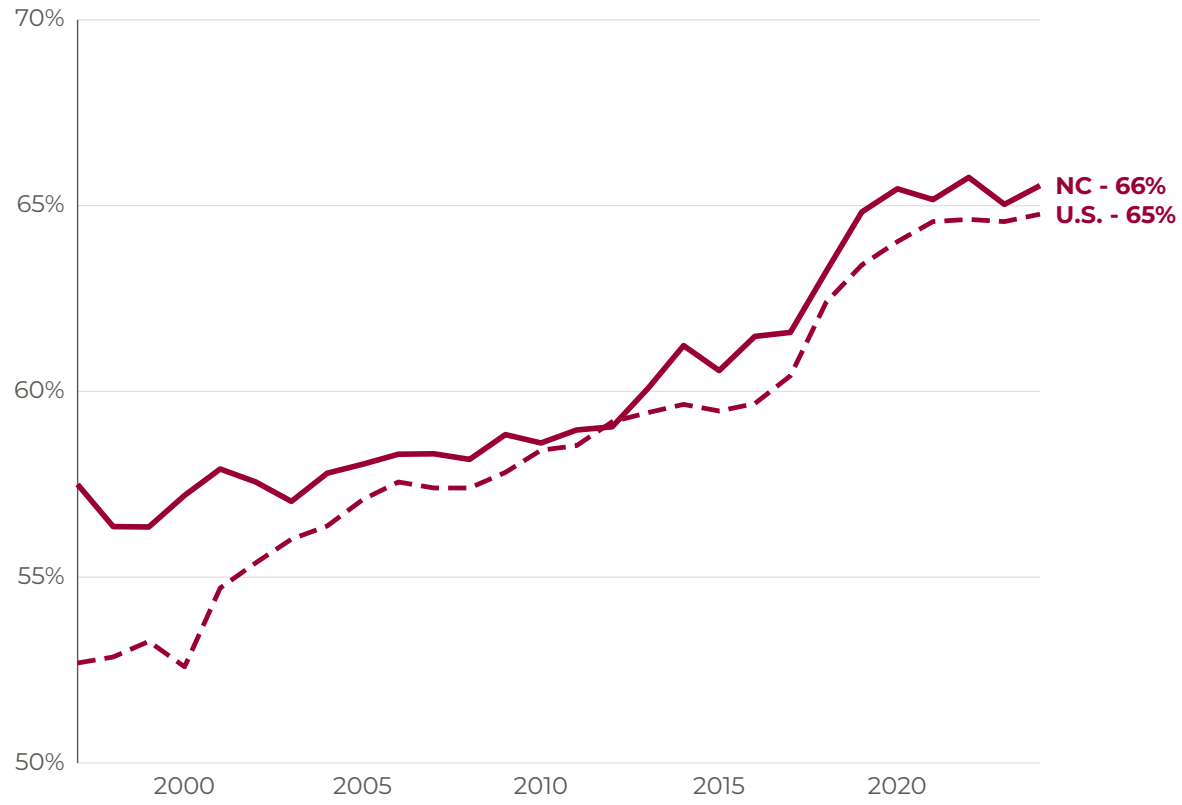
SECTION 14 · STUDENT OUTCOMES

How Many of North Carolina's Students Graduate?

The line follows the share of first-time, full-time bachelor's-degree students at North Carolina's four-year institutions who graduated from the institution where they started within six years (150% of normal time). Each year is the cohort that entered six years earlier. The rate leaves out students who start part-time (39% of the state's students attend part-time, against 39% nationally) and students who transfer and finish elsewhere. The state is solid and the United States dashed.

Six-year graduation rate. The rate was 66% in 2024, against 65% for the U.S., up 0.5 points from 65% in 2023 and up 5.0 points since 2015, when it was 61%.

Figure 38 Six-Year Bachelor's Graduation Rate



Four-year institutions; state solid, U.S. dashed. Data from NCES, IPEDS Graduation Rates.

66%
Six-year graduation rate, 2024
U.S. **65%** · **22nd** of 51

41%
Two-year completion rate, 2024
U.S. **41%** · **26th** of 50

11.8
Bachelor's per 100 undergraduates, 2025
U.S. **12.3** · **37th** of 51

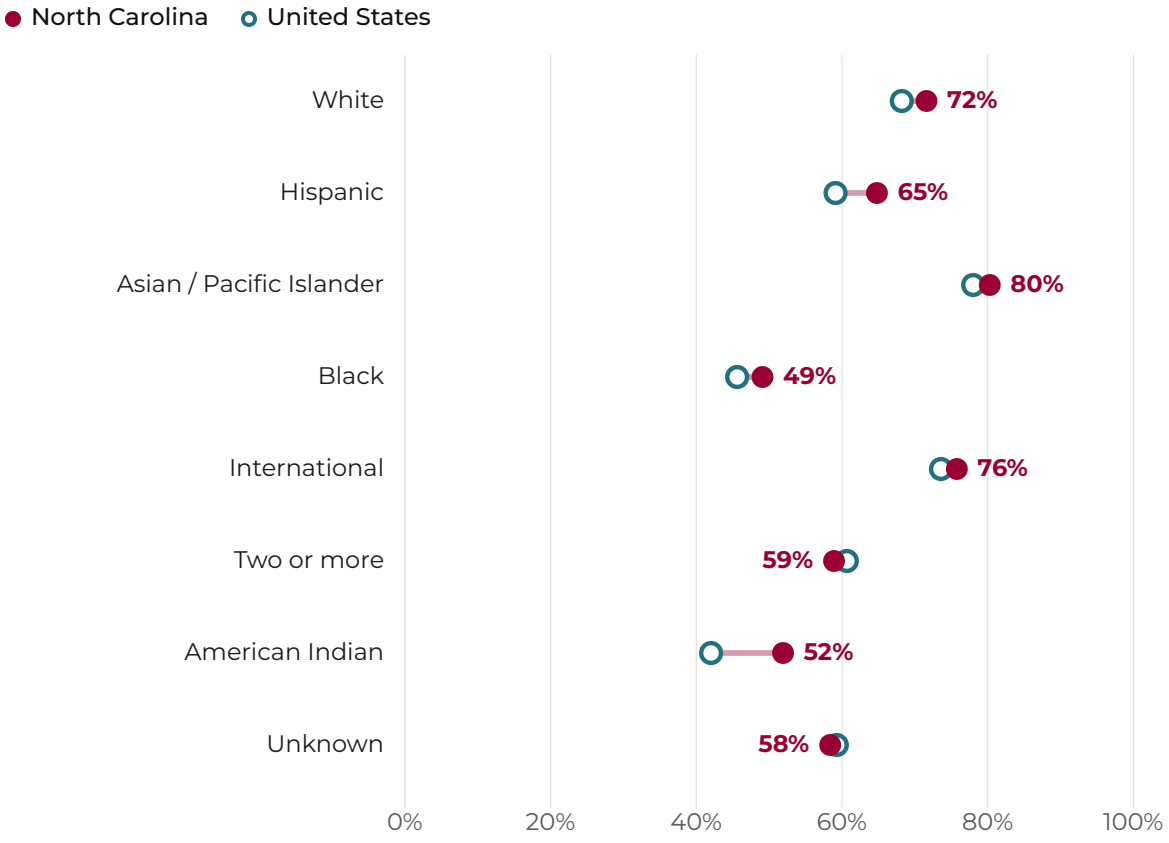
Each row is the six-year bachelor's graduation rate for one racial or ethnic group of students at North Carolina's four-year institutions (filled dot), against the national rate for the same group (open dot). Groups with fewer than 100 students in the cohort are left out.

White. The six-year rate was 72% in 2024, against 68% for the U.S.

Hispanic. The six-year rate was 65% in 2024, against 59% for the U.S.

Asian / Pacific Islander. The six-year rate was 80% in 2024, against 78% for the U.S.

Figure 39 How Do Graduation Rates Differ by Race and Ethnicity?



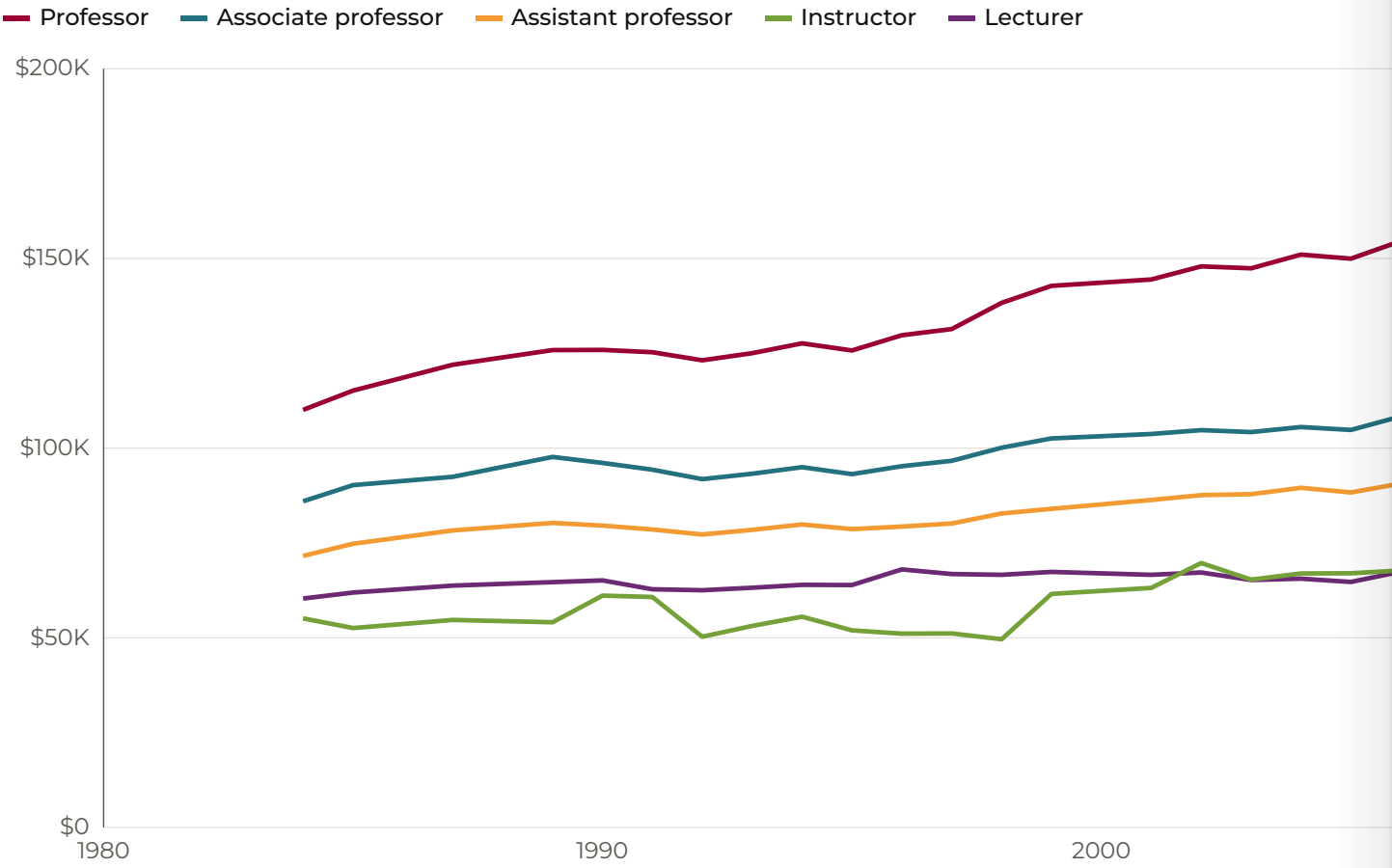
Six-year rate, 2024; groups with 100+ students. Data from NCES, IPEDS.

SECTION 15

Faculty

North Carolina ranks 28th of 51 on the average full-time faculty salary, at $95.1K in 2024 against $106K for the U.S.

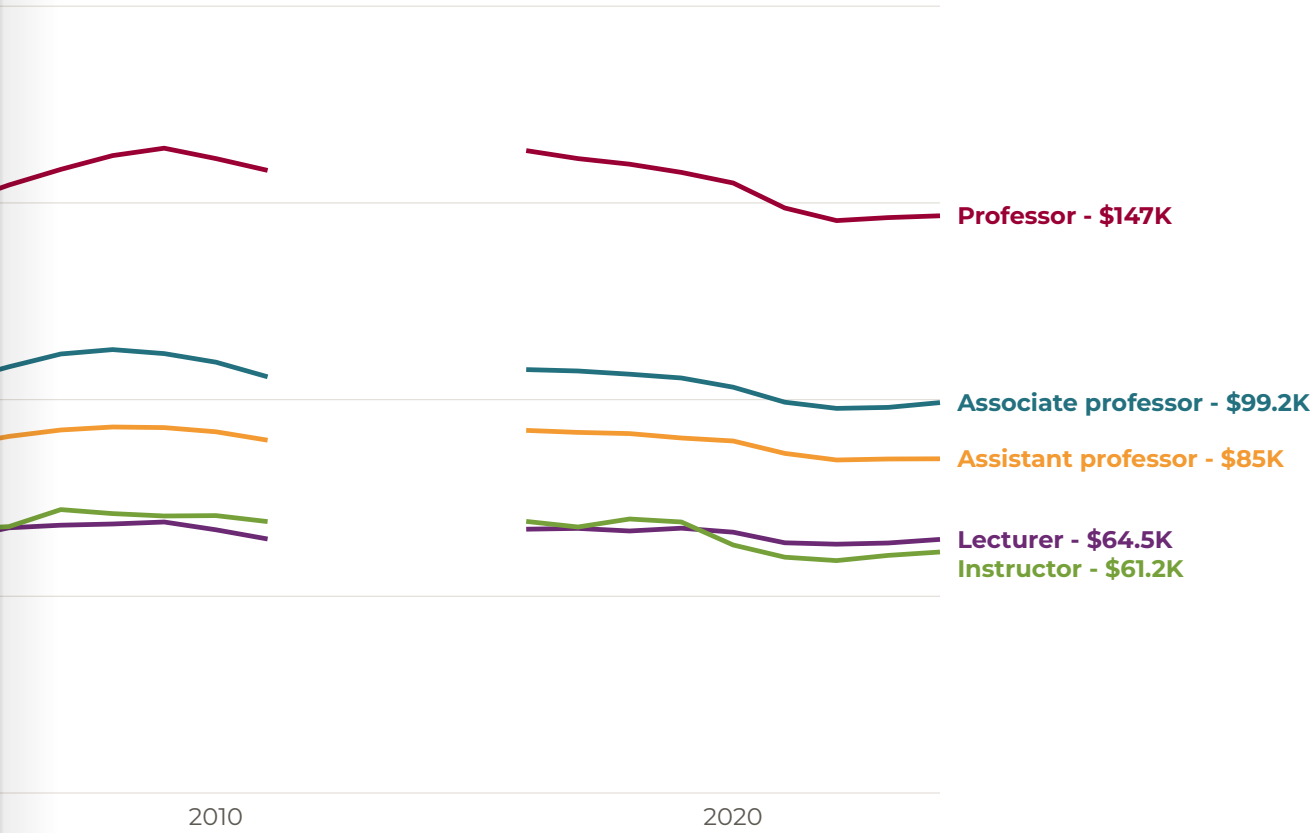
Figure 40 What Do North Carolina's Faculty Earn, by Rank?



Average salary of full-time instructional faculty on 9/10-month contracts, by academic rank, 2024 dollars (CPI-U). Data fro

Full-time instructional faculty on 9/10-month contracts at North Carolina's institutions earned an average of $95.1K in 2024 (2024 dollars), 10% below the U.S. ($106K); the state ranks 28th of 51. By rank (2024): professors $147K, associate professors $99.2K, assistant professors $85K, instructors $61.2K, lecturers $64.5K.

North Carolina's institutions had 29.4K full-time instructional staff in 2024, up 10% since 2015. Average salary across all ranks is down 6% since 2016.



m NCES, IPEDS Human Resources.

$95.1K

Average faculty salary, 2024

U.S. **$106K** · **28th** of 51 · −6.3% since 2016

SECTION 15 · FACULTY

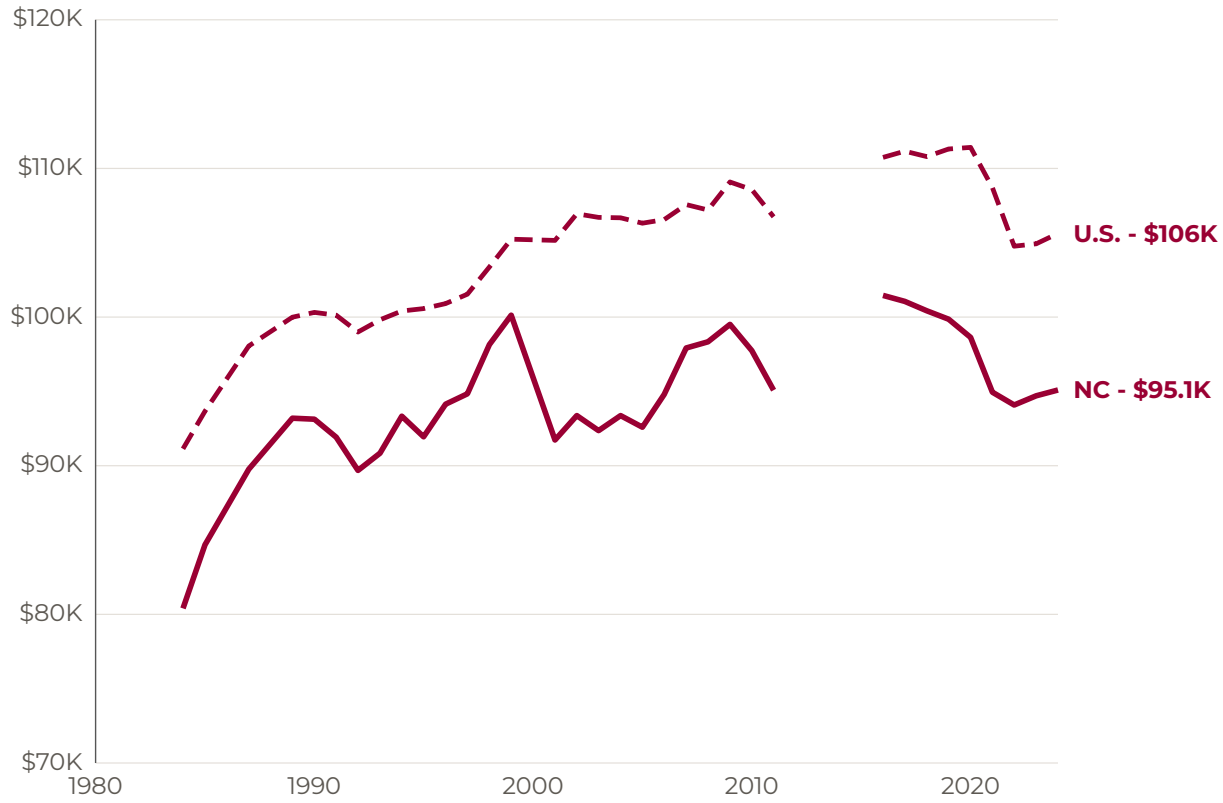
What Do North Carolina's Faculty Earn, and How Many Hold Tenure?

The line follows the average salary of full-time instructional faculty on 9- or 10-month contracts at North Carolina's degree-granting institutions, all ranks combined, in 2024 dollars (CPI-U). The state is solid and the United States dashed.

Average salary. The figure was $95.1K in 2024, against $106K for the U.S., 28th among the states, up 0.4% from $94.7K in 2023 and down 6.3% since 2016, when it was $101K.

Full-time instructional staff. The number was 29.4K in 2024, up 2.7% from 28.6K in 2023 and up 10% since 2015, when it was 26.7K.

Figure 41 Average Faculty Salary



Full-time instructional faculty, 9/10-month, 2024 dollars. Data from NCES, IPEDS.

$95.1K
Average faculty salary, 2024
U.S. **$106K** · **28th** of 51

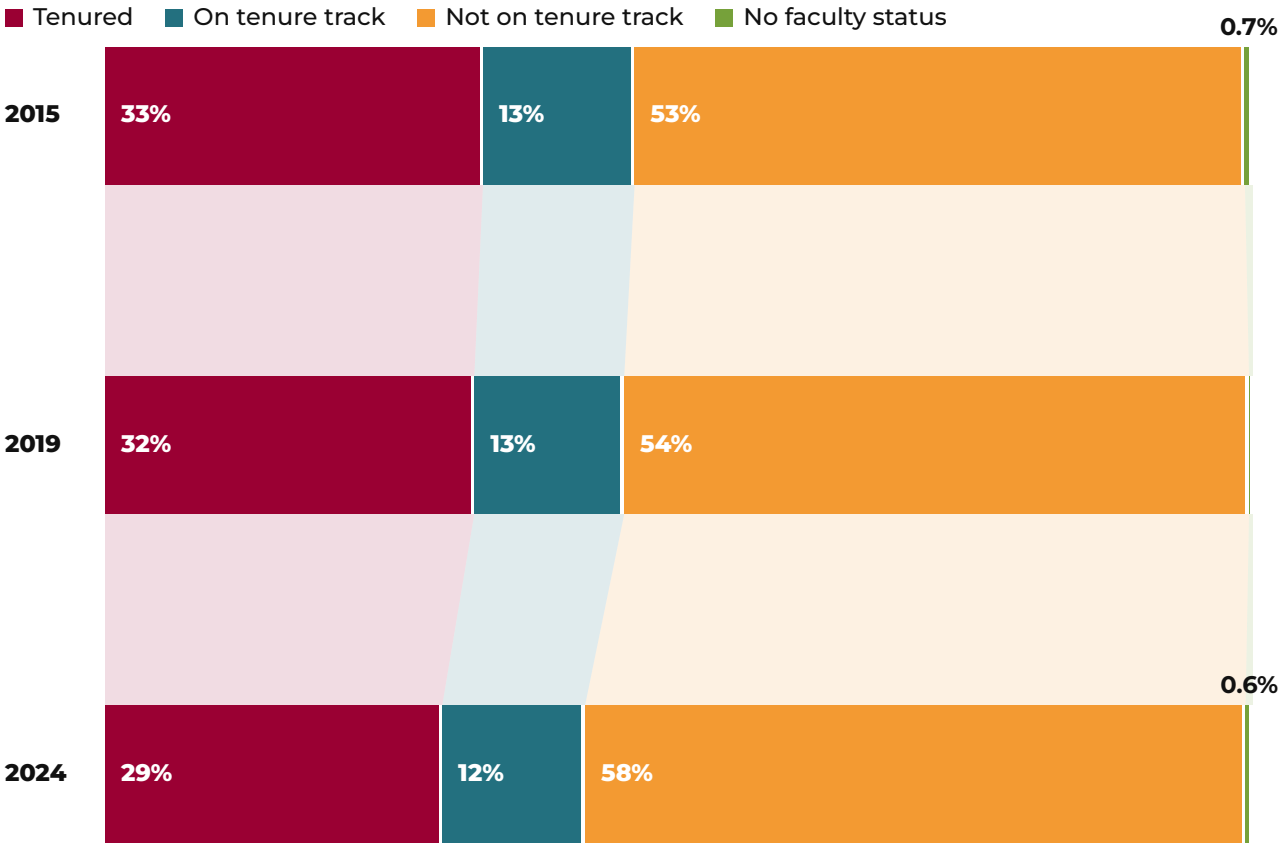
29.4K
Full-time instructional staff, 2024

29%
Tenured, 2024

The bars divide North Carolina's full-time instructional staff by tenure status in 2015, 2019 and 2024: tenured, on the tenure track, not on the tenure track (for example, lecturers on fixed-term contracts), and staff at institutions with no tenure system.

Tenured, share of full-time instructional staff. The share was 29% in 2024, down 1.1 points from 31% in 2023 and down 3.5 points since 2015, when it was 33%.

Figure 42 Tenure Status



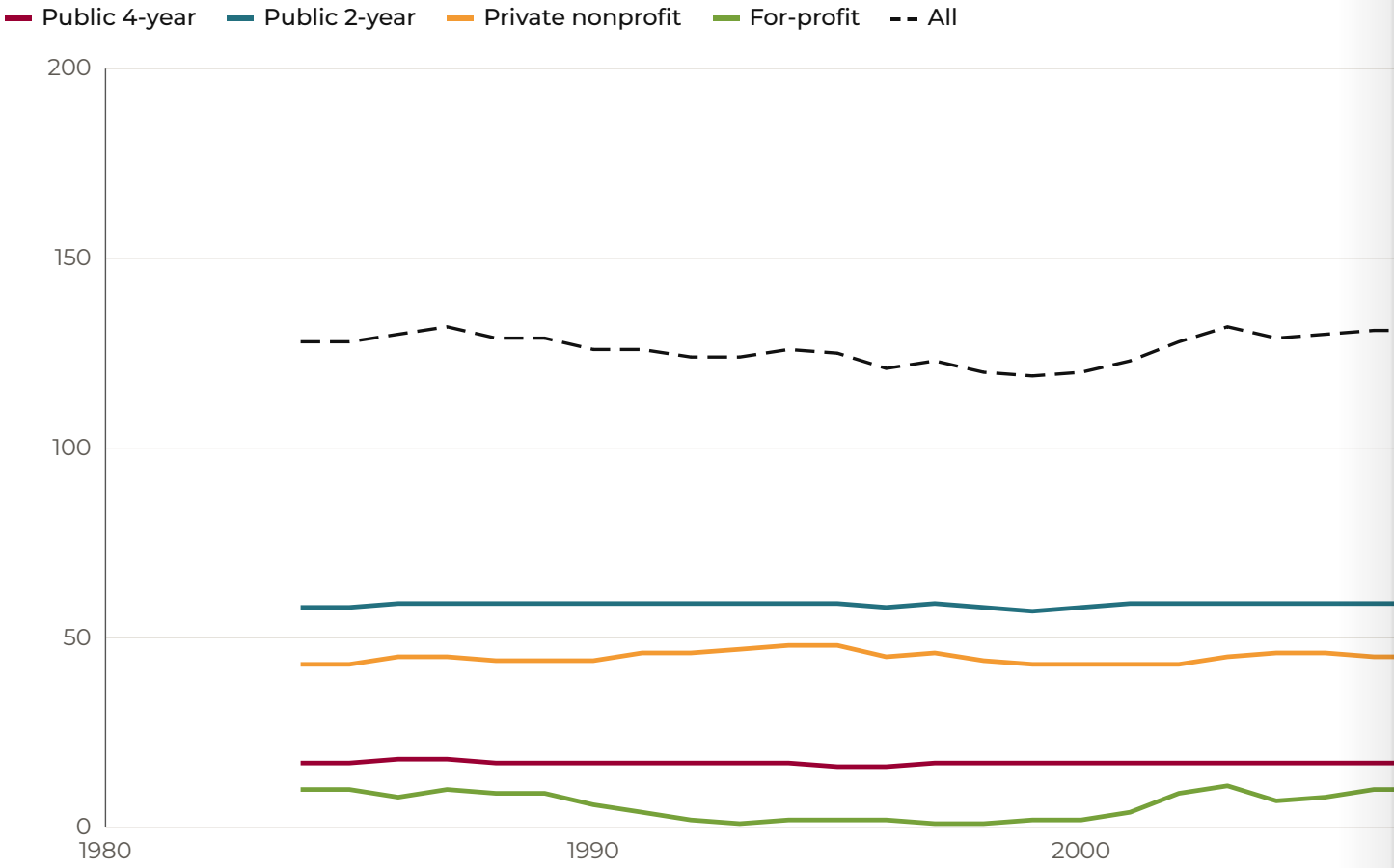
Full-time instructional staff by tenure status, 2015, 2019, 2024. Data from NCES, IPEDS.

SECTION 16

Institutions

North Carolina ranks 36th of 51 on the share of enrollment at for-profit institutions, at 0.8% in 2024 against 5.5% for the U.S.

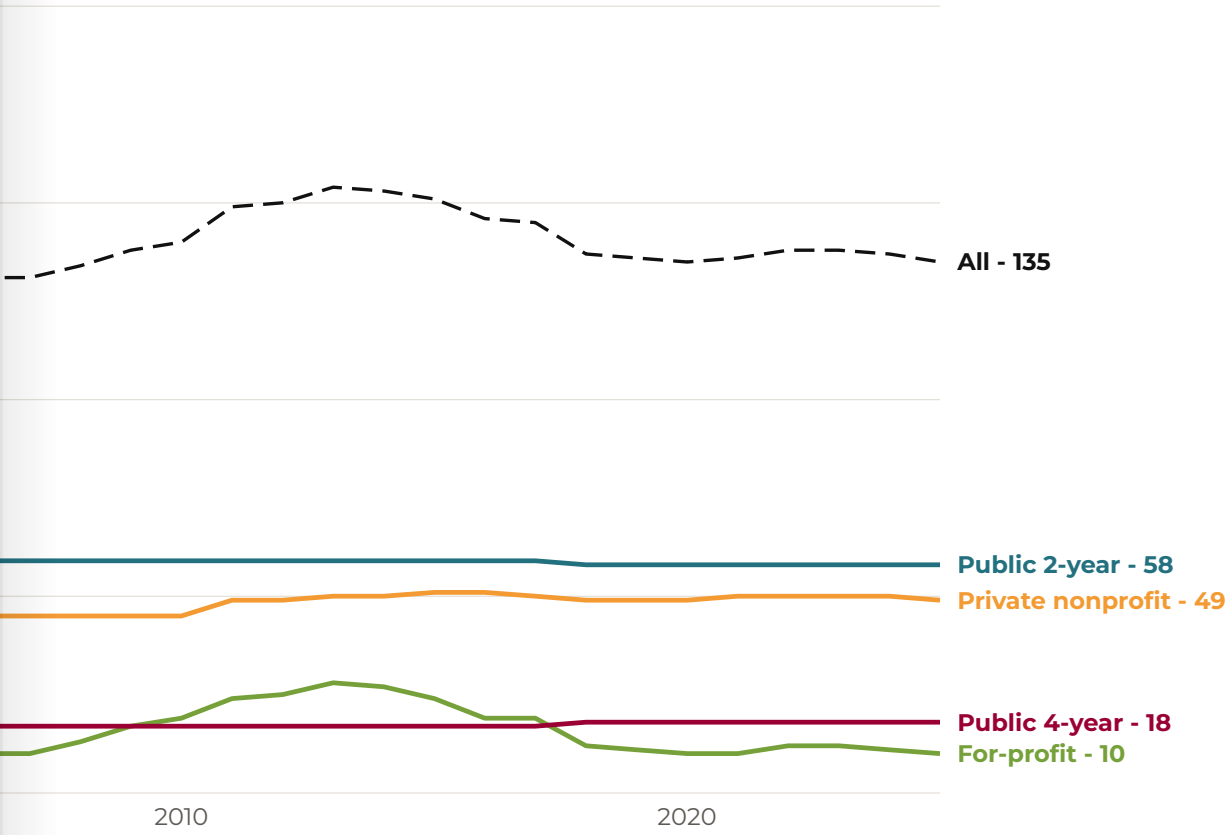
Figure 43 How Many Institutions Does North Carolina Have?



Degree-granting institutions by sector. Data from NCES, IPEDS.

North Carolina has 135 degree-granting institutions:18 public 4-year, 58 public 2-year, 49 private nonprofit, 10 for-profit (2025).

The number of institutions is down 11% since 2015. 84% of students attend public institutions and 0.8% attend for-profit ones.



84%

Students at public institutions, 2024

U.S. **73%** · **18th** of 51 · +4.0 pp since 2015

0.8%

Students at for-profit institutions, 2024

U.S. **5.5%** · **36th** of 51 · −2.2 pp since 2015

27%

Students fully online, 2024

U.S. **27%** · **20th** of 51 · +12.5 pp since 2015

SECTION 16 · INSTITUTIONS

Which Institutions Lead in North Carolina?

The table ranks institutions in North Carolina by the STEM master's degrees and research doctorates they awarded to international students in 2025, the two counted together. The share columns give international students' share of each institution's STEM degrees at that level.

Largest. North Carolina State University at Raleigh (811 international STEM master's degrees and 187 doctorates), Duke University (688 international STEM master's degrees and 123 doctorates), and University of North Carolina at Charlotte (640 international STEM master's degrees and 40 doctorates) awarded the most.

Table 7 Which Institutions Train the Most International STEM Graduates?

Institution	Intl STEM master's	Int. share	Intl STEM doctorates	Int. share
North Carolina State University at Raleigh Raleigh · Public 4-year	**811**	46%	187	47%
Duke University Durham · Private nonprofit	**688**	61%	123	41%
University of North Carolina at Charlotte Charlotte · Public 4-year	**640**	69%	40	71%
University of North Carolina at Chapel Hill Chapel Hill · Public 4-year	**82**	33%	58	21%
North Carolina A & T State University Greensboro · Public 4-year	**75**	38%	33	58%
University of North Carolina at Greensboro Greensboro · Public 4-year	**64**	48%	8	36%
Wake Forest University Winston-Salem · Private nonprofit	**11**	21%	14	28%
University of North Carolina Wilmington Wilmington · Public 4-year	**18**	14%	1	20%

Degrees awarded in 2025. Data from NCES, IPEDS.

135
Degree-granting institutions, 2025

84%
Students at public institutions, 2024
U.S. **73%** · **18th** of 51

0.8%
Students at for-profit institutions, 2024
U.S. **5.5%** · **36th** of 51

27%
Students fully online, 2024
U.S. **27%** · **20th** of 51

The table lists North Carolina's largest degree-granting institutions by fall 2024 headcount enrollment, with the international share of graduate students, STEM bachelor's degrees awarded in 2025, the six-year graduation rate and the admission rate (a dash where the institution admits all applicants or does not report one).

Largest. The three largest were North Carolina State University at Raleigh (38.5K students), University of North Carolina at Chapel Hill (32.4K students), and University of North Carolina at Charlotte (31.1K students).

Table 8 What Are North Carolina's Largest Institutions?

Institution	Enrollment	Int. share, grad	STEM bachelor's	Six-year grad rate	Admit rate
North Carolina State University at Raleigh	**38.5K**	30%	3.24K	85%	42%
University of North Carolina at Chapel Hill	**32.4K**	12%	2.02K	91%	15%
University of North Carolina at Charlotte	**31.1K**	27%	2.36K	69%	80%
East Carolina University	**26.9K**	2.1%	1.07K	63%	89%
Wake Technical Community College	**24.6K**	–	–	–	–
Appalachian State University	**21.6K**	1.5%	897	75%	90%
Central Piedmont Community College	**20.4K**	–	–	–	–
University of North Carolina Wilmington	**18.8K**	2.6%	650	71%	64%
University of North Carolina at Greensboro	**18K**	9.7%	539	56%	89%
Duke University	**17.5K**	35%	1K	97%	5.7%

Fall 2024 enrollment; degrees 2025. Data from NCES, IPEDS.

NOTES

About These Data

International students

U.S. nonresidents on temporary visas, as IPEDS defines them. IPEDS does not record country of citizenship. Permanent residents count with U.S. citizens.

STEM

CIP families 03, 04, 11, 14, 15, 26, 27 and 40. Doctorates are research and scholarship doctorates; professional doctorates appear only in the degree-count figure. Degrees are first majors only.

Universe

Degree-granting institutions located in the state; Title IV participants from 1996. Geography is the institution's location, not the student's home. Enrollment is fall headcount.

Migration

First-time degree-seeking undergraduates by home state, reported in even years before 2001 and every year since; odd years with incomplete reporting are omitted. Net migration is students arriving from other states minus residents leaving, per 100 resident first-time students.

Money

Tuition, appropriations, revenue and faculty salaries are in 2024 dollars using the CPI-U. Research funding (HERD, Federal Funds, USAspending), wages and the NAFSA economic contribution are nominal dollars, and are labeled so. Appropriations and revenue are per full-time-equivalent student at public institutions; the tuition share divides net tuition by net tuition plus state and local appropriations.

Race and ethnicity

IPEDS did not collect race and ethnicity in every year before 2010, and categories changed in 2008–2010; “Two or more” exists only from then.

Countries of citizenship

From DHS SEVIS by the Numbers: active F-1 and M-1 student records in the month named, at every level from primary school to doctorate, by the state of the school. IPEDS records only whether a student is a U.S. nonresident, not where they come from. SEVIS counts are monthly snapshots and are not directly comparable with IPEDS fall enrollment or degree counts. Year-to-year comparisons use one fall snapshot per year (the month closest to November).

Peers

Comparison states are the four closest in total fall enrollment, measured proportionally: a state 1.5 times as large counts as near as one two-thirds the size.

Sources

IPEDS (National Center for Education Statistics) is the backbone: Institutional Characteristics, Fall Enrollment, Completions, Graduation Rates, Finance, Human Resources and Admissions, 1980–2025, harmonized into a state panel. The profile also draws on NAEP, NSF's Survey of Graduate Students and Postdoctorates, Survey of Earned Doctorates, HERD and Federal Funds surveys, USAspending.gov, DHS SEVIS by the Numbers and ICE SEVIS releases, the Department of Labor's Office of Foreign Labor Certification and BLS wage data, the USCIS H-1B Employer Data Hub, the Census Bureau's American Community Survey and Post-Secondary Employment Outcomes, USPTO PatentsView, and other federal sources; every figure names its source, and the Sources page lists them in full. Generated November 2026.

NOTES

How the Numbers Are Counted

Ranks

Ranks are among the 50 states and the District of Columbia (1 = highest value, whether or not a high value is good). Where fewer than 51 report, the count is given ("23rd of 33").

Sectors

Sectors are as IPEDS classifies institutions each year. IPEDS moves a whole institution to "public, 4-year or above" once it awards any bachelor's degree, so community colleges that began offering applied bachelor's degrees carry all of their students into the public 4-year sector. Sector series (enrollment shares, public 4-year tuition, graduation rates by sector) reflect these moves.

Research doctorates

Two counts appear. IPEDS counts degrees institutions report as "doctor's degree, research/scholarship", which includes some practice-oriented degrees such as many Ed.D.s; the NSF Survey of Earned Doctorates counts research doctorate recipients and is smaller. Each figure names which it uses. "STEM master's per STEM research doctorate" divides IPEDS STEM master's degrees by IPEDS STEM research doctorates.

Engineering

"Engineering" alone is CIP 14. Where engineering technology (CIP 15) is added, the label says "engineering & eng. technology".

Graduation rates

IPEDS graduation rates follow first-time, full-time students and count only those who finish at the institution where they started, so they leave out part-time starters and students who transfer and finish elsewhere.

International students in SEVIS

SEVIS counts active F-1 and M-1 records, which include graduates on Optional Practical Training (up to 36 months for STEM graduates, 12 for others), so STEM shares run high. STEM follows the DHS STEM Designated Degree Program List, which is broader than the eight CIP families above, and DHS draws its STEM pages on a different basis from its all-student pages, so a country's STEM count can exceed its all-level count.

Visas and green cards

DOL labor condition applications (H-1B positions) are counted by worksite and include renewals and job changes; USCIS approvals are counted by the employer's address and split into new and continuing. PERM certifications are the employer's step toward a green card.

Patents

Patents are counted by the owner's (assignee's) location, not where inventors live; a company headquartered in another state takes its patents there.

Years

Each measure uses its latest year, so years differ across a page; every figure, table and statement gives its year, and lines that stop early are labeled with their last year.

North Carolina – By the Numbers

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
K-12 PREPARATION							
Grade 8 mathematics, average score	281	**276**[2024]	−5.5 ▼	272	−9.0 ▼	#20	[5]
Grade 8 mathematics: at or above Proficient	33%	**31%**[2024]	−1.1 pp ▼	27%	−5.3 pp ▼	#14	[5]
Grade 4 mathematics, average score	244	**239**[2024]	−4.9 ▼	237	−2.5 ▼	#19	[5]
Grade 4 mathematics: at or above Proficient	44%	**41%**[2024]	−3.1 pp ▼	39%	0.0 pp	#19	[5]
Grade 8 reading, average score	261	**255**[2024]	−5.7 ▼	257	−7.3 ▼	#35	[5]
Grade 8 reading: at or above Proficient	30%	**27%**[2024]	−3.4 pp ▼	29%	−4.0 pp ▼	#32	[5]
Grade 4 reading, average score	226	**213**[2024]	−12.5 ▼	214	−7.1 ▼	#32	[5]
Grade 8 science, average score, 2009 framework	–	**150**[2015]	–	153	–	#34	[5]
Current spending per public-school pupil	$11.2K	**$12.4K**[2021]	+11% ▲	$16.8K	+12% ▲	#45	[3]
Public high school 4-year graduation rate, ACGR	86%	**86%**[2022]	0.0 pp	87%	+4.0 pp ▲	#25	[3]
Public high school graduates	97.4K	**109K**[2024]	+12% ▲	3.49M	+8.8% ▲	#9	[3]
Average public-school teacher salary	$62.5K	**$58.9K**[2022]	−5.9% ▼	$72.9K	−3.0% ▼	#41	[3]
Pupils per teacher, public schools	15.5	**15.4**[2022]	−0.1 ▼	15.4	−0.6 ▼	#18	[3]
Public elementary and secondary enrollment	1.54M	**1.56M**[2024]	+0.7% ▲	49.4M	−2.1% ▼	#9	[3]
Public-school students eligible for free or reduced-price lunch	57%	**63%**[2023]	+5.3 pp ▲	53%	+1.5 pp ▲	#7	[3]
English learners as a share of public-school students	6.3%	**8.0%**[2022]	+1.7 pp ▲	11%	+1.0 pp ▲	#26	[3]
Charter-school share of public-school enrollment	4.6%	**9.0%**[2023]	+4.4 pp ▲	7.6%	+2.2 pp ▲	#13	[3]
High schools offering calculus	56%[2016]	**61%**[2024]	+4.4 pp ▲	51%	−5.8 pp ▼	#16	[14]
High schools offering physics	50%[2016]	**52%**[2024]	+2.6 pp ▲	64%	−2.4 pp ▼	#38	[14]
High schools offering chemistry	79%[2016]	**86%**[2024]	+7.4 pp ▲	81%	+0.7 pp ▲	#18	[14]
High schools offering Algebra II	92%[2016]	**96%**[2024]	+4.4 pp ▲	87%	+1.3 pp ▲	#6	[14]
High schools offering computer science	48%[2021]	**76%**[2024]	+28.0 pp ▲	55%	+2.0 pp ▲	#10	[14]
High schools offering any AP course	67%[2016]	**68%**[2024]	+0.9 pp ▲	56%	0.0 pp	#19	[14]
High schools offering AP computer science	31%[2021]	**31%**[2024]	+0.1 pp ▲	25%	+2.6 pp ▲	#18	[14]

Superscript years mark the year observed; changes since 2015 start from 2015 or the first later year with data. Rank is among the 50 states and DC (1 = highest). "pp" is percentage points. Dollars are 2024 dollars (CPI-U) unless the measure is labeled nominal. Counts and dollars are rounded to three significant figures (K, M, B). Arrow colors show direction only. Source numbers refer to the list of sources that follows the tables.

NORTH CAROLINA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
K-12 PREPARATION (CONTINUED)							
High schools offering dual enrollment	64%[2016]	**84%[2024]**	+20.8 pp ▲	71%	+11.1 pp ▲	#13	[14]
AP course enrollments per 100 high-school students	15.2[2016]	**15.7[2024]**	+0.5 ▲	18.4	+1.0 ▲	#23	[14]
Calculus enrollments per 100 high-school students	2.3[2016]	**1.9[2024]**	−0.4 ▼	3.3	−1.0 ▼	#39	[14]
Physics enrollments per 100 high-school students	2.4[2016]	**1.6[2024]**	−0.8 ▼	9.4	−0.4 ▼	#50	[14]
SAT or ACT takers per 100 high-school students	20.2[2016]	**17.5[2024]**	−2.6 ▼	18.1	−2.5 ▼	#31	[14]
Students in gifted and talented programs per 100 students	11.2[2016]	**12.5[2024]**	+1.3 ▲	7.0	+0.4 ▲	#7	[14]
ENROLLMENT AND INSTITUTIONS							
Pell Grant share of undergraduates	43%	**37%[2025]**	−5.9 pp ▼	36%	−0.8 pp ▼	#18	[15]
Fall enrollment	562K	**582K[2024]**	+3.4% ▲	19.7M	−1.7% ▼	#10	[4]
Share of enrollment at public institutions	80%	**84%[2024]**	+4.0 pp ▲	73%	0.0 pp	#18	[4]
Share of enrollment at for-profit institutions	3.0%	**0.8%[2024]**	−2.2 pp ▼	5.5%	−1.2 pp ▼	#36	[4]
Share of students enrolled part-time	36%	**39%[2024]**	+2.7 pp ▲	39%	+0.7 pp ▲	#29	[4]
Share of students who are graduate students	13%	**14%[2024]**	+0.9 pp ▲	17%	+2.0 pp ▲	#34	[4]
Share of students enrolled exclusively online	14%	**27%[2024]**	+12.5 pp ▲	27%	+12.1 pp ▲	#20	[4]
DEGREES AND FIELDS OF STUDY							
Research doctorates awarded	1.7K	**1.97K[2025]**	+15% ▲	59K	+7.3% ▲	#8	[7]
Science & engineering research doctorates, excluding health	1.27K	**1.46K[2025]**	+14% ▲	44.4K	+14% ▲	#9	[7]
Share of U.S. research doctorates	3.1%	**3.3%[2025]**	+0.2 pp ▲	–	–	#8	[7]
Engineering research doctorates	295	**352[2025]**	+19% ▲	10.9K	+9.9% ▲	#12	[7]
Mathematics & computer science research doctorates	147	**184[2025]**	+25% ▲	5.07K	+33% ▲	#8	[7]
STEM master's degrees per research doctorate	3.3	**4.1[2025]**	+0.8 ▲	6.4	+2.1 ▲	#37	[4]
STEM share of bachelor's	20%	**26%[2025]**	+6.4 pp ▲	24%	+5.3 pp ▲	#13	[4]
Share of bachelor's in computer & information sciences	2.6%	**6.0%[2025]**	+3.4 pp ▲	6.9%	+3.7 pp ▲	#24	[4]
Share of bachelor's in engineering & eng. technology	5.5%	**6.8%[2025]**	+1.3 pp ▲	6.8%	+0.7 pp ▲	#23	[4]
Share of bachelor's in mathematics & statistics	1.3%	**1.4%[2025]**	+0.1 pp ▲	1.2%	0.0 pp	#9	[4]

Superscript years mark the year observed; changes since 2015 start from 2015 or the first later year with data. Rank is among the 50 states and DC (1 = highest). "pp" is percentage points. Dollars are 2024 dollars (CPI-U) unless the measure is labeled nominal. Counts and dollars are rounded to three significant figures (K, M, B). Arrow colors show direction only. Source numbers refer to the list of sources that follows the tables.

NORTH CAROLINA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
DEGREES AND FIELDS OF STUDY (CONTINUED)							
Share of bachelor's in biological & biomedical sciences	6.7%	**8.6%**[2025]	+1.9 pp ▲	6.6%	+0.8 pp ▲	#3	[4]
Share of bachelor's in physical sciences	2.0%	**1.7%**[2025]	−0.4 pp ▼	1.3%	−0.3 pp ▼	#8	[4]
Share of bachelor's in health professions	9.5%	**9.9%**[2025]	+0.4 pp ▲	12%	+1.1 pp ▲	#36	[4]
Share of bachelor's in business	17%	**18%**[2025]	+1.2 pp ▲	19%	−0.1 pp ▼	#33	[4]
STEM share of master's	19%	**22%**[2025]	+2.4 pp ▲	23%	+6.7 pp ▲	#24	[4]
Share of master's in computer & information sciences	5.0%	**7.2%**[2025]	+2.3 pp ▲	10%	+6.3 pp ▲	#30	[4]
Share of master's in engineering & eng. technology	7.0%	**7.9%**[2025]	+0.9 pp ▲	6.7%	−0.2 pp ▼	#11	[4]
Share of master's in mathematics & statistics	1.4%	**1.2%**[2025]	−0.3 pp ▼	1.3%	+0.3 pp ▲	#24	[4]
Share of master's in biological & biomedical sciences	2.2%	**2.4%**[2025]	+0.2 pp ▲	2.5%	+0.6 pp ▲	#26	[4]
Share of master's in physical sciences	1.3%	**1.1%**[2025]	−0.2 pp ▼	0.8%	−0.1 pp ▼	#19	[4]
Share of master's in health professions	15%	**16%**[2025]	+1.1 pp ▲	15%	+1.7 pp ▲	#18	[4]
Share of master's in business	26%	**27%**[2025]	+0.6 pp ▲	21%	−3.2 pp ▼	#5	[4]
STEM share of research doctorates	46%	**46%**[2025]	−0.1 pp ▼	39%	−2.2 pp ▼	#14	[4]
Share of research doctorates in computer & information sciences	3.1%	**3.6%**[2025]	+0.5 pp ▲	4.2%	+1.4 pp ▲	#24	[4]
Share of research doctorates in engineering & eng. technology	14%	**15%**[2025]	+1.1 pp ▲	14%	−1.0 pp ▼	#18	[4]
Share of research doctorates in mathematics & statistics	3.8%	**3.9%**[2025]	0.0 pp	2.4%	−0.2 pp ▼	#2	[4]
Share of research doctorates in biological & biomedical sciences	16%	**15%**[2025]	−1.4 pp ▼	10%	−1.1 pp ▼	#10	[4]
Share of research doctorates in physical sciences	6.8%	**7.5%**[2025]	+0.7 pp ▲	7.1%	−1.1 pp ▼	#28	[4]
Share of research doctorates in health professions	11%	**12%**[2025]	+1.0 pp ▲	12%	+3.5 pp ▲	#21	[4]
Share of research doctorates in business	1.3%	**2.4%**[2025]	+1.0 pp ▲	5.6%	+1.3 pp ▲	#37	[4]
Bachelor's degrees per 100 undergraduates	10.6	**11.8**[2025]	+1.2 ▲	12.3	+1.4 ▲	#37	[4]
INTERNATIONAL STUDENTS							
First-time students from outside the U.S.	1.6%[2016]	**1.7%**[2024]	+0.2 pp ▲	3.1%	+0.1 pp ▲	#39	[4]
STEM master's degrees earned by international students	1.25K	**2.42K**[2025]	+94% ▲	117K	2.18× ▲	#17	[4]
Share of all U.S. international STEM master's degrees	2.3%	**2.1%**[2025]	−0.3 pp ▼	–	–	#17	[4]

Superscript years mark the year observed; changes since 2015 start from 2015 or the first later year with data. Rank is among the 50 states and DC (1 = highest). "pp" is percentage points. Dollars are 2024 dollars (CPI-U) unless the measure is labeled nominal. Counts and dollars are rounded to three significant figures (K, M, B). Arrow colors show direction only. Source numbers refer to the list of sources that follows the tables.

NORTH CAROLINA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
INTERNATIONAL STUDENTS (CONTINUED)							
International share of undergraduate enrollment	2.1%	**2.1%**[2024]	0.0 pp	3.0%	−0.3 pp ▼	#32	[4]
International share of graduate enrollment	13%	**14%**[2024]	+1.8 pp ▲	17%	+2.8 pp ▲	#24	[4]
International share of all bachelor's	2.1%	**3.3%**[2025]	+1.2 pp ▲	4.1%	+0.1 pp ▲	#24	[4]
International share of STEM bachelor's	3.2%	**4.3%**[2025]	+1.0 pp ▲	6.6%	+0.8 pp ▲	#33	[4]
International share of all master's	14%	**19%**[2025]	+4.4 pp ▲	22%	+6.2 pp ▲	#21	[4]
International share of STEM master's	38%	**49%**[2025]	+11.4 pp ▲	52%	+9.1 pp ▲	#20	[4]
International share of computer & information sciences master's	48%	**64%**[2025]	+16.3 pp ▲	65%	+10.1 pp ▲	#25	[4]
International share of engineering & eng. technology master's	47%	**54%**[2025]	+6.8 pp ▲	50%	+0.9 pp ▲	#10	[4]
International share of mathematics & statistics master's	45%	**46%**[2025]	+0.7 pp ▲	55%	+6.1 pp ▲	#17	[4]
International share of biological & biomedical sciences master's	15%	**18%**[2025]	+2.5 pp ▲	23%	+6.2 pp ▲	#27	[4]
International share of physical sciences master's	14%	**31%**[2025]	+17.2 pp ▲	33%	+3.6 pp ▲	#21	[4]
International share of all research doctorates	24%	**25%**[2025]	+0.8 pp ▲	25%	−1.0 pp ▼	#30	[4]
International share of STEM research doctorates	38%	**39%**[2025]	+1.0 pp ▲	43%	−0.5 pp ▼	#30	[4]
International share of computer & information sciences research doctorates	66%	**64%**[2025]	−1.8 pp ▼	56%	+2.0 pp ▲	#8	[4]
International share of engineering & eng. technology research doctorates	53%	**54%**[2025]	+1.6 pp ▲	55%	−1.0 pp ▼	#22	[4]
International share of mathematics & statistics research doctorates	42%	**47%**[2025]	+4.8 pp ▲	53%	+2.9 pp ▲	#20	[4]
International share of biological & biomedical sciences research doctorates	24%	**18%**[2025]	−5.4 pp ▼	23%	−4.3 pp ▼	#32	[4]
International share of physical sciences research doctorates	23%	**33%**[2025]	+9.4 pp ▲	38%	−1.1 pp ▼	#31	[4]
Active F-1/M-1 students, all levels	19K	**23.2K**[2026]	+22% ▲	1.19M	−0.1% ▼	#16	[20]
Share of U.S. active F-1/M-1 students	1.6%	**1.9%**[2026]	+0.4 pp ▲	–	–	#16	[20]
Share of F-1/M-1 students from India	21%	**23%**[2026]	+2.0 pp ▲	27%	+11.4 pp ▲	#22	[20]
Share of F-1/M-1 students from China	29%	**23%**[2026]	−6.0 pp ▼	18%	−11.7 pp ▼	#8	[20]
Share from the two largest countries of origin	50%	**46%**[2026]	−4.0 pp ▼	45%	−0.3 pp ▼	#19	[20]

Superscript years mark the year observed; changes since 2015 start from 2015 or the first later year with data. Rank is among the 50 states and DC (1 = highest). “pp” is percentage points. Dollars are 2024 dollars (CPI-U) unless the measure is labeled nominal. Counts and dollars are rounded to three significant figures (K, M, B). Arrow colors show direction only. Source numbers refer to the list of sources that follows the tables.

NORTH CAROLINA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
INTERNATIONAL STUDENTS (CONTINUED)							
Countries of citizenship represented	177	**183**[2026]	+3.4% ▲	219	−2.7% ▼	#9	[20]
Active STEM F-1/M-1 students	9.29K	**17.6K**[2026]	+89% ▲	749K	+61% ▲	#14	[20]
Active STEM master's students	4.44K	**9.54K**[2026]	2.15× ▲	393K	+90% ▲	#14	[20]
Active STEM doctoral students	2.84K	**4.82K**[2026]	+70% ▲	158K	+61% ▲	#12	[20]
Share of STEM master's students from India	67%	**64%**[2026]	−2.9 pp ▼	58%	−1.8 pp ▼	#18	[20]
U.S. students studying abroad for credit, from the state's institutions	13.3K[2017]	**13.7K**[2024]	+3.5% ▲	–	–	#7	[1]
Economic contribution of international students, nominal dollars	$722M[2018]	**$886M**[2024]	+23% ▲	$42.9B	+4.6% ▲	#18	[2]
Jobs supported by international students' spending	9.04K[2018]	**8.11K**[2024]	−10% ▼	356K	−22% ▼	#15	[2]
Share of U.S. student and exchange-visitor admissions, by destination	1.3%	**1.8%**[2024]	+0.5 pp ▲	–	–	#13	[16]
Share of U.S. F-1 student admissions, by destination	1.1%	**1.5%**[2024]	+0.4 pp ▲	–	–	#16	[16]
Share of U.S. J-1 exchange-visitor admissions, by destination	1.8%	**2.6%**[2024]	+0.8 pp ▲	–	–	#11	[16]
J-1 exchange visitors starting a program in the state	7.35K	**8.8K**[2025]	+20% ▲	275K	−14% ▼	#10	[18]
GRADUATE EDUCATION AND RESEARCH							
Graduate students in science & engineering	17.9K	**17.3K**[2016]	−3.4% ▼	620K	+0.4% ▲	#12	[9]
S&E graduate students on temporary visas	5.84K	**5.92K**[2016]	+1.4% ▲	241K	+2.3% ▲	#14	[9]
Temporary-visa share of S&E graduate students	33%	**34%**[2016]	+1.6 pp ▲	39%	+0.7 pp ▲	#25	[9]
Temporary-visa share of engineering graduate students	52%	**54%**[2016]	+2.2 pp ▲	57%	0.0 pp	#23	[9]
S&E postdoctoral appointees	1.46K	**1.45K**[2016]	−0.3% ▼	45.7K	+1.0% ▲	#8	[9]
Temporary-visa share of S&E postdocs	48%	**43%**[2016]	−4.7 pp ▼	56%	−0.9 pp ▼	#43	[9]
University R&D spending, nominal dollars	$2.82B	**$4.65B**[2024]	+65% ▲	$118B	+71% ▲	#7	[6]
Share of U.S. university R&D	4.1%	**4.0%**[2024]	−0.1 pp ▼	–	–	#7	[6]
Federally funded share of university R&D	57%	**59%**[2024]	+1.9 pp ▲	55%	−0.2 pp ▼	#14	[6]
Business-funded share of university R&D	12%	**8.4%**[2024]	−3.6 pp ▼	5.4%	−0.4 pp ▼	#3	[6]
State- and locally funded share of university R&D	5.2%	**5.2%**[2024]	0.0 pp	5.2%	−0.4 pp ▼	#26	[6]
University R&D in mathematics and computer science, nominal dollars	$91.3M	**$158M**[2024]	+73% ▲	$5.12B	+97% ▲	#10	[6]

Superscript years mark the year observed; changes since 2015 start from 2015 or the first later year with data. Rank is among the 50 states and DC (1 = highest). "pp" is percentage points. Dollars are 2024 dollars (CPI-U) unless the measure is labeled nominal. Counts and dollars are rounded to three significant figures (K, M, B). Arrow colors show direction only. Source numbers refer to the list of sources that follows the tables.

NORTH CAROLINA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
GRADUATE EDUCATION AND RESEARCH (CONTINUED)							
University R&D in engineering, nominal dollars	$257M	**$471M[2024]**	+83% ▲	$19.3B	+74% ▲	#15	[6]
University R&D personnel, headcount	28.7K	**39.4K[2024]**	+37% ▲	1.09M	+20% ▲	#10	[6]
Federal R&D obligations to the state, nominal dollars	$1.89B	**$3.28B[2024]**	+74% ▲	$193B	+53% ▲	#17	[8]
Federal R&D obligations to universities in the state, nominal dollars	$1.22B	**$1.84B[2024]**	+51% ▲	$44.4B	+64% ▲	#7	[8]
Defense R&D obligations to the state, nominal dollars	$205M	**$350M[2024]**	+71% ▲	$97.2B	+60% ▲	#27	[8]
Health (HHS/NIH) R&D obligations to the state, nominal dollars	$1.25B	**$2.24B[2024]**	+80% ▲	$48.5B	+62% ▲	#6	[8]
NSF, NIH, NASA and DOE Science grants performed in the state, nominal dollars	$1.18B	**$1.72B[2025]**	+46% ▲	$45B	+47% ▲	#6	[19]
Federal R&D contracts performed in the state, nominal dollars	$422M	**$440M[2026]**	+18.0 ▲	$59.3B	+22,105.9 ▲	#20	[19]
COLLEGE COSTS AND OUTCOMES							
Median earnings 10 years after entering college	$39.9K[2020]	**$43K[2022]**	+7.7% ▲	$47.5K	+11% ▲	#46	[15]
Median earnings 1 year after completing	–	**$43.9K[2025]**	–	$47.5K	–	#49	[15]
Median debt of completers, nominal dollars	$20.5K	**$19.5K[2024]**	−4.9% ▼	$17.6K	−4.6% ▼	#18	[15]
STEM bachelor's graduates: median earnings 1 year out	–	**$65.3K[2025]**	–	$68.2K	–	#31	[15]
Non-STEM bachelor's graduates: median earnings 1 year out	–	**$50.1K[2025]**	–	$51.9K	–	#29	[15]
Bachelor's graduates: median earnings 5 years out	–	**$65.7K[2025]**	–	$68.1K	–	#30	[15]
STEM master's graduates: median earnings 1 year out	–	**$86.6K[2025]**	–	$91.6K	–	#25	[15]
Six-year bachelor's graduation rate, 4-year institutions	61%	**66%[2024]**	+5.0 pp ▲	65%	+5.3 pp ▲	#22	[4]
Completion within 150% of normal time, 2-year institutions	19%	**41%[2024]**	+22.2 pp ▲	41%	+12.1 pp ▲	#26	[4]
In-state tuition & fees, public 4-year	$9.17K	**$7.54K[2023]**	−18% ▼	$9.71K	−14% ▼	#45	[4]
State & local appropriations per FTE, public institutions	$13.9K	**$15.6K[2024]**	+12% ▲	$11.4K	+22% ▲	#9	[4]
Net tuition share of tuition plus state and local appropriations, public institutions	32%	**28%[2024]**	−4.3 pp ▼	43%	−5.9 pp ▼	#46	[4]
Average full-time faculty salary (9/10-month)	$101K[2016]	**$95.1K[2024]**	−6.3% ▼	$106K	−4.6% ▼	#28	[4]
WHERE STUDENTS AND GRADUATES GO							

Superscript years mark the year observed; changes since 2015 start from 2015 or the first later year with data. Rank is among the 50 states and DC (1 = highest). "pp" is percentage points. Dollars are 2024 dollars (CPI-U) unless the measure is labeled nominal. Counts and dollars are rounded to three significant figures (K, M, B). Arrow colors show direction only. Source numbers refer to the list of sources that follows the tables.

NORTH CAROLINA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
WHERE STUDENTS AND GRADUATES GO (CONTINUED)							
International graduates on OPT working in the state where they studied	32%	**33%**[2016]	+1.0 pp ▲	44%	−1.8 pp ▼	#30	[21]
OPT retention, bachelor's graduates	46%	**41%**[2016]	−5.4 pp ▼	58%	+0.3 pp ▲	#32	[21]
OPT retention, master's graduates	28%	**30%**[2016]	+1.7 pp ▲	39%	−2.5 pp ▼	#28	[21]
OPT retention, STEM graduates	30%	**32%**[2016]	+1.9 pp ▲	37%	−1.9 pp ▼	#22	[21]
OPT workers employed in the state per OPT graduate of its schools	0.81	**0.85**[2016]	0.0	1.00	0.0	#27	[21]
STEM OPT extensions approved, by state of the school	783	**924**[2016]	+18% ▲	41.2K	+51% ▲	#13	[21]
Bachelor's graduates working in the state 1 year after graduating (PSEO)	77%[2016]	**79%**[2019]	+1.3 pp ▲	72%	−0.1 pp ▼	#4	[11]
Bachelor's graduates working in the state 5 years after graduating (PSEO)	–	**71%**[2016]	–	65%	–	#6	[11]
Bachelor's graduates working in the state 10 years after graduating (PSEO)	–	**67%**[2010]	–	64%	–	#6	[11]
STEM bachelor's graduates working in the state 1 year out (PSEO)	74%[2016]	**74%**[2019]	+0.4 pp ▲	67%	+1.5 pp ▲	#4	[11]
Master's graduates working in the state 1 year out (PSEO)	–	**69%**[2016]	–	66%	–	#11	[11]
Doctoral graduates working in the state 1 year out (PSEO)	–	**55%**[2016]	–	49%	–	#3	[11]
Bachelor's graduates' median earnings 1 year out (PSEO)	$44K[2016]	**$45.1K**[2019]	+2.5% ▲	$48.1K	+2.3% ▲	#23	[11]
Bachelor's graduates' median earnings 5 years out (PSEO)	–	**$62.5K**[2016]	–	$67.3K	–	#24	[11]
STEM bachelor's graduates' median earnings 1 year out (PSEO)	$55K[2016]	**$56.1K**[2019]	+2.0% ▲	$59.3K	+0.1% ▲	#20	[11]
Net domestic migration of bachelor's-plus adults, per 1,000 residents	+11.3	**+5.4**[2025]	−5.9 ▼	–	–	#11	[12]
Net domestic migration of graduate-degree holders	+10.2K	**+196**[2025]	−10,014.0 ▼	–	–	#26	[12]
Bachelor's-plus adults who moved in from another state	4.1%	**3.3%**[2025]	−0.8 pp ▼	–	–	#17	[12]
Bachelor's-plus adults who moved out to another state	2.9%	**2.7%**[2025]	−0.2 pp ▼	–	–	#25	[12]
Bachelor's-plus adults who arrived from abroad in the past year	13.6K	**20.3K**[2025]	+49% ▲	–	–	#9	[12]
First-time students who are in-state residents	82%[2016]	**79%**[2024]	−3.0 pp ▼	–	–	#11	[4]
Net student migration (per 100 resident first-time students)	+3[2016]	**+4**[2024]	+1.2 ▲	–	–	#29	[4]

Superscript years mark the year observed; changes since 2015 start from 2015 or the first later year with data. Rank is among the 50 states and DC (1 = highest). "pp" is percentage points. Dollars are 2024 dollars (CPI-U) unless the measure is labeled nominal. Counts and dollars are rounded to three significant figures (K, M, B). Arrow colors show direction only. Source numbers refer to the list of sources that follows the tables.

NORTH CAROLINA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
WHERE STUDENTS AND GRADUATES GO (CONTINUED)							
Recent high school graduates who go to college out of state	13%[2016]	**14%[2022]**	+1.8 pp ▲	22%	+2.5 pp ▲	#41	[3]
WORK VISAS AND GREEN CARDS							
Certified H-1B positions, by worksite	34.2K	**29.3K[2025]**	−14% ▼	914K	−24% ▼	#7	[17]
Share of U.S. certified H-1B positions	2.9%	**3.2%[2025]**	+0.3 pp ▲	–	–	#7	[17]
STEM share of certified H-1B positions	91%	**88%[2025]**	−2.8 pp ▼	76%	−7.8 pp ▼	#3	[17]
Median offered H-1B wage, nominal dollars, nominal dollars	$71.1K	**$113K[2025]**	+59% ▲	$119K	+66% ▲	#12	[17]
H-1B positions at H-1B-dependent employers	58%	**30%[2025]**	−28.2 pp ▼	22%	−25.7 pp ▼	#4	[17]
Green-card labor certifications	1.67K	**3.77K[2025]**	2.27× ▲	138K	+75% ▲	#10	[17]
Share of PERM certifications for citizens of India	48%	**46%[2024]**	−2.7 pp ▼	47%	−10.6 pp ▼	#17	[17]
PERM certifications for workers on F-1 student visas	2.9%	**6.6%[2024]**	+3.7 pp ▲	8.7%	+4.9 pp ▲	#39	[17]
PERM workers with a master's or doctorate	40%	**35%[2024]**	−5.0 pp ▼	47%	−4.9 pp ▼	#29	[17]
STEM share of PERM certifications	57%	**51%[2025]**	−5.5 pp ▼	49%	−25.8 pp ▼	#13	[17]
Median offered PERM wage, nominal dollars, nominal dollars	$85.3K	**$96.6K[2025]**	+13% ▲	$100K	+9.9% ▲	#18	[17]
New lawful permanent residents (green cards), by state of residence	18.5K	**28.1K[2024]**	+52% ▲	1.36M	+30% ▲	#13	[16]
Share of U.S. new green-card holders	1.8%	**2.1%[2024]**	+0.3 pp ▲	–	–	#13	[16]
People naturalized as U.S. citizens, by state of residence	11.6K	**17.8K[2024]**	+54% ▲	819K	+12% ▲	#13	[16]
Share of U.S. naturalizations	1.6%	**2.2%[2024]**	+0.6 pp ▲	–	–	#13	[16]
Share of U.S. temporary-worker admissions, by destination	1.6%	**1.7%[2024]**	+0.2 pp ▲	–	–	#12	[16]
Share of U.S. H-1B worker admissions, by destination	1.7%	**2.0%[2024]**	+0.3 pp ▲	–	–	#11	[16]
Share of U.S. L-1 intracompany-transferee admissions, by destination	1.6%	**2.1%[2024]**	+0.5 pp ▲	–	–	#11	[16]
Share of U.S. O-1 extraordinary-ability worker admissions, by destination	0.4%	**0.7%[2024]**	+0.3 pp ▲	–	–	#18	[16]
New H-1B approvals (initial), by employer state	3.06K	**3.12K[2026]**	+2.0% ▲	127K	+14% ▲	#12	[13]
Continuing H-1B approvals, by employer state	3.94K	**6.18K[2026]**	+57% ▲	268K	+52% ▲	#14	[13]
Share of U.S. new H-1B approvals	2.7%	**2.5%[2026]**	−0.3 pp ▼	–	–	#12	[13]

Superscript years mark the year observed; changes since 2015 start from 2015 or the first later year with data. Rank is among the 50 states and DC (1 = highest). "pp" is percentage points. Dollars are 2024 dollars (CPI-U) unless the measure is labeled nominal. Counts and dollars are rounded to three significant figures (K, M, B). Arrow colors show direction only. Source numbers refer to the list of sources that follows the tables.

NORTH CAROLINA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
WORK VISAS AND GREEN CARDS (CONTINUED)							
Denial rate, new H-1B petitions	4.7%	**3.1%**^{2026}	−1.5 pp ▼	3.4%	−3.0 pp ▼	#34	[13]
New H-1B approvals at colleges and schools (NAICS 61)	8.8%	**11%**^{2026}	+2.1 pp ▲	8.6%	−1.7 pp ▼	#35	[13]
New H-1B approvals in professional, scientific & technical services	28%	**58%**^{2026}	+30.7 pp ▲	44%	−12.5 pp ▼	#5	[13]
Five largest employers' share of new H-1B approvals	68%	**33%**^{2026}	−34.7 pp ▼	9.1%	−6.7 pp ▼	#25	[13]
STEM WORKFORCE AND INNOVATION							
Workers in STEM occupations	267K	**452K**^{2024}	+69% ▲	13.7M	+51% ▲	#9	[12]
STEM workers as a share of all workers	5.8%	**8.2%**^{2024}	+2.5 pp ▲	8.1%	+2.1 pp ▲	#15	[12]
Foreign-born share of STEM workers	16%	**20%**^{2024}	+4.4 pp ▲	24%	+1.6 pp ▲	#15	[12]
Foreign-born share of STEM workers with a graduate degree	31%	**39%**^{2024}	+7.4 pp ▲	41%	+2.2 pp ▲	#12	[12]
Foreign-born share of STEM workers with a doctorate	26%	**38%**^{2024}	+12.2 pp ▲	47%	+0.9 pp ▲	#15	[12]
Noncitizen share of STEM workers	8.8%	**11%**^{2024}	+1.8 pp ▲	11%	+0.7 pp ▲	#11	[12]
STEM workers born abroad who arrived in the last 10 years	6.0%	**6.4%**^{2024}	+0.4 pp ▲	7.5%	+0.3 pp ▲	#14	[12]
Share of foreign-born STEM workers born in India	45%	**45%**^{2024}	+0.6 pp ▲	28%	+0.7 pp ▲	#1	[12]
Foreign-born share of residents	7.9%	**9.9%**^{2025}	+2.0 pp ▲	15%	+1.3 pp ▲	#21	[12]
Foreign-born share of graduate-degree holders, 25+	12%	**16%**^{2025}	+3.3 pp ▲	21%	+2.8 pp ▲	#19	[12]
STEM jobs, employer survey	227K	**368K**^{2025}	+62% ▲	10.4M	+26% ▲	#8	[10]
STEM jobs concentration vs the U.S., location quotient	0.92	**1.12**^{2025}	+0.2 ▲	1.00	0.0	#10	[10]
Mean annual STEM wage, nominal, nominal dollars	$82.5K	**$113K**^{2025}	+37% ▲	$120K	+38% ▲	#18	[10]
Median software developer wage, nominal, nominal dollars	$103K^{2019}	**$135K**^{2025}	+31% ▲	$136K	+26% ▲	#12	[10]
Software developer jobs	32.7K	**58.5K**^{2025}	+79% ▲	1.69M	+48% ▲	#7	[10]
Mean annual wage, all occupations, nominal, nominal dollars	$44.2K	**$64.3K**^{2025}	+46% ▲	$69.8K	+44% ▲	#29	[10]
Patents granted to assignees in the state	2.54K	**4.33K**^{2025}	+70% ▲	135K	−0.6% ▼	#8	[22]
Share of U.S.-assignee patents	1.9%	**3.2%**^{2025}	+1.3 pp ▲	–	–	#8	[22]
Share of the state's patents in digital technologies	37%	**42%**^{2025}	+5.2 pp ▲	43%	+1.6 pp ▲	#11	[22]
Share of the state's patents in life sciences	16%	**10%**^{2025}	−5.3 pp ▼	17%	+0.4 pp ▲	#42	[22]

Superscript years mark the year observed; changes since 2015 start from 2015 or the first later year with data. Rank is among the 50 states and DC (1 = highest). "pp" is percentage points. Dollars are 2024 dollars (CPI-U) unless the measure is labeled nominal. Counts and dollars are rounded to three significant figures (K, M, B). Arrow colors show direction only. Source numbers refer to the list of sources that follows the tables.

NORTH CAROLINA – BY THE NUMBERS (CONTINUED)

Measure	2015	Latest	Change since 2015	U.S., latest	U.S. change since 2015	Rank	Source
STEM WORKFORCE AND INNOVATION (CONTINUED)							
Share of the state's patents in semiconductors	5.2%	**1.7%**[2025]	−3.5 pp ▼	2.9%	−1.7 pp ▼	#12	[22]
Women's share of inventors on the state's patents	9.6%	**14%**[2025]	+4.2 pp ▲	14%	+3.4 pp ▲	#15	[22]

Superscript years mark the year observed; changes since 2015 start from 2015 or the first later year with data. Rank is among the 50 states and DC (1 = highest). “pp” is percentage points. Dollars are 2024 dollars (CPI-U) unless the measure is labeled nominal. Counts and dollars are rounded to three significant figures (K, M, B). Arrow colors show direction only. Source numbers refer to the list of sources that follows the tables.

Sources

[1] Institute of International Education. *Open Doors Report on International Educational Exchange: State Fact Sheets*. Editions 2020–2026. New York: Institute of International Education. https://opendoorsdata.org/. Accessed November 2026.

[2] NAFSA: Association of International Educators. *NAFSA International Student Economic Value Tool*. Academic years 2018–19 to 2024–25. Washington, DC: NAFSA. https://www.nafsa.org/isev. Accessed November 2026.

[3] National Center for Education Statistics. *Digest of Education Statistics*. Editions 1995–2025. Washington, DC: U.S. Department of Education, Institute of Education Sciences. https://nces.ed.gov/programs/digest/. Accessed November 2026.

[4] National Center for Education Statistics. *Integrated Postsecondary Education Data System (IPEDS)*. Survey files, 1980–2025. Washington, DC: U.S. Department of Education, Institute of Education Sciences. https://nces.ed.gov/ipeds/. Accessed November 2026.

[5] National Center for Education Statistics. *National Assessment of Educational Progress (NAEP)*. NAEP Data Service, 1990–2024. Washington, DC: U.S. Department of Education, Institute of Education Sciences. https://www.nationsreportcard.gov/. Accessed November 2026.

[6] National Center for Science and Engineering Statistics. *Higher Education Research and Development Survey*. Data tables, FY2003–2024. Alexandria, VA: National Science Foundation. https://ncses.nsf.gov/surveys/higher-education-research-development. Accessed November 2026.

[7] National Center for Science and Engineering Statistics. *Survey of Earned Doctorates*. Data tables, 1994–2025. Alexandria, VA: National Science Foundation. https://ncses.nsf.gov/surveys/earned-doctorates. Accessed November 2026.

[8] National Center for Science and Engineering Statistics. *Survey of Federal Funds for Research and Development*. Data tables, FY1985–2024. Alexandria, VA: National Science Foundation. https://ncses.nsf.gov/surveys/federal-funds-research-development. Accessed November 2026.

[9] National Center for Science and Engineering Statistics. *Survey of Graduate Students and Postdoctorates in Science and Engineering*. Data tables, 1994–2024. Alexandria, VA: National Science Foundation. https://ncses.nsf.gov/surveys/graduate-students-postdoctorates-s-e. Accessed November 2026.

[10] U.S. Bureau of Labor Statistics. *Occupational Employment and Wage Statistics*. State estimates, 1999–2025. Washington, DC: U.S. Department of Labor. https://www.bls.gov/oes/. Accessed November 2026.

[11] U.S. Census Bureau, Center for Economic Studies. *Post-Secondary Employment Outcomes (PSEO)*. Release R2025. Washington, DC: U.S. Department of Commerce. https://lehd.ces.census.gov/data/pseo_experimental.html. Accessed November 2026.

[12] U.S. Census Bureau. *American Community Survey*. 1-year estimates and Public Use Microdata Sample, 2010–2025. Washington, DC: U.S. Department of Commerce. https://data.census.gov/. Accessed November 2026.

[13] U.S. Citizenship and Immigration Services. *H-1B Employer Data Hub*. Fiscal years 2009–2026. Washington, DC: U.S. Department of Homeland Security. https://www.uscis.gov/tools/reports-and-studies/h-1b-employer-data-hub. Accessed November 2026.

[14] U.S. Department of Education, Office for Civil Rights. *Civil Rights Data Collection*. School years 1980–81 to 2023–24. Washington, DC: U.S. Department of Education. https://civilrightsdata.ed.gov/. Accessed November 2026.

[15] U.S. Department of Education. *College Scorecard*. Institution and field-of-study data, 1996–2025. Washington, DC: U.S. Department of Education. https://collegescorecard.ed.gov/data/. Accessed November 2026.

[16] U.S. Department of Homeland Security, Office of Homeland Security Statistics. *Yearbook of Immigration Statistics*. Fiscal years 1996–2024. Washington, DC: U.S. Department of Homeland Security. https://ohss.dhs.gov/topics/immigration/yearbook. Accessed November 2026.

[17] U.S. Department of Labor, Employment and Training Administration, Office of Foreign Labor Certification. *LCA and PERM Disclosure Data*. Fiscal years 2008–2026. Washington, DC: U.S. Department of Labor. https://www.dol.gov/agencies/eta/foreign-labor/performance. Accessed November 2026.

[18] U.S. Department of State, Bureau of Educational and Cultural Affairs. *BridgeUSA: Exchange Visitor Program Participants by State*. 2015–2025. Washington, DC: U.S. Department of State. https://j1visa.state.gov/. Accessed November 2026.

[19] U.S. Department of the Treasury, Bureau of the Fiscal Service. *USAspending.gov*. Award data, FY2008–2026. Washington, DC: U.S. Department of the Treasury. https://www.usaspending.gov/. Accessed November 2026.

[20] U.S. Immigration and Customs Enforcement, Student and Exchange Visitor Program. *SEVIS by the Numbers: Data Mapping Tool*. Monthly data, 2014–2026. Washington, DC: U.S. Department of Homeland Security. https://studyinthestates.dhs.gov/sevis-data-mapping-tool. Accessed November 2026.

[21] U.S. Immigration and Customs Enforcement. *Student and Exchange Visitor Information System (SEVIS) Data*. Freedom of Information Act releases, 2008–2016. Washington, DC: U.S. Department of Homeland Security. https://www.ice.gov/foia/library. Accessed November 2026.

[22] U.S. Patent and Trademark Office. *PatentsView*. Annualized patent data, grant years 1976–2025. Alexandria, VA: U.S. Patent and Trademark Office. https://patentsview.org/. Accessed November 2026.

This page intentionally left blank.